Apple — CapEx Systems Onboarding Data Integrity Program Manager

30-minute call packet · Rajesh Rishiyur Chandrasekaran · interviewer: Capital Expenditure Systems Lead · 76 pages

Read in this order if short on time: Part 1 pages 1–3 (cram sheet, tell me about yourself, why Apple) → Part 4 (the use case, so the asset story is crystal clear) → Part 1 solution design → Part 3 Elementum shortage alert → Part 6 Looker flow → Part 2 as a question bank.

1. Main packet — cram sheet, TMAY, why Apple, solution design, resume map, 15 STAR answers, questions to ask	p. 2	16 pp
2. Recruiter's 14 sample questions — answered	p. 18	15 pp
3. Elementum stories — Class A shortage alert (8 domains, nightly + delta feeds, the calculation) and procure-to-pay EDI	p. 33	11 pp
4. Real-world use case — asset management at a contract manufacturer	p. 44	6 pp
5. How a BI tool works — Looker primer	p. 50	7 pp
6. Looker end-to-end technical flow on the use case	p. 57	9 pp
7. Looker hands-on lab guide (data + SQL + LookML are in looker-lab.zip)	p. 66	11 pp

Say-aloud rules for this packet.

- Stories come from four places only: Sephora S/4 HANA, Sephora UDC dual inventory, Elementum, Flexport. No other migration stories.
- Everything describing Apple's CapEx tool or process is **your model** built from public knowledge and the job description — say "here is how I understand it; correct me where I'm wrong."
- Numbers carry a basis label: record / record-from-docs / user anchor / illustrative. Placeholders in [brackets] are yours to fill or omit; never guess them aloud.
- BI proficiency: you wrote the seven metric definitions and specified Looker SQL Runner queries for the Zendesk-to-Snowflake pipeline; you built weekly-review metrics in Excel and SQL; you have not built dashboards hands-on in Looker, Tableau or Power BI — say so plainly.
- The placed-in-service date is a financial control, not a formality.

30-MINUTE CALL · PRINT AND MARK UP · SPEAK ONLY WHAT IS IN THE SAY BOXES

Apple CapEx — Systems Onboarding Data Integrity Program Manager

You. Rajesh Rishiyur Chandrasekaran

Them. Capital Expenditure Systems Lead, Apple, Cupertino, hybrid. In role since June 2025; before that, CapEx Operations Program Manager (Feb 2022 – Jun 2025), including OPM on the first pilot in the line of business.

The seat. CapEx Systems Onboarding Data Integrity Program Manager. The team's stated purpose is change management: move people off "we have always done it this way," close the work without escalation, and make every input have a why.

Your position in minute one. You have moved live operations off side files onto a system of record, and you run data integrity as counted rows with an owner. You have not seen Apple's proprietary CapEx tool, and you will not pretend you have.

A. One-page cram sheet

CLOCK	WHAT YOU DO
0–2 Rapport	One human beat, then their language: system of record, pilot into a playbook, Procurement / Engineering / contract manufacturer. Do not pitch.
2–5 Tell me about yourself	75-second version in section B. Land four beats and stop: system of record, counted-row integrity, change without escalation, getting into the weeds. If they nod early, use the 45-second cut.
5–20 Their questions	Four answers cold, in this order of likelihood: **(1)** off spreadsheets onto a system of record — UDC dual inventory. **(5)** budget, schedule, and supplier in tension — Utah conversion, Locus vendor, council July 31. **(10)/(11)** a decision from someone who does not report to you, and a review that actually decides — architecture council, 44 rows. **(14)** a gate you missed — functional design planned July 26, signed August 14. Section H has the other eleven compact.
20–26 Product idea	Only if they invite "how would you improve this." Otherwise hold section G and stay on their thread. The idea is a question, not a critique.
26–30 Your questions + close	The three questions in section J, in that order. Close on one sentence: one instrument, one owner per field, no escalation of a row the owner can close.

Five numbers never to get wrong

- **7 → 68 customers = 9.7×**, targets landed two months early. `record` Never a promotional multiplier.
- **$0.66 → $0.03 per container, about 95%.** `record` The hand-carried start near a dollar comes up only if they ask.
- **S/4 exit bars:** no SKU with two published AFS values; **Negative AFS clean at T+1**. January go-live, reserve rules live against demand; May go-live, execution. The order system reads ISM AFS and never computes. `record` design. Oversell, recurring, to zero through both.
- **836-row log, 714 closed = 85.4%.** Council pack: **44 requirement rows across 23 systems**; **seven system families, about two dozen named systems**. **Three reserve buckets**, not a longer list. `record / record-from-docs`
- **Functional design planned July 26, signed August 14.** Never say the gate chain held to the day. The other miss, if asked: five users passed ten scenarios; concurrent access failed two days before sign-off. `record`

Three phrases in their language

- **System of record.** "The side file stops being the decision once the forum will only read the system."
- **Data integrity, and the why behind every input.** "I can tell you who owns the field, what breaks if it is wrong, and which control catches it."
- **Change without escalation, playbook from the pilot.** "I pre-wire the owner. The pilot writes the playbook. I do not escalate a row the owner can close."

Three things never to say

1. Any promotional customer multiplier, and the resume's aggregate savings line. Use 9.7× and the program-level noun, one program at a time.
2. Inside knowledge of Apple's proprietary CapEx tool, or that your language-model triage agent was deployed at an employer. It was an independent build.
3. "The gate chain held to the day." The slide said July 26; the signature was August 14. Also never: nine reserve buckets; a device-failure ratio; a bill-of-material explosion; a six-site North America claim.

B. Tell me about yourself

Built for this interviewer. Four beats, in their order of care: system of record, counted integrity, change without escalation, weeds. Resume-faithful. No Apple tool, no aggregate savings.

75-SECOND VERSION — SAY THIS, THEN STOP

I run the operating system around a live supply-chain system, and I get into the data myself.

First, I have moved live operations off spreadsheets and side files onto a system of record. At Sephora a Utah building ran as two inventory books. We put it on one pool, with two selling identities and three reserve buckets, and we refused to convert historical transactions. On the inventory publish, S/4 HANA and ISM became the one number. The order system reads available-for-sale and never computes its own.

Second, I run data integrity as a program with counted rows. The bars on that publish were no SKU with two published AFS values, and Negative AFS clean at T+1. The Utah program carried an 836-row issue log; 714 closed, 85.4 percent. On a European transportation program the report said 40 percent data quality; I pulled the history myself, the true figure was about 55 percent, and it reached about 99 percent within a quarter.

Third, I move teams who have always done it this way without escalating. I pre-wire every row with the person who owns the input. On a cadence change, the skeptic wrote the gate.

The weeds are the work. I have read application logs to find a missing trace id, and I count the rows myself before I call a go or a no-go.

45-SECOND VERSION — IF THEY CUT YOU OFF OR THE ROOM IS WARM

I move live operations off side files onto a system of record, and I treat data integrity as the program. At Sephora we took a Utah building from two inventory books to one pool with three reserve buckets, and we did not convert history. The order system reads one published available-for-sale and never computes its own. The bars were no SKU with two published AFS values, and Negative AFS clean at T+1. I keep an issue log with an owner on every row — 836 rows, 85.4 percent closed — and I get teams to change by pre-wiring the owner, not by escalating. I count the rows myself before I call a gate.

- **If they ask “what did you personally do?”** “I wrote the scorecard, the gates, and the cadence, and I made the cutover call. Engineering built the code. DC Ops measured the 15 percent processing efficiency.”
- **Do not add** the aggregate savings line, a customer multiplier, or any sentence that starts “at Apple you do it by…”

C. Why Apple

60-SECOND VERSION — SPECIFIC TO THIS TEAM

You are moving CapEx off Excel-heavy, person-dependent tracking onto a system of record, so that later AI has a foundation it can trust. That sequence is the one I run. I will not automate a field I cannot reconcile. The only language-model tool I have built was an independent, eval-first exercise. It was never deployed at an employer.

The pattern on your profile — a first pilot in the line of business, then a playbook other programs can launch — is how I scale. I standardize the gate and the owner of each input, and I leave local whatever the floor truly varies.

What I want at Apple is the operations problem at this company’s manufacturing and supply-chain scale: capital equipment through contract manufacturers, where a placed-in-service date and the link from purchase order to asset are both a launch fact and a book fact. The job is change management with teams who have always done it this way, closed without escalation. That is work I have actually done, in warehouse systems and in data programs, not inside your tool.

I have not seen the proprietary system. If you want, I can walk the operating model I would bring, and I want you to correct it.

Grounding. The Excel-to-system migration, the centralized foundation for future AI, the pilot-to-playbook loop, and the Procurement / Engineering / contract-manufacturer alignment are the interviewer’s own public profile, as you have it. They are

not a claim about how the tool is built. The Apple line is scale and operations rigor in supply chain and manufacturing. No internal tool name, no invented metric, no “I already know your process.”

D. Solution design — how a CapEx system works at a company like Apple

My model, not inside knowledge. This section is my model of the domain, built from public knowledge of capital-equipment programs, the job description, and the interviewer’s public profile. It is not knowledge of Apple’s proprietary tool. In the call I say: “Here is how I understand it; correct me where I’m wrong.”

The interviewer’s profile describes a migration of multi-million-dollar programs off Excel-heavy, person-dependent tracking onto one system of record, with Procurement, Engineering, and contract manufacturers in the same launch, and with pilot lessons turned into a playbook. The picture below is how I would expect that operating model to be structured. Every dollar, station count, and date in section E is labelled illustrative.

IF THEY ASK YOU TO WALK IT — ABOUT 90 SECONDS, THEN HAND IT BACK

Here is how I understand a capital-equipment program on a system of record. Correct me where Apple does it differently.

It starts with a program need and a capacity model: units per hour, yield, shifts, lines. That becomes a list of tools and fixtures, then an appropriation with a cost center and a program code. Global Supply Management runs the quote and the purchase order. The supplier builds. Factory acceptance and site acceptance are different dates from placed-in-service. The contract manufacturer installs, tags, and says where the tool sits. Finance capitalizes: asset id, location, custodian, in-service date, depreciation start. Then ramp, then reuse or retrofit against a source asset, then disposition.

The seat I am interviewing for does not own those decisions. It makes sure every input has an owner, a reason, and a control that catches a bad value before the book closes. The join I would watch first is purchase order to asset. I would onboard one program as the pilot, write the playbook from the misses, and scale on the same gates.

If they turn “first ninety days” back on you

- **Days 1–20.** One live program. Count the side file against the system, field by field. The gap is a list, not a vibe.
- **Days 21–45.** An owner on each of the ten fields. A blank owner is the first exception, and it does not wait for a tool.
- **Days 46–70.** One reconciliation, purchase order to asset, denominator fixed, published before any cleanup.
- **Days 71–90.** One playbook page from that pilot: what we standardized, what stayed local. No new system.

That plan is my model of the seat. It is not a claim that Apple’s first ninety days are empty.

1. Lifecycle of capital equipment at a contract manufacturer

Program need → capacity model (units per hour, yield, shifts, lines) → equipment list, the bill of tools and fixtures → budget and approval (capital appropriation, cost center, program) → sourcing (GSM: RFQ, quote, purchase order) → supplier build → FAT then SAT → ship and install at the CM → asset tagging and capitalization (asset master, placed-in-service date, depreciation start, location, custodian) → ramp → sustaining, including retrofit and reuse across programs → disposition (transfer, idle, or scrap).

	1 Need	2 Buy	3 Accept	4 Books	5 Ramp	6 Reuse
TPM / AME	Capacity + tool list	spec	FAT / SAT	consulted	yield	retrofit decision
GSM	informed	RFQ, quote PO	supplier	PO open	—	transfer
CM site	shifts, lines	—	install	tag + location	runs line	physical
Finance / Systems / DI	program code	cost center on the PO	acceptance date	in-service + asset id	watch	source asset link

Dark cells are the lane that enters the input. Pale cells are consulted or informed. DI = Data Integrity PM.

The join that must not break: purchase order line → asset id, with one owner, before capitalization.

Model only. Correct me where Apple's lanes are different.

Swimlane. Six stages, four lanes. Read left to right. The books lane does not invent the physical fact; it records the fact the site and the GSM already own.

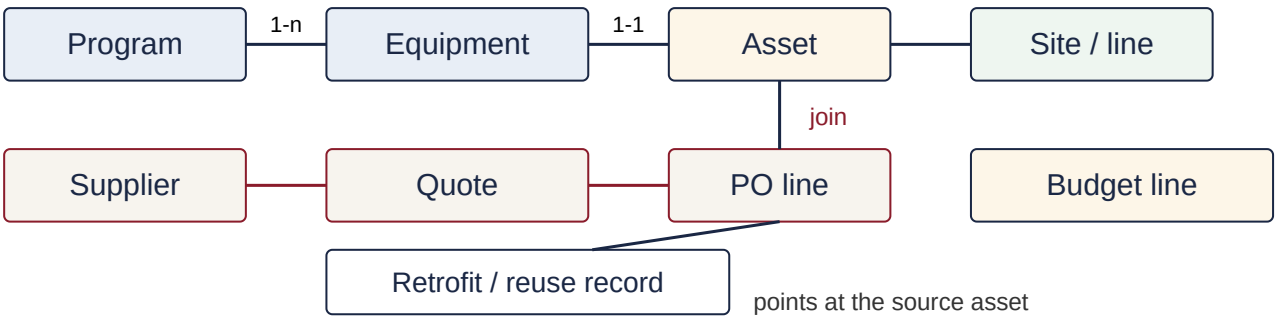
2. Who owns which input

One accountable owner per input. The Data Integrity PM does not take the business decision. The Data Integrity PM makes the missing or conflicting input visible, aged, and closable by that owner.

INPUT	GSM	TPM	AME	CM	FINANCE	SYS PM	DI PM
Capacity (UPH, yield, shifts, lines)	C	A	R	C	I	I	C
Tool and fixture list, internal ID	C	A	R	C	I	C	R
Appropriation, cost center, program	C	R	C	I	A	C	C
RFQ, quote, PO, currency	A	C	I	I	C	R	C
FAT / SAT and acceptance date	C	A	R	C	I	I	C
Install location, tag, custodian	I	C	C	R	C	C	A
Placed-in-service, depreciation start	I	C	I	C	A	R	R
Shared-tool program allocation	C	A	C	I	C	C	R
Reuse source asset	C	A	R	C	C	C	R
Exception queue for a broken join	R	C	C	R	C	C	A

R does the entry. A is the single accountable owner. C is consulted before the value is trusted. I is informed. On a shared tool, TPM is accountable for the split and Finance is accountable for the book.

3. Data model, and the why behind ten fields



Entity sketch. The purchase-order line is the join between what was bought and what is on the book. A reuse record points at an existing asset and must not create a second one.

ENTITY	FIELDS THAT HAVE TO BE TRUE
Program	Program code, owner, CM sites, appropriation id
Equipment / tool	Internal ID, supplier part number, serialized or quantity, station

ENTITY	FIELDS THAT HAVE TO BE TRUE
Supplier / quote / PO	Supplier id, quote currency, PO line, quantity, cost center
Asset	Asset id, tag, location, custodian, placed-in-service date, depreciation start
Site / line	CM, line, station
Budget line	Appropriation, program allocation if the tool is shared
Retrofit / reuse	Source asset id, receiving program, disposition of the old use

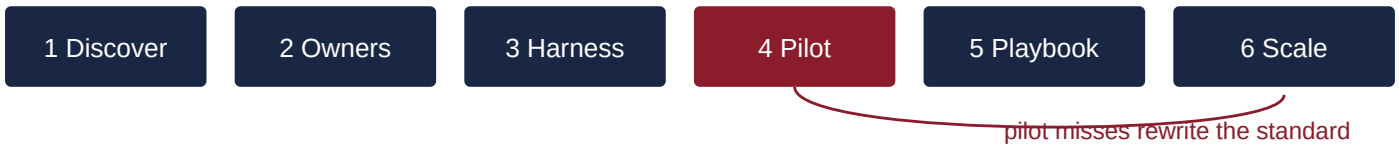
FIELD THAT GOES WRONG	WHY THE INPUT EXISTS	WHAT A BAD VALUE DOES
PO ↔ asset linkage	The buy and the book are one fact.	Orphan asset or an open PO that looks received.
Placed-in-service date	Depreciation starts on a real date.	Expense too early, or a live tool still “in transit” on the book.
Quantity vs serialized units	A station and a fixture are not the same grain.	40 ordered becomes 1 asset, or 40 tags for one line.
Site / location	The CM that holds the tool.	Wrong ramp, wrong custodian, failed physical count.
Cost center	Which org carries the asset.	FP&A cannot see the program.
Program allocation of a shared tool	One tool, more than one program.	Double count, or a hole in both programs.
Reuse source asset	Stops a second buy of a tool already on the book.	A new PO and a ghost asset.
FX / currency	The quote and the PO must be the same money.	Appropriation over or under, silent.
Acceptance date	FAT or SAT passed. It is not in-service.	Depreciation starts before the tool can run.
Supplier PN vs internal ID	The vendor’s number will not join the asset master.	Duplicates on every refresh from the CM or the supplier.

4. Where integrity breaks, and the control

WHERE IT ENTERS	TYPICAL BAD INPUT	CONTROL
Excel → system migration	Duplicate IDs, quantity pasted over serials, blank owner	Pre-load harness: schema, duplicate, referential checks, counted rows. No load on a red count.
Vendor quote and PO	Currency, supplier part number	Validation at entry. Currency required. A crosswalk from supplier part number to internal ID, owned by GSM + Data Integrity.
CM status update	Location, acceptance date typed as in-service	Owner per field. SAT pack required before acceptance date can close. CM cannot edit the depreciation date.
Manual status	Placed-in-service, program split, “someone knows”	PO ↔ invoice ↔ asset reconciliation. Exception queue with owner and age. Periodic physical-to-book count. Audit trail on every override.

The go or no-go is a count of rows, the way I read the S/4 publish: no SKU with two published AFS values, and Negative AFS clean at T+1. A gate that can be rewritten after the file arrives is not a gate. Beside that, the issues log is the triage — 331 go-live, 77 fast-follow, 36 enhancement — so a red row has an owner before it becomes a speech. Person-dependent fields — the ones only one analyst can explain — are flagged on the scorecard until they have an owner in the system.

5. Onboarding a new program — six gates, then the playbook loop



GATE	ENTRY	EXIT BAR
1 Discover	Named sponsor and program code	Equipment list counted and signed by TPM and AME. Row count written down.
2 Owners	Signed list	An owner on each of the ten fields above. Blank owner blocks the gate.
3 Harness	The Excel or extract, not production	Schema, duplicate, serial, and referential checks green. Counted rows match the signed list.
4 Pilot load	Harness green, one line or one CM	Counted-rows go or no-go, same shape as the S/4 bars: no record with two published values of the field, and the exception report clean at the next check. Every open row has an owner and an age.

GATE	ENTRY	EXIT BAR
5 Playbook	The pilot's exception log	Standard work the next program can run without the pilot team in the room. What stayed local is written, not implied.
6 Scale	Playbook plus the same gates	First-time-right on the ten fields. The side file is not what the forum reads.

What I standardize from a pilot: the field owner, the exit bar, the exception grammar, and the reconciliation. What I leave local: the CM's station naming and the supplier's pack format, mapped through the crosswalk, not forced into one spreadsheet layout. That split is the same one I use when a pilot becomes a rollout.

E. Worked example — a new test-station line at a CM

> **Illustrative, end to end.** One made-up program: a new test-station line at a contract manufacturer for a product ramp. Every count and dollar in this section is ILLUSTRATIVE . None of it is an Apple program, an Apple site, or a figure from my record.

Shape of the example: ILLUSTRATIVE 40 stations at an illustrative unit cost of $75,000, so an illustrative equipment cost of $3,000,000. Two CM sites. Three suppliers. Twelve fixtures reused from a prior program, so those twelve must point at source assets and must not become twelve new purchase orders.

STEP	INPUT	WHO ENTERS	WHAT GOES WRONG	CONTROL AND EXCEPTION PATH
Need	Ramp needs a test line	TPM	Scope written as "tools" with no program code	No list starts without a program code. Data Integrity sends it back.
Capacity	UPH, yield, shifts, lines	AME, CM confirms shifts	CM shift pattern differs from the model	AME owns the number. A variance is a row, not a quiet edit.
Tool list	40 stations + fixtures; 12 marked reuse	AME drafts, TPM accountable, DI counts	12 reused fixtures pasted in as new buys	Reuse flag required. Source asset id required or the row is red.
Budget	Appropriation and cost center	Finance accountable, TPM requests	Shared metrology tool allocated 100% to this program	Shared-tool rule: allocation must sum to 100% across programs.
Source	3 suppliers, quotes, POs	GSM	One quote in a different currency	Currency required. Mismatch parks the line before the PO is cut.
Build / FAT	Supplier build, FAT date	Supplier via GSM; AME witnesses	FAT date later typed as in-service	Acceptance date and in-service date are different fields. CM cannot edit the second.
Ship / install	Site, line, station	CM	A station arrives at site 2 with site 1 on the PO	Location owner is the CM. Mismatch opens an exception; the asset is not capitalized.
Tag / capitalize	Asset id, tag, in-service, custodian	Finance accountable; Systems PM posts; DI reconciles	See the exception below	Counted reconciliation before the book closes.
Ramp	Station actually running	CM and AME	Book says in service; line is not yielding	In-service requires SAT plus CM confirmation, not FAT alone.
Reuse / disposition	12 source assets	TPM accountable, Finance updates the book	Old asset still "in use" on the prior program	Reuse record closes the old use or the new row stays open.

One reconciliation, all the way through

ILLUSTRATIVE At capitalization week the counted reconciliation is: 40 purchase-order lines in scope, 38 assets tagged, 2 lines with no asset. The two are not "missing." The carrier event says they are in transit to site 2, and the CM has not received them.

- **Log.** Two exception rows, same day, on the program queue. Type: PO without asset. Grain: PO line. Count written on the row: 40 expected, 38 tagged, 2 open.
- **Owner.** Data Integrity PM is accountable for the queue. GSM owns the PO fact. The CM owns the receipt. Finance does not capitalize either line.
- **Age.** The clock starts at the promised receipt date. I would set the age bar with Finance during the pilot. I am not assuming Apple's number.
- **Close.** A row closes only when an asset id, a tag, a location, and a placed-in-service date post, or when GSM confirms the line is still a legitimate in-transit with a carrier event newer than the age bar. "We think it arrived" does not close it.
- **What I refuse to do.** I do not let the 38 post and drop the 2 off the scorecard so the program looks at 100%. The denominator stays 40 until the two rows close or the scope is formally cut.

That is the same mechanism as the S/4 go/no-go: the exit bar and the denominator are written before the file arrives — no record with two published values, the exception report clean at the next check — classify the miss, name the owner, and do not change the denominator to go green. The Utah issues log is how I triaged that kind of queue: 331 go-live, 77 fast-follow, 36 enhancement.

F. Resume → this role

Every experience bullet on the Apple-tailored resume, in the inventory already mapped in the Apple delta add-on. The raw resume text file was not on disk for this build; that add-on states the table is complete. The summary line is a guardrail, not a story.

RESUME BULLET	JD LINE IT PROVES	STORY AND NUMBER	20-SECOND VERSION
S/4HANA ↔ ISM; one definition of available; legacy inventory system retired; zero incidents.	Accuracy of the input. One meaning the whole chain uses.	S1. Order system reads ISM AFS and never computes. Exit bars: no SKU with two published AFS values; Negative AFS clean at T+1. January, reserve rules live against demand; May, execution. Oversell recurring → zero. **record**	I owned the policy and the go/no-go recommendation. ERP engineers owned configuration. The AFS correction report had a screen, a role, and an owner.
Portfolio system: RAID, 836-item log, numeric rollback, 60+ stakeholders.	Track, prioritize, and close many tasks. Protocols.	836 rows, 714 closed = 85.4%. Categories 331 go-live, 77 fast-follow, 36 enhancement. 60+ stakeholders. **record**	One log for the umbrella program, not a side status. Owner, age, and a close. A percent closed was not the gate. Rollback triggers were numeric and never fired.
UDC omni launch on Blue Yonder plus robotics; staged gates; efficiency up.	Manufacturing and supply-chain program work. Milestones. Learn the system.	+15% processing efficiency, DC Ops' measure. Two go-lives, zero customer-facing incidents. **record**	I had not configured that WMS. I ran gates. DC Ops measured the 15 percent. I do not claim the code.
Re-sequenced an 18-month roadmap; 66% less effort; value in six months.	Influence. Analytical. Change a plan people already believed.	18 months to value in 6. 66% less effort. About $5M of avoided cost, once. **record**	I brought the cost-to-serve comparison to the person who owned the roadmap. Avoided cost, not an annual reduction.
Scaled 4PL visibility from 7 customers; cost-to-serve; a separate accuracy program.	Pilot, then a wider launch. Data integrity. Learn software fast.	7 → 68 = 9.7×, two months early. $0.66 → $0.03 per container, about 95%. Accuracy 60% → 90% is the other story. **record**	Six graduation criteria, not a date. I keep the customer ramp and the accuracy pipeline as two stories.
Utah omni conversion with a Blue Yonder upgrade on the same path, and a Locus vendor under a statement of work.	Budget, schedule, and supplier in one program. High accuracy. The upgrade is part of this conversion.	Council signed July 31. Functional design planned July 26, signed August 14. Locus: 94% against a 98% gate; full automation in week three. Two go-lives, zero customer-facing incidents, +15% by DC Ops' measure. **record**	I will not quote a program-budget variance here. The date moved in the open. The gate did not move. I do not tell the warehouse upgrade as its own migration.
VP-sponsored move of a 20-person group onto a four-week cadence; defects down.	Change management without escalation. People skills.	20 people. Six-week to four-week cadence in six weeks. Critical defects −50%. **record**	Nobody reported to me. The skeptic in QA wrote the gate and defended it.
SAP supply and demand view; daily reconciliation; forecast accuracy; safety stock.	Analytical, high accuracy, adherence to a protocol. Excel-grade hygiene on a system.	Forecast accuracy 70% → 85%. About $2M safety stock released. Planner collaboration up 50%. **record**	I say collaboration up, not effort down. The release rule was the one Planning and Finance signed.
About 1,000 supplier and carrier routes; valid ASN at or above 95%; completeness up.	Vendor onboarding. Data integrity as the toll to go live.	A thousand-lane program across 25 European locations. Completeness 60% → 95%. **record** Delivered-count split is not settled.	Phased by value. If asked what went live first, the 597 figure is my earlier anchor, not the settled count.
MetricStream: dual-hat PM and solution lead. SOWs, estimates, budgets, onshore and offshore, through UAT, cutover, hypercare.	Vendor relationships. Timelines. Do what the delivery needs.	No safe client, headcount, or dollar in the source set.	I owned the SOW and the delivery. I will not invent a client or a value.

RESUME BULLET	JD LINE IT PROVES	STORY AND NUMBER	20-SECOND VERSION
Manhattan OMS across 3 DCs and 100 stores; stockouts, fill, protected sales.	Get into the weeds. Cover the live operation. Supply chain.	99% fill vs a 95% bar. Stockouts −20%. About $6M on the line when paged; about $8M of that day's sales protected. **record**	Those are different nouns. On the line is not the same figure as protected.
Summary line only: an aggregate savings claim.	Do not use it to prove anything.	No reconciled ledger. Do not say the aggregate.	I will use each program's number with its noun rather than blend savings, avoided cost, released inventory, and protected revenue.

What I have not done. I have not used Apple's proprietary CapEx tool, and I have not run a factory capital-equipment launch. Tableau and Power BI do not appear in the mapped experience bullets. **Safe sentence:** "I build the scorecard in the tool the team already reads. I am not a dashboard stylist. I can interrogate the data underneath it, including in SQL, which I have done when a reported quality number did not match the rows. If this team's view is Tableau, I will learn the view. The grain, the owner, and the exit bar are the part I already run." Excel is a must-have on the JD and is how the side files I replace are actually kept; I do not need to pretend it is the system of record.

G. Product improvement idea — ask it, do not pitch it

THE QUESTION, ABOUT 45 SECONDS, ONLY IF INVITED

The profile says the system of record is the foundation for later AI. I would not put a model on a field I cannot reconcile, because a wrong join compounds. The thing I would want in the first pilot is a weekly data-integrity scorecard, per program and per input owner: completeness and validity on the fields that actually move the book, exception aging, purchase-order to asset reconciliation with the denominator held still, and a flag on any field that only one person can explain. Onboarding playbooks would be measured on a first-time-right rate, not on a go-live date. Does that grain already exist, and is the gap the owner — or is the gap the view?

Three metrics, defined before the first file arrives.

- **Reconciliation.** Tagged assets divided by purchase-order lines in scope. In-transit stays in the denominator until the row closes or scope is cut.
- **First-time-right.** Share of the ten high-risk fields that pass validation on first load, by program and by owner.
- **Exception aging.** Share of open exceptions older than the age bar Finance and the Data Integrity PM set in the pilot. I will not invent that bar in the room.

MVP in six weeks, one pilot program. Weeks 1–2: field dictionary and a named owner for each of the ten fields. Weeks 3–4: one reconciliation extract, PO to invoice to asset, and an exception queue with owner and age. Weeks 5–6: one weekly view the forum actually reads, even if the refresh is a query. A polished visual can wait.

What I would not build first. An anomaly model, a contract-manufacturer portal, a backfill of every historical program, or a leaderboard. The second, smaller idea is the pre-migration harness: schema, duplicate and serial checks, referential checks, and a counted-rows go or no-go before any Excel file touches the system — the same shape as Negative AFS clean at T+1, and as the issues-log triage that separated go-live from fast-follow. That harness is the entry bar to the scorecard, not a second product.

One risk. If the view shames owners, they go back to the side file. I would design it as that owner's queue, reviewed in their forum, with Data Integrity accountable for the queue and the business owner accountable for the value.

Still my model. I am describing a scorecard I would propose. I am not describing a gap I have seen inside Apple. The AI line is the interviewer's own public profile: a centralized data foundation for future AI applications. My independent triage agent — eval set first, a citation on every proposal, unit cost from $25 to $0.40 — is the proof I know how to refuse automation. It was not deployed at an employer.

H. Likely questions — STAR

Cold answers are 1, 5, and 10 with 11. The other eleven are the 45-second cut. Every answer is first person. A number carries its basis. One failure line. One data-integrity line: what was wrong, who owned it, what control fixed it.

1. A time you took a process off spreadsheets onto a system of record — what broke in month one, what you refused to migrate

S At Sephora the Utah building ran as two systemic warehouses, Dotcom and Retail, with transfers and side tracking between the books. **record**

T Put one physical unit on one system of record, with no customer-facing miss, and without migrating history I could not defend.

A Discovery, then an architecture council, one row per impacted system, pre-wired with its owner. The model we signed was one pool, two selling identities, and three reserve buckets, with no conversion of historical transactions. Purchasing and inbound went first; the combined cutover came after. In the interim, every report had to name which book it read.

R Two go-lives, zero customer-facing incidents. Processing efficiency +15%, DC Ops' own measurement. **record** The issue log for that umbrella program is 836 rows.

Failure. Month one of the interim broke for the analyst, not the shopper: two truths, hand reconciliation. I had left that reporting experience off the first deck. What I refused to migrate was history.

Data integrity. The bad input was "which book is this unit on." The council owned the definition. The control was one location, three named buckets, and a written refusal to convert history.

2. How you got people to stop keeping a side file

S A weekly review was four systems plus the extract someone trusted, because the systems disagreed. **record**

T Make the side file stop being the decision, without banning it by mandate.

A One scorecard, one grain, one date. The forum would not read a number that did not match it. Cutover waited for a parallel run within 1%. **record**

R System accuracy moved from 60% to 90%. **record** The extract died because the meeting stopped accepting it.

Failure. Usage is not adoption. Adoption is the old path losing its seat in the decision.

Data integrity. The bad input was a metric with no grain. I owned the instrument. The control was one scorecard the forum would actually use.

3. A pilot you ran before a wider rollout — what you standardized, what you left local

S Shipment Visibility started as hand-carried pilots. Seven customers. **record**

T Scale only what the pilot had proved.

A I standardized six graduation criteria and a calibration bar on the customer's own containers. I left carrier-specific mapping local. I do not name the first two customers.

R 7 → 68 customers, 9.7×, and 170 users, two months early. **record** A public shipment target stayed a target.

Failure. I will not say every public target landed. The customer count and the user count did.

Data integrity. The bad input would have been our demo file standing in for theirs. I owned the entry bar. The control was no graduation without the customer's own data.

4. When the system and the floor disagreed, which did you change

S Locus robots, Salt Lake City DC, week two: 94% pick-accurate against a 98% gate published before the bots took volume. **record**

T Choose the floor or the gate. Not both, quietly.

A I did not lower the gate. I throttled about 30% of volume onto manual carts while zone logic was fixed.

R Full automation in week three. Combined go-live held. Zero customer-facing incidents. **record**

Failure. Physical readiness was not its own row, so the earlier slip arrived as a vendor message.

Data integrity. The input was pick-accurate throughput. The program owned the gate; the floor lived the consequence. The control was the dial, with the denominator unchanged.

5. A multi-team launch where budget, schedule, and supplier were all in tension

S Utah, one building, two inventory books, a Blue Yonder upgrade on the same path, and Locus robots on retail outbound. The architecture council signed July 31. Peak, October through December, was not mine to spend. I did not own the systems or the vendor. `record`

T Hold two go-lives — purchasing and inbound in January, the combined flip in May — with the vendor's readiness honest, and without a customer-facing incident. I will not invent a program-budget variance for this story.

A The council pack asked for separable decisions: the architecture, one pool with two selling identities and three reserve buckets, the staged timeline, and a written fence. Functional design was planned for July 26 and signed August 14. I do not say the chain held to the day. Locus sat under a statement of work. The gate I published before a bot took volume was 98 percent pick-accurate throughput. Retail testing slipped three weeks. Steering got the choice that week: compress the test, or move retail end-to-end to the end of January and keep the test whole. We moved the date. Week two of go-live read 94 percent against that gate. I throttled about 30 percent of volume to carts. Full automation was week three. On the inventory publish, go or no-go was counted rows: no SKU with two published AFS values, and Negative AFS clean at T+1. January put the reserve rules live against demand. May was execution.

R Both go-lives landed. Zero customer-facing incidents. Processing efficiency +15 percent, DC Ops' measure. The retail date moved in public. The May date did not. `record`

Failure. I carried the vendor's January 15 as if it were the floor's date. The slip was handled in the open, and it was visible later than it should have been.

Data integrity. The bad input was a vendor date treated as a floor date, and a zone map that did not match our induction rules. The control was a gate published before volume, and a go/no-go read from counted rows. The issues log triaged what could not wait: 331 go-live, 77 fast-follow, 36 enhancement.

6. Engineering wanted a capability finance would not fund

S The release people wanted was "forecast improved, so release the safety stock." Finance would not fund a release on one error. `record`

T Land a real release the book could defend.

A Planning and Finance signed a both-errors-improved rule. I did not ask engineering to build the looser one.

R Forecast accuracy 70% → 85% on the scoped set. About $2M of safety stock released. Planner collaboration up 50%. `record`

Failure. The wrong noun would have been "planner effort down." Collaboration went up.

Data integrity. The bad input was a single accuracy number standing in for two errors. Finance owned the release rule. The control was both errors improved before a dollar moved.

7. Where you saved real money by reusing an asset or avoiding a buy

S I do not have a capital-equipment reuse dollar in my record. I will not borrow a retrofit figure I did not earn.

T Answer with a real reuse, then the avoided-cost noun if they want a dollar.

A On the Canada bulk-ship design we reused the existing cross-dock path and two mailbox locations. The system change was six items, five of them small. `record-from-docs` That outcome is the plan at discovery exit. I will not quote a post-launch saving.

R The dollar I will say, once, is different: about $5M of avoided cost on the forwarding-app sequence, value in six months instead of eighteen. `record` Avoided cost, not a machine reused.

Failure. If I skip the noun, a listener adds it to something else. I say the noun first.

Data integrity. The bad input would have been a new location created when one already posted. Design owned that choice. The control was "point at the path that exists."

8. A supplier or vendor who slipped

S Locus needed three more weeks of retail testing. My council timeline said retail end-to-end on January 15. `record`

T Keep the slip visible. Do not hide it in a shorter test.

A Steering got the choice the week it surfaced: compress acceptance testing, or move the date. We moved it. The network work sat under a written SOW, with photo evidence before anything powered on.

R Retail end-to-end completed at the end of January. The combined go-live did not move. Zero customer-facing incidents. `record`

Failure. That readiness row should have existed when the wiring SOW was signed, not a month later.

Data integrity. The bad input was a date that belonged to the vendor's calendar and sat on my slide as if it were mine. I owned it once it was on the slide. The control was a revision in the open.

9. Three teams wanted three outcomes

S Retail wanted store fill, Dotcom wanted a sellable unit, Finance wanted one book. Same building. None of them reported to me.

T One design all three would sign.

A One pool, two selling identities, three reserve buckets. History conversion, a reserve-math rewrite, and a reporting rebuild were written down as not built.

R Two go-lives, zero customer-facing incidents. `record` 44 requirement rows were the manifest.

Failure. Finance's analyst still lived with two books for four months. That was my miss.

Data integrity. Three teams were about to write three versions of on-hand. The control was one signed definition and an owner per bucket.

10. A decision from someone who did not report to you, and who was not in the meeting

S Architecture council, sixty-plus stakeholders. Digital, finance, and merchandising did not report to me, and not every owner was going to be in the room. `record`

T Get the decision from the owner of the row before the meeting, or do not pretend the meeting made it.

A I pre-wired every row with its owner. The room was asked only for what was still open: the architecture, the location and reserve model, the staged timeline, and the fence. The pack was 44 requirement rows across 23 systems, seven families, about two dozen named systems. `record-from-docs`

R The council signed the architecture on July 31. The estimates on the rows were the owners' estimates.

Failure. I printed a functional-design date the owners had not finished. Pre-wire the date the same way you pre-wire the row.

Data integrity. Each row was an input. A row with no named owner did not enter the pack. That was the control.

11. How you run a review so it makes a decision

S The same council would have failed if the agenda had been status.

T Leave with a decision, an owner, and a date — or an explicit no.

A A row reaches the room only after its owner has seen it. I ask for a decision or a fence. I read the red item myself. There is no second deck.

R The architecture, the model, the timeline, and the out-of-scope fence were signed. The go-lives later held. `record`

Failure. A review that cannot say no is a status meeting. The date on that slide was me decorating a gate.

Data integrity. The artifact is the row: field, owner, estimate. If the review cannot name those, it is not ready to decide.

12. A playbook from a messy pilot

S The first visibility customers were hand-carried. The mess was calibration, connectivity, and a pull to customize each one.

T Write the standard the next wave could run without me in the room.

A Six graduation criteria became the playbook. Carrier mapping stayed local. A later gate — 500 containers in 20 minutes at 99.6% connectivity before customer testing — is the same rule. `record-from-docs`

R 7 → 68, two months early. `record`

Failure. An earlier test passed for five users across ten scenarios and crashed two days before sign-off, because I had tested my shape. The playbook now requires the customer's shape.

Data integrity. The input I refused to skip was the customer's own data. I owned the bar. No graduation without it.

13. Someone you mentored

S I have no direct-report mentoring story in the record, and I will not invent a name. The closest is a QA lead on a 20-person cadence change who did not report to me.

T Leave the gate in their hands.

A They authored the ceremony. I published whether work actually reached QA and demo. The QA lead wrote the definition of done and defended it. Neighboring teams copied it from the user guide.

R Six-week cadence to four-week, in six weeks. Critical defects −50%. `record`

Failure. If "mentor" means a person on my org chart, I say I have not had that seat. What I have done is make a skeptic the owner.

Data integrity. I was the person-dependent field. The control was the gate, owned in writing by QA.

14. A program gate you missed

S On July 31 I presented a council timeline that said functional-design sign-off on July 26. It was not signed. Build on the upstream rows had been open since June 9. `record-from-docs` for the slide, `record` for the signature.

T Close the spec without smuggling scope, and without moving the go-lives.

A I named the slip the week it was clear. Rows whose owners had already started stayed fenced. The downstream rows got the re-read they asked for. We signed on August 14. I do not say the chain held to the day.

R About three weeks late on the spec. Both go-lives held. Zero customer-facing incidents. `record` What changed: I count the signers by name before a date is printed, and discovery results close with the functional design as one gate.

Failure. An optimistic date costs more trust than the slip. If they want the other miss: on an Elementum track-and-trace application, five key users passed all ten end-to-end scenarios, and concurrent logins crashed the system two days before sign-off. I first sent screenshots. The date was missed. Load is now an entrance criterion.

Data integrity. The bad input was a date with no signer list. I owned it. The control is a named list before the date exists.

15. An AI use case on the CapEx system — which data you would refuse to automate until it is trustworthy

S The profile treats the system of record as the foundation for later AI. I agree with the sequence, and I have not seen the tool. `model`

T Name the fields I would not automate.

A I would refuse placed-in-service date, the purchase-order to asset join, and the reuse-source link until the reconciliation is a counted go or no-go and every exception has an owner. My only language-model build was independent: eval set first, a citation on every proposal, unit cost from $25 to $0.40. `record` It was not deployed at an employer.

R The result I want credit for is the refusal. A narrative draft can be assisted. A book field cannot be guessed.

Failure. I once scoped a pipeline for correctness and discovered the access review mid-build. Same lesson: do not automate a field whose lineage you have not named.

Data integrity. Those three fields compound into depreciation and a second buy. A person owns them until the scorecard is green. That is the control.

I. Behavioral specifics the job description implies

Covered someone else's work so the team succeeded

S At an Austria transportation site the report said data quality was 40%, and the room blamed carriers. `record`

T Get a number the team could act on, even though the census was not "my job."

A I pulled three years of history in SQL myself. True quality was about 55%, not 40%. The defect was our measurement.

R About 99% within a quarter. Manual reconciliation from 8–10 hours a week to 1–2. Austria about $500K a year on a Finance-signed, executed-only method; about $8M across 25 European sites on that same method. `record`

Failure. I carried the census alone longer than I should have. The design of the measure belonged in the room earlier.

Data integrity. The reported 40% was a single-field proxy. I owned the recount. The control was a defect census before anyone blamed a carrier.

Learned a new software fast

S I had not configured S/4, and I had not configured the warehouse system on the Utah conversion. The Blue Yonder upgrade was part of that program, not a migration I tell on its own. `record`

T Learn each system well enough to set gates, not well enough to pretend I configured it.

A On S/4 I walked one unit from the book to the number the order system reads, and I wrote the exit bars before the night: no SKU with two published AFS values, and Negative AFS clean at T+1. On a visibility program I learned the event path through six systems well enough to find a missing trace id, counting events in versus events out at each boundary. Bugs went from hours to locate to traceable per system. `record-from-docs` I will not quote a time-cut percentage.

R Oversell went from recurring to zero through the January and May go-lives. The trace id is what made monitoring possible.

Failure. I learned the trace-id hole during testing, not during design. Next time the identifier is a design row.

Data integrity. The missing input was a trace id. Engineering owned the pipe. I owned the test that proved the hole. The control is an identifier before the first customer test.

Closed an issue without escalation

S QA and engineering were headed for a standoff on a cadence a VP wanted changed. I had no authority to settle it by rank.

T Close it in the room.

A The skeptic authored the gate. I brought evidence of whether work reached QA, not an escalation email.

R Four-week cadence in six weeks, on a 20-person group. Critical defects −50%. `record`

Failure. Early on I was tempted to ask the VP to rule. The gate would have been mine, and it would have died when I left.

Data integrity. The contested input was "is this story done." QA owned it after the change. The control was their written definition of done.

Kept many tasks tracked and closed

S The umbrella program's log is 836 rows over the life of the Utah conversion, not a private list. `record`

T Prioritize and close, in the open.

A One row, one owner, an age, a severity. The grammar the log itself used: 331 go-live, 77 fast-follow, 36 enhancement. Just over half the book was Blocker or Critical.

R 714 of 836 closed, 85.4%. `record` Go or no-go still required the severe rows open at zero. A percent closed was not the gate.

Failure. A log without an owner is a diary. I have let a row sit when the owner was a team name. Team names do not close rows.

Data integrity. The row is the input. A named person owns it. The control is the close rate on a denominator we do not shrink.

Adhered to a protocol when it was slower

S A pipeline I had scoped carried customer content. I had planned correctness and not the access review. **record**

T Keep the weekly review alive without bypassing the review.

A I did not ask for an exception. The manual path stayed up. Cutover still waited for parity within 1%.

R The access review cost two weeks. The Monday review never missed. **record**

Failure. The protocol was slower because I discovered it. It now sits on the kickoff checklist, which is cheaper than discovering it mid-build.

Data integrity. The bad input was “this extract is fine to join.” Security owned access. I owned the scope that forgot them. The control is access lead time planned at the start, and no cutover inside the parity bar.

J. Questions to ask

Ask these three, in this order, in the last four minutes. Then stop. The backup replaces (2) if they have already told you where the side file still wins.

1 — SAY THIS

On the pilot that became the playbook, which lesson did teams actually follow, and which one did you have to keep enforcing?

2 — SAY THIS

Where does the CapEx system still lose to a side spreadsheet, and what would you want a program manager to do about that in the first ninety days?

3 — SAY THIS

When Engineering and Finance disagree on a launch, who makes the call today, and what evidence do they need in the room?

BACKUP — ONLY IF (2) IS ALREADY ANSWERED

What does “data integrity” mean on this team in numbers — which field, which rate, measured where?

Close, one breath. “I baseline the number, I put one instrument in the room, and I do not escalate a row the owner can close. I would like the seat that makes that true for onboarding.”

Listen for: which field they name, who they say owns it, and whether the pilot lesson is a document or a gate that has actually stopped a program. Write their words down. Do not improve them.

K. Verify list, safe sentences, source trace

Say the safe sentence if you cannot confirm. Do not upgrade a label in the room.

#	FACT	SAFE SENTENCE
V1	Canada bulk-ship results after discovery exit are not in the sources.	“That was the plan at discovery exit. I will not quote a post-launch workload or brand count.”
V2	How many of the thousand lanes actually went live. 597 is an earlier anchor, not the settled count.	“A thousand-lane program across 25 European locations. I phased by value.”
V3	Austria quality timing: sources disagree on two months versus about three.	“About 99 percent within a quarter.”
V4	Elementum UAT. Five key users, ten scenarios, concurrent access.	“Five key users passed ten scenarios. Concurrent access crashed the system two days before sign-off. The date was missed. Load is now an entrance criterion.”

#	FACT	SAFE SENTENCE
V5	Names of the first visibility customers disagree across documents.	“The first two hand-carried customers.”
V6	MetricStream: responsibilities are on the resume. No client, headcount, or outcome is.	“I owned SOWs, estimates, budgets, and delivery through hypercare. I will not attach a client or a dollar.”
V7	The resume summary adds unlike savings, avoidance, released inventory, and protected revenue.	“I will use each program’s number with its noun. I will not quote the rolled-up total.”
V8	Tableau and Power BI are not in the mapped experience bullets. The resume text file was not on disk; the bullet inventory is the Apple delta add-on, which states it is complete. SQL is a practice in the stories, not a skill line I could re-read off the resume file.	“I am not a dashboard stylist. I can interrogate the rows, including in SQL. If the view is Tableau, I will learn the view.”
V9	No direct report and no named mentee in the sources.	“I have not had that seat. The person I grew into an owner was the QA lead who wrote the gate.”
V10	No capital-equipment reuse dollar in the record.	“I don’t have that dollar. The reuse I can show is a path we did not rebuild. The avoided-cost figure is a different noun.”
V11	Why the functional design slipped is reconstructed. The dates are sourced: slide said July 26, signed August 14.	“The slide said July 26. We signed August 14. I would rather own the date than the excuse.”
V12	The language-model triage agent was an independent build.	“I built it independently. It was not deployed at an employer.”
V13	Anything about Apple’s CapEx tool, age bars, or who signs a launch.	“Here is how I understand it. Correct me where I’m wrong.”
V14	Trace-id time saved. A half-reduction appears only as your own note.	“Bugs went from hours to locate to traceable per system. I will not quote a percentage.”
V15	S/4 annual dollar is a split, not a record total.	“If they ask, about one million of support cost retired is my program anchor, and the rest of that annual figure is my model. I do not open with it, and I do not say a bare total.”

Source trace — numbers in sections B, F, and H

NUMBER	WHERE	STORY	SOURCE FILE	MATCH
Three reserve buckets; no historical conversion	B, H1, H9	UDC dual inventory	udc-canada-story-pack (council deck p.4)	Y
No SKU with two published AFS values; Negative AFS clean at T+1; OMS reads ISM AFS and never computes; January reserve rules live against demand; May execution; oversell recurring → zero	B, F, H5, I	S1 S/4 HANA ↔ ISM	goforth-s4-end-to-end; panel S1 cutover phase	Y, record design
836 rows; 714 closed = 85.4%	B, F, H1, I	UDC issues log	numbers-reconciliation S2.03; apple-delta	Y
Reported 40%; true about 55%; about 99% within a quarter	B, I	S10 Austria	cabreros-stories-b S10; numbers-reconciliation S10	Y on the levels; months = within a quarter
Zero customer-facing incidents; +15% DC Ops	F, H1, H4	UDC C1	apple-delta resume map; udc pack	Y
60+ stakeholders; issues-log triage is the close mechanism	F, I	UDC council / issues log	udc pack; numbers-reconciliation S2	Y. Severe-open at zero is the gate, not percent closed
18 months → value in 6; 66%; about $5M avoided cost	F, H7	S12 / B2	apple-delta numbers check	Y, avoided-cost noun
7 → 68 = 9.7×; two months early; 170 users	F, H3, H12	S11 / A1	apple-delta; numbers-reconciliation S11	Y, 9.7× is 68/7
$0.66 → $0.03 per container, about 95%	F	S11 / A1	apple-delta; numbers-reconciliation S11	Y, rounded from 95.45%
System accuracy 60% → 90%; parity within 1%	F, H2, I	S16 / A3 and G3	apple-delta A3 / G3	Y
Council July 31; FSD planned July 26, signed August 14; Locus 94% vs 98%; about 30% to carts; full automation week three; retail date moved, May held	F, H5, H8	UDC + Locus	udc pack; goforth-pgm-stories P3	Y. No program-budget variance on this story

NUMBER	WHERE	STORY	SOURCE FILE	MATCH
20 people; six-week → four-week in six weeks; critical defects −50%	F, H13, I	S15 / D3	apple-delta; cabreros-stories-b S15	Y, defect result via resume
Forecast 70% → 85%; about $2M safety stock; collaboration up 50%	F, H6	S6 / F3	apple-delta numbers check	Y
Thousand-lane program; 25 European locations; completeness 60% → 95%	F	S9 / F1	apple-delta; numbers-reconciliation S9	Y on the program; delivered split unresolved
3 DCs + 100 stores; 99% vs 95%; stockouts −20%; about $6M on the line; about $8M protected	F	S13 / M1	apple-delta M1	Y, nouns kept apart
94% vs 98% gate; about 30% throttled; full automation week three	H4	Locus Salt Lake City	goforth-pgm-stories P3; udc pack	Y
Six changes, five small; cross-dock path reused	H7	Canada discovery	udc-canada-story-pack B2	Y as the plan at discovery exit
Retail E2E planned Jan 15, completed end of January; combined go-live held	H8, H14	Locus slip / council timeline	udc pack B5, B6	Y
44 rows / 23 systems; seven families; about two dozen named systems	H9, H10	Council pack pp.9, 13–14	udc-canada-story-pack conflict row 6	Y, source doc over “about a dozen” alone
FSD planned Jul 26; signed Aug 14; council Jul 31	H10, H14	B5	udc pack B5; apple-delta does not say the chain held	Y
500 containers / 20 min / 99.6%; 5 users × 10 scenarios; crash 2 days before sign-off	H12, H14, V4	A5 and Elementum UAT	apple-delta A02	Y. The sign-off date was missed
$25 → $0.40; citations on every proposal	H15, G	I1 independent agent	apple-delta A01	Y, not an employer deployment
8–10 hours/week → 1–2; about $500K Austria; about $8M across 25 European sites	I	S10	cabreros-stories-b S10	Y, executed-only method
331 go-live / 77 fast-follow / 36 enhancement; just over half Blocker or Critical	I	UDC log categories	numbers-reconciliation S2.04, S2.05	Y, 441/836 rounded in speech
Access review cost two weeks	I	G3	apple-delta G3	Y
Four months with two systems of record	H1, H9	UDC interim	apple-delta JΔ5; udc pack	Y
Illustrative 40 stations, $75,000, $3,000,000, 2 sites, 3 suppliers, 12 fixtures, 40 / 38 / 2	E only	Worked example	This packet	ILLUSTRATIVE, not a source

Files behind the short names: amazon/panel/amazon-nacf-apple-delta-addon-star-kimi-k3.html; amazon/numbers-reconciliation.md; amazon/panel/amazon-nacf-udc-canada-story-pack-star-kimi-k3.html; amazon/panel/amazon-nacf-panel-3-goforth-s4-end-to-end-kimi-k3.html; amazon/panel/amazon-nacf-panel-3-goforth-pgm-stories-star-kimi-k3.html (Locus, P3 only); amazon/panel/amazon-nacf-panel-4-cabreros-stories-b-star-kimi-k3.html. Apple workbook v11 and the Apple-tailored resume were not on disk as text; every resume bullet above is the bullet the delta add-on says it mapped, except the warehouse-migration bullet, which is omitted, and every number is one that add-on or the reconciliation already checked. No new client, headcount, or outcome was added.

COMPANION · RECRUITER SAMPLE QUESTIONS · APPEND AFTER THE MAIN 30-MINUTE PACKET

CapEx Systems Onboarding — answers to the questions the recruiter sent

Who this is for. Rajesh Rishiyur Chandrasekaran, speaking to the Capital Expenditure Systems Lead. Thirty minutes. These fourteen answers sit beside the main packet (cram sheet, 75-second introduction, solution design). Same facts, tighter where the main packet is longer.

How to use a page. The serif box is what to say. Under a story, S / T / A / R is the full version if they stay on it. Every number has a basis: record Apple workbook or resume, record-from-docs a program artifact, user anchor confirm before the exact figure, my division arithmetic, not banked.

Integrity. No inside knowledge of Apple's tools. Public claims are labelled. The resume's aggregate savings line is not used.

1. Tell me about yourself and background.

60 seconds. Built for this interviewer — system of record, data integrity, change without escalation. Same facts as the 75-second version in the main packet; this one leaves out the Flexport customer-scale line and the council date chain so the minute lands on those three habits.

SAY THIS — ABOUT 60 SECONDS

I am a supply-chain program manager. At Sephora I have taken live warehouse and inventory work off side files and onto one system of record. A Utah distribution center moved from two systemic warehouses inside one building to one omni pool, while the warehouse system underneath it was upgraded. Two go-lives, zero customer-facing incidents, and DC Operations measured processing efficiency up 15 percent — their number. On the inventory book, S/4 HANA and ISM became the one publisher of available-for-sale. The order system reads that number and never computes its own. Go or no-go was a count of rows. The bars I would not move were no SKU with two published AFS values, and Negative AFS clean at T+1.

Three habits are what I would bring. I run data integrity as a program: counted rows, an owner on the field, and an exit bar written before the go or no-go. I move people who have always done it this way without escalating — the person who owns the row reads it before the room does. And I get into the weeds. I have read application logs to find a missing trace id, and I count the rows myself before a go or no-go.

I have not worked in Apple's CapEx system. The job, as I understand it, is to make the inputs trustworthy enough that the system of record replaces the spreadsheet, and to turn one pilot into a playbook the next program can run.

Basis: zero incidents and +15% — record, DC Ops' measurement. S/4 publish-switch and the two exit bars — record design, ISM functional specification. Trace-id work — record-from-docs, Shipment Visibility ingestion note.

2. Explain your day-to-day responsibilities while working on your recent project.

Most recent role on the Apple-tailored resume: Sephora, inventory and warehouse programs (S/4HANA inventory integration, the Utah omni conversion, the portfolio log and the staged cutover). The week below is the Utah dual-inventory program — discovery through hypercare — as the workbook's "How I run a program" and the council record describe it. It is a representative week, not a reconstructed calendar.

SAY THIS — ABOUT 50 SECONDS, THEN STOP FOR THE TABLE IF THEY WANT IT

On that program my week had one heartbeat. I opened the issues log myself, sorted by severity and age, and I did not walk into the weekly review with a number I had not counted. The review was one two-block status: thirty minutes

with IT, thirty with the business, one dashboard, one risk register, no side deck. I said the worst number out loud, with an owner and a date. Steering was DC Operations, IT, and Supply Chain Strategy. Between reviews I sat with the owner of a row — a system owner on a level of effort, DC Operations on what the floor would actually do, Finance on the location re-point, the robotics vendor on the readiness gate. What I counted was open blockers, rows closed, and whether an end-to-end date was still the date. I owned the requirement table, the issues log, and the decision log. I did not own the systems.

A REPRESENTATIVE WEEK — ARTIFACTS, PEOPLE, WHAT WAS COUNTED

CADENCE	WHAT I DID	WHAT I COUNTED
Issue-log review, before the weekly	I read the umbrella program log myself. One grammar: type, severity, operation, owner. Oldest open rows went to the owner before the meeting, not after.	836 rows; 714 closed, 85.4% record. Blocker plus Critical, 441, just over half the book record. Categories include go-live 331, fast-follow 77, enhancement 36 — those are category counts, not a split of 836 record.
Weekly two-block status	Thirty minutes IT, thirty minutes business. One dashboard, one risk register. I said the worst number. Steering: DC Operations, IT, Supply Chain Strategy.	Open blockers, the end-to-end chain (inbound, dotcom, retail), and any date that had moved since last week record, Apple workbook.
RAID and decision log	A slip or a scope fence was written the week it surfaced, with a price, in front of steering. Examples: the functional-design date, and the robotics retail re-slot.	Functional design planned for July 26, signed August 14 record. Retail end-to-end planned January 15, completed at the end of January after a three-week re-slot record.
Owner working sessions	System owners for their own level of effort. DC Operations on floor impact. The SAP and finance track on the location change. Locus on the gate. Blue Yonder on the warehouse-system upgrade. Architecture Council only at its two touchpoints, not every week: May 15 challenge, July 31 sign-off.	44 requirement rows across 23 named systems; seven families, about a dozen domains, about two dozen named systems on the impact table record-from-docs. Last level of effort July 2 record.
Near a gate or in hypercare	Daily blocker count against a written exit. Go or no-go off counted rows: severe issues open at zero, not a percent complete. On the inventory publish, no SKU with two published AFS values, and Negative AFS clean at T+1. The issues log was the triage — go-live, fast-follow, or enhancement.	Issues log 836 rows; go-live category 331, fast-follow 77, enhancement 36 record. Robotics gate 98% pick-accurate throughput, published before a bot took volume record. Hypercare spike: 271 log rows in June, then 65, 21, 14, 4 record-from-docs.

VERIFY — weekday and clock. The record is a weekly two-block status and one risk register. Safe sentence: “I will not invent a meeting name or a clock time; the cadence I can point to is that weekly two-block review, and a daily blocker read once we were in hypercare.” Whether the decision log was a separate file from the risk register is not spelled out as two artifacts. Safe sentence: “One risk register carried the decisions; I will not claim a second system.”

Five-part method behind the week, Apple workbook Part 0: baseline the number, one scorecard, a gate that can stop the work, root cause before spend, the same number proves the fix. About sixty stakeholders on the steering path; the discovery deck listed about eighty names across about fifteen functions record.

3. Why do you want to join Apple?

45 seconds. This team’s purpose: change management, a data foundation, pilot to playbook. No product poetry.

SAY THIS — ABOUT 45 SECONDS

I want this team because the work is the work I already do, on inputs where a wrong field is a real asset. You are moving CapEx off person-dependent spreadsheets onto one system of record, and you have described that data

foundation as what later applications, including AI, will sit on. I have watched a weekly review steer on a number one person could produce and nobody else could check. I will not put a model on top of a field that has not been reconciled.

The day-to-day you wrote down is change management: teams who have always done it this way, and closure without escalation. Those people are not wrong. The workbook was the system. My job is to make the new path easier to trust than the side file, and to close the old path without taking it up the chain. The pattern on your team — a first pilot in the line of business, then a playbook — is how I already run a program. One site, a numeric gate, then the same control on the next program. I want to do that where the fields are a purchase order, a placed-in-service date, and the asset that should match them.

4. What do you know about Apple?

PUBLIC KNOWLEDGE — VERIFY CURRENT FIGURES BEFORE THE CALL

I know Apple from the outside. I do not know the internal CapEx tool, and I will not guess its name.

Operations is not a support function there. The chief executive came up through operations. That shows up in how the company talks about manufacturing: product, process, and the factory move together. Public knowledge. No figure in that sentence.

Manufacturing is largely outsourced to contract manufacturers. Public filings describe Apple as owning a substantial share of the specialized tooling and test equipment that sits at those partners. That is why capital-equipment records across contract manufacturers are a discipline, not a spreadsheet habit: the tool is Apple's, the floor is the partner's, and the record has to survive both. Public knowledge — verify the latest 10-K wording before the call. I will not quote a tooling dollar.

The supplier footprint has been diversifying. China remains central, and India and Vietnam are part of the public story alongside it. I will not quote a country share. Public knowledge — verify current figures before the call if a share comes up.

Every year Apple publishes a supply-chain responsibility report — the Supplier Responsibility / People and Environment in Our Supply Chain report. It is the public window on how suppliers are measured. I have read that it exists and what it is for. I have not memorized a year. Public knowledge — verify the current report title and year before the call.

Reported capital expenditures have been on the order of ten billion dollars a year in recent years. Public knowledge — verify the latest 10-K before the call. I will say "order of ten billion," not a precise year.

New product introduction is the cadence the operations roles sit on. The names I will use — Global Supply Manager, Technical Program Manager, Advanced Manufacturing Engineer, Operations Program Manager — are the names in this requisition and on the interviewer's public profile. They match how Apple's operations roles are described from the outside. I do not have an org chart, and I will not invent one.

What I will not say. No internal tool name, no unpublished split of capital spend, no headcount, no claim that I have seen a playbook. If they want the lifecycle, I point at the model in the main packet and I ask them to correct it. That answer is question 14.

5. Can you walk me through your experience managing program or project management initiatives, particularly within supply chain, manufacturing, or operations?

Story: Sephora, Utah DC dual inventory with the Blue Yonder warehouse-system upgrade. Supply-chain operations, not a slide program. About ninety seconds if they let it run; the serif box is the forty-five-second cut.

SAY THIS FIRST

The program I would walk is a live distribution center. One building was running as two systemic warehouses, Dotcom and Retail, with duplicate receiving and transfers that the customer never saw. We converted it to one omni pool and

upgraded the warehouse system on the same path, with robotics on that path. I owned the program from discovery through two go-lives and hypercare. I owned none of the systems. Purchasing and inbound went first, in January 2025; the inventory flip was the first week of May. Both landed with zero customer-facing incidents. DC Operations measured processing efficiency up 15 percent. The method was a requirement row per change, an owner on every row, and a council that was asked for decisions, not a status.

S — Sephora, 2024. The Utah DC shipped Dotcom from inventory location 801 and Retail from 901, inside one building, with stock kept physically apart. Leadership wanted one omni facility, and the warehouse system underneath it was being upgraded to Blue Yonder on the same path. That upgrade is part of this conversion, not a migration I tell on its own. The Architecture Council was the challenge gate. Two constraints I did not get to negotiate: zero customer-facing incidents, and the October-to-December peak left untouched. record

T — I owned discovery, the functional design, the council case, the staged timeline, the issues log, and the hypercare exit. About sixty stakeholders sat on the steering path; the discovery deck later listed about eighty names across about fifteen functions. record

A — I ran discovery with every impacted team from the March 29 kickoff. Every system owner estimated their own row; the last level of effort landed July 2. The council pack, signed off July 31, asked for separable decisions: the future-state architecture, the location model — one pool, two selling identities, three reserve buckets, no conversion of historical transactions — the staged timeline, and a written fence. Reserve math stayed untouched. Decommission of the old merchandising stack stayed with the global ERP program. Upstream build that discovery had already stabilised opened in June, fenced; the warehouse-conversion, outbound, finance, and reporting tracks stayed closed until August 1. After sign-off the heartbeat was the weekly two-block status. Go or no-go read counted rows on the issues log: Blocker plus Critical open at zero, not a percent complete. record and record-from-docs

R — Two go-lives, zero customer-facing incidents. Processing efficiency +15 percent, DC Operations' measurement. 714 of 836 issues closed, 85.4 percent, with the tail still owned. The approved boundaries held; there was no mid-build argument about what the program was. record

Data-integrity angle. The input that goes wrong in a conversion like this is "which location is the inventory, and which reserve may promise it." I refused a historical conversion, and I kept the reserve model to the three buckets that already worked. Three buckets — Dotcom reserve, Retail reserve, uncommitted — replicated from the Canada logic that already worked. During the four-month interim, one system auto-copied on-hand and on-order from the old location to the new one, and a daily reconciliation watched for drift. The owner of each requirement row owned the input. The control was the row plus the dated end-to-end test that proved it.

Failure / learning. I printed July 26 for the functional-design sign-off and we signed it August 14. I had called a date the owners had not finished re-reading. I do not say the gate chain held to the day. Since then a date on a slide is a plan until the owner signs it.

6. Tell me about a systems onboarding project you managed. How did you ensure accurate data inputs and successful adoption by end users?

Story: Sephora S/4 HANA and ISM — one published available-for-sale. This is the onboarding story: a go/no-go read from counted rows, a publish switch the order system does not recompute, and exit bars written before the night. The Utah issues log is the triage beside those shared go-lives. It is not an ISM-only log, and it is not a second migration.

SAY THIS FIRST

At Sephora two inventory books were each publishing their own available — the legacy merchandising system and S/4 HANA — and every organization meant something different by the word. I owned the signed definition and the one engine that publishes it. I did not configure S/4. The order system reads ISM available-for-sale and never computes its

own. Go or no-go was a recommendation I read from counted rows: the reserve copy equal to the plan, and every old-versus-new difference reason-coded. The bars I would not relax were no SKU with two published AFS values, and Negative AFS clean at T+1. January put the reserve rules live against demand. May was execution. Oversell went from recurring to zero through both go-lives. Adoption was a screen with an owner: Inventory Control, Internal Audit, and Finance on Negative AFS; the ISM application team on the AFS correction report.

S — Two books, three node types, a promise that could not hold. Finance meant the posted book, dotcom meant what it could promise, the distribution center meant what it could pick. Recurring oversell, and a negative available-to-sell that showed up as an order that could not ship. record

T — One signed definition, one publisher, and a go/no-go the forum could read without taking my confidence as evidence. Engineers built the extracts and the screens. I owned the definitions, the keys, the owner map, and the recommendation. record

A — S/4 owns the physical book. ISM takes the lower of that book and the warehouse floor, subtracts what is committed or sold and not yet posted, and publishes available-for-sale per SKU, node, and channel. The publish switch was zero on the old location, then the new one, so no SKU was promisable from two places at once. The reserve-copy invariant: the reserve in ISM equals the reserve in the plan at every nightly cut. I split the cutover. January was planning — reserve rules live against demand. May was execution. The recommendation listed counted rows: severe issues open at zero on the umbrella log, reconciliation clean, rollback rehearsed, control owners named. The forum decided. record on the design and the two go-lives. The 836-row log is the Utah program's log, which those shared go-lives were gated on.

R — Customer-facing oversell incidents, recurring, to zero through both go-lives. Finance closing on the same on-hand Operations reads. No SKU with two published AFS values. Negative AFS clean at T+1. I do not put a cutover dollar on this answer. record

Data-integrity angle. The bad input was the word available, owned by nobody and computed in more than one place. The control is one publisher, a reserve copy that must match the plan, and an AFS correction report with a screen, a role, and an audit row. The Utah issues log — 331 in the go-live category, 77 fast-follow, 36 enhancement — is how I triaged what could not wait for the next release.

Failure / learning. The adoption miss I will tell is on a different program, and it is the one I got wrong. On an Elementum track-and-trace application, five key users had passed all ten end-to-end scenarios. Two days before sign-off the sponsor opened access more broadly and concurrent logins crashed the system. I reran my scenarios, they passed, and I sent screenshots. I was defending the test plan. I owned the miss. The sign-off date passed. Every critical scenario I write now has numeric entrance and exit criteria for concurrent load at the customer's user count, volume, and network. record

7. How do you identify, investigate, and resolve data discrepancies when working across multiple systems or teams?

Story: Elementum, Austria routing-guide data quality. The mechanism I still use when a number disagrees across systems is the Shipment Visibility rule: events in versus events out at every boundary.

SAY THIS FIRST

I do not start from the explanation already in the room. At a transportation customer, on-time delivery could not be measured. The report said data quality was about 40 percent, and everyone, including my sponsor, believed the carriers were late. The team was spending eight to ten hours a week reconciling promised dates against actuals by hand. I pulled three years of shipment history in SQL myself. True data quality, at the grain we would actually score, was about 55 percent, not 40. The defects were ours: missing rates, lanes at the wrong grain, one carrier under five spellings, brochure transit times sitting where a promise should be. We burned those classes down by reason code. Data quality

reached about 99 percent within a quarter. The carriers never disputed the scorecard, because Finance had signed the method before I claimed a dollar.

S — Austria site. Reported routing-guide data quality about 40 percent. Manual reconciliation eight to ten hours a week. No enforceable on-time standard. The room's cause was "carriers." record

T — An on-time number a carrier could not argue with, and any dollar claimed only on a method Finance co-owned. I owned the analysis and the rules. The data platform team built the engine. record

A — I classified every blank. The rules engine looked up ship-from, ship-to, mode, volume, Incoterm, and the service level. A no-hit paged a person. It never defaulted a carrier. The scorecard could not go out until the automated on-time figure matched a manual sample for that lane cohort. Savings were contracted rate minus paid rate, executed shipments only. Austria went first, inside a month, about 97 percent agreement between the automated figure and the manual sample, about five hundred thousand dollars a year at that site on the signed method. The same rules file then went to the other European locations. record

R — About 55 percent true, against 40 percent reported, to about 99 percent within a quarter. About eight million dollars of validated freight savings across 25 European locations, executed shipments only. I do not quote a two-month or a three-month bar; "within a quarter" is the line both of my own write-ups support. record

Data-integrity angle. The wrong input was the routing guide itself — identity, grain, and a promise that was really a brochure. The owner of each defect class burned it down weekly. The control was a no-hit that stops, plus a pre-release sample, plus a Finance-signed formula. When the discrepancy is between systems rather than inside one file, I use the same idea at the boundary: count events in versus events out at every hop, and give the load a trace id. On Shipment Visibility, data crossed six system layers and a missing trace id was costing two to four hours per blocker just to find the row. That design change is record-from-docs. A note I wrote then claimed about half the time to resolve; I treat that half as my estimate, not a measured result.

Failure / learning. I disproved the brochure times lane by lane, which was slow, and my sponsor was unhappy for weeks. I would start from a benchmark, measure before I blame a partner, and sign the method before the number. The answer to "why is this taking so long" was the defect classes trending down, not a speech.

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8. Can you describe a situation where you had to manage multiple stakeholders who were resistant to adopting a new process or system? How did you drive change?

Story: Sephora, a 20-person Product, Engineering, and QA team, six-week hybrid cadence to four-week. No line authority. The gate was owned by the skeptic. This is the "they have always done it this way" answer, and it closes without escalation.

SAY THIS FIRST

A VP asked me to move a twenty-person team from a six-week cadence to four weeks. QA was a contracting firm. They were heading for a standoff with in-house engineering, and they were right to be afraid: a shorter cadence often means a shorter test window. I had no authority over anyone in the room. I did not write the definition of done and impose it. I asked the most skeptical person, the QA lead, to author it. I coached what a gate has to be able to say — no, on the day of the release, with the room watching — and she wrote the content. In the tracker, a story could not enter QA until its entry criteria were met, and every sprint had a drop date with a reason code, so a heavy story landing in the last two days was a visible re-scope, not a heroics night. We converted in six weeks. Critical defects per release fell by about half. She defended the gate after I stepped back. I did not escalate the standoff. I gave the resistance the artifact.

S — Newly in the role. VP of Supply Chain wanted a four-week agile cadence. QA feared compressed test windows. Engineering wanted the date. Neither reported to me. record

T — Convert the cadence in weeks, not quarters, without the standoff becoming the story, and without a mandate. record

A — One-on-ones with every person. One influencer named per function. The skeptic authored the definition of done; engineering reviewed and did not get a veto. Entry-criteria fields and a QA drop date with reason codes made the rule visible in the tool, so adoption was not a speech. I covered some of the low-impact stories the drop-date rule pushed out, so the team did not pay for my process with a silent miss. record on the mechanism; covering dropped stories is how I kept the window — I will not invent a count of those stories.

R — Six weeks from kickoff to a cadence the team ran without me. Critical defects per release down about 50 percent — that result is on the resume; the workbook chapter is thinner and does not restate the defect figure, so I cite the resume. No direct reports. The gate survived my leaving, which was the test. record

Data-integrity angle. The input people were protecting was “this story is ready for QA.” It had been a conversation. I made it a field: benefit rank, review status, entry criteria met or not. A story with a blank entry could not move. The owner of the field was the skeptic, which is why the control outlived me. A process change that lives only in my slides dies when I leave the meeting.

Failure / learning. I went to engineering first and QA second. The person who can stop the release should be in the room on day one, with her name on the review line. A gate I write myself is decoration the week I am not there.

9. How proficient are you with Excel? Can you walk me through how you've used it for data analysis, reporting, tracking, and reconciliation?

Truthful. The artifacts I can point to are workbooks and Google Sheets, plus SQL where the grain was wrong for a sheet. Tableau and Power BI are not in the resume, the Apple workbook, or the program artifacts I have. Looker views sat on a pipeline the infrastructure team built; I owned the metric definitions, not that dashboard.

SAY THIS — THEN THE FOUR USES IF THEY WANT DEPTH

I am proficient at the spreadsheet work this job is actually about: a reconciliation to a written exit bar, a weekly scorecard, an issue log I can age, and a variance table by defect class. I have done that in Google Sheets and in workbook logs, and the mechanics are the same in Excel — pivots, lookups, a counted row against a bar I wrote before the go or no-go. When the sheet is the wrong grain I write SQL. I have not built Tableau or Power BI. I would pick them up on the metric definitions I already know how to freeze. I learn tools quickly; that is how I became useful on systems I had not configured before. I do not confuse “I can analyze the extract” with “I configured the system.”

USE	WHAT I ACTUALLY DID	NUMBER
Reconciliation S/4 go/no-go	Old available-for-sale versus the new publish, by SKU, node, and channel. A difference was a counted row with a reason code, or it was unexplained and the gate stayed shut. The reserve copy had to equal the plan. The order system was not allowed to compute its own number. I will not claim a file named Excel if the extract lived in the reconciliation the team shared; the work was row-level, and I counted it before I recommended go.	Exit bars: no SKU with two published AFS values; Negative AFS clean at T+1. January, reserve rules live against demand; May, execution. record Beside it, the Utah issues log triaged 331 go-live, 77 fast-follow, 36 enhancement. record
Tracking Utah issues log	One workbook, one grammar: type, severity, operation, owner. I aged it myself before the weekly review. The hypercare read was the same log, by month, not a new dashboard.	836 rows; 714 closed, 85.4%. June 271, then 65, 21, 14, 4. Go-live 331, fast-follow 77, enhancement 36, as category counts. record and record-from-docs. The consolidated log in the source set is a workbook.

USE	WHAT I ACTUALLY DID	NUMBER
Reporting Monday review, Flexport	Four systems did not join. I stitched the weekly review in Google Sheets: ticket data plus a manual container pull, matched by hand. Seven frozen metrics. I later specified the pipeline that replaced the sheet. The review format — input to output, a red/green band — is the one I copied from the LSA weekly-review workbook.	1.5 to 2 hours per snapshot. System accuracy 60% to 90%. Parallel run within 1% for consecutive weeks. First automated catch about a 4% ticket-sync drop. The business-requirements document priced the old process at $103,680 a year on its assumptions — priced, not a saving I claim. record and record-from-docs
Analysis Austria routing guide	Three years of shipments in SQL, then a defect census I worked down weekly: missing rates, wrong-grain lanes, five spellings of one carrier, brochure times posing as promises. Until the rules engine existed, the reconciliation of promised date against actual was manual.	Reported about 40%, true about 55%, then about 99% within a quarter. Eight to ten hours a week of hand reconciliation before. record

VERIFY — Tableau / Power BI. Not evidenced in the resume, the Apple workbook, or the program artifacts. Looker is evidenced only as the view layer on the Snowflake pipeline the infrastructure team built. Safe sentence: “I have built the metrics behind weekly reviews in Excel and SQL; Tableau I would pick up quickly — I learn tools fast, which is how I became the go-to on three enterprise systems.” The three I mean, if asked, are the inventory logic on S/4 HANA and ISM, the Blue Yonder upgrade inside the Utah conversion, and Manhattan at peak — programs I led, not modules I configured. A solver workbook I built later for a marketplace design case is a case, not a client result, and I will not offer it as job history.

10. Tell me about a complex project where you had to manage multiple timelines, milestones, dependencies, and competing priorities. How did you ensure everything stayed on track?

Same program as question 5, schedule lens. If they already heard the omni story, do not retell the council. Open with the bridge and go to the chain. If they have not, this answer can stand alone.

SAY THIS FIRST

The Utah conversion had more than one date, on purpose. Purchasing and inbound went live in mid-January. The inventory flip was the first week of May. Between them the floor still had to ship peak, October through December, untouched. Inbound end-to-end was late November, dotcom late December, retail planned for January 15 and finished at the end of January after robotics testing slipped three weeks. About thirty build tracks sat under one log. The business wanted one date. I refused it, because two defect classes in one week is how you get a customer-facing incident. What kept it on track was a written chain, a weekly review that read the chain, and a rule that a slip is re-slotted in the open the week it is real. I did not absorb a miss into weekends and then call the old date “held.”

S — One building, a warehouse-system upgrade, a location-model change, and a robotics rollout, with peak freeze as a hard band and a council that had not signed when the first merchandising tracks opened. record

T — Hold two go-lives — January 15 for purchasing and inbound, May 5 for the combined flip, inventory moved on May 1 under the conversion script — without spending the peak, and without a surprise in the room. record and record-from-docs on the days.

A — I published the chain as the only deadlines that mattered: inbound November 22, dotcom December 20, retail January 15. Every track’s exit was one of those dates, not a percent complete. Upstream location and purchase-order work ran from June 9, fenced to what discovery had stabilised; warehouse conversion, outbound, finance, and reporting waited until August 1. When retail robotics testing slipped three weeks, I took the choice to the weekly steering read that week: compress acceptance testing, or move retail end-to-end to the end of January and keep the test whole. Steering took the re-slot. The dependency list was the requirement table — 44 rows across 23 systems, each with an owner — and the issues log, not a Gantt nobody opened. The inventory publish on those same two

dates followed the S/4 plan: January, reserve rules live against demand; May, execution. Competing asks that lost: a single go-live, a retune of reserve math, and folding an unrelated decommission into this release. record

R — Both staged go-lives landed. Zero customer-facing incidents. The retail date moved in public and the May date did not. Processing efficiency +15 percent is DC Operations' number on the building, not a schedule metric. record

Data-integrity angle. A date is an input. The July 26 functional-design marker on the council slide was a plan; the signature was August 14. I tracked plan versus signed the same way I track a counted row versus a tolerance. The owner of a milestone was the owner of the rows that proved it. A change with no row did not get a date.

Failure / learning. January 15 on the retail robotics track was the vendor's calendar, not the floor's. A readiness row with its own owner would have moved that date in November, not December. I handled the slip in the open. I was late to see it.

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11. Have you worked with external vendors or suppliers during system implementations or onboarding? How did you ensure they followed the required processes and data protocols?

> **Story:** Locus robotics at the Salt Lake City DC, inside the Utah conversion. Vendor statement of work, a readiness gate, the miss handled in the open. Blue Yonder's architects sat on the same program; the vendor I governed with a document was Locus and the wiring vendor under the statement of work.

> **SAY THIS FIRST**
>
> Yes. At the Salt Lake City DC we added Locus robots to retail outbound while the building was also changing warehouse systems. Locus brought the bots and the software. A wiring vendor expanded the network under a versioned statement of work: two new closets, thirty-one new access points, a fiber ring, photo evidence before anything powered on. I did not manage the vendor by status meetings. I managed them by a gate I published before a bot took volume: 98 percent pick-accurate throughput, with manual carts as the fallback and the bot-versus-cart split built as a dial. Week two of go-live read 94 percent. I throttled about 30 percent of volume to carts, Locus and our engineers fixed the zone logic, and we were fully automated in week three. The three-week slip in retail testing, earlier, was re-slotted in the open. I did not compress the test to protect a vendor date.

S — Expansion of an existing Locus installation, not a greenfield. Design and build July through December 2024. Physical testing February and March 2025. Combined go-live May 2025. The network scope was written down: statement of work version 1.4, November 19, 2024. record and record-from-docs

T — Land the robotics without adding a customer-facing risk to the conversion, and make the vendor's process checkable by someone who does not work for the vendor. record

A — Five readiness categories on one track page: physical, hardware and network, software, integration, testing. Discovery was three working days with Locus, not an email thread. The statement of work was the physical baseline; acceptance was photos of the rack, the closets, and the access points, and a rule that switches did not power on until the fiber was connected. Interfaces were unit-tested with simulated data before a physical bot rolled. The integration decision on the track was sockets rather than REST, written down, with an owner. Retail end-to-end had been planned for January 15; after the three-week slip it completed at the end of January, and user acceptance stayed full length. The 98 percent gate, 99 percent stretch, was counted weekly. record

R — Full automation in week three. Zero customer-facing incidents through the combined go-live. Locus-side rows were 86 of the program's 836, about a tenth of the book. DC Operations' +15 percent is the building's number, not a robotics-only claim. record

Data-integrity angle. The vendor's map and our induction rules disagreed about which zone owned the work, which is a data protocol failure, not a hardware failure. The network checks had already passed. The control was the published gate plus the dial: at 94 percent the rule fired, I did not improvise, and volume moved to carts until the zone data was fixed. Photo evidence was the acceptance test for the physical data — closet, access point, fiber — so "installed" meant a picture, not a verbal done.

Failure / learning. I carried the vendor's retail date as if it were the floor's date. The slip was handled in the open, which was right, and it was visible later than it should have been. A vendor milestone without its own readiness owner is a wish.

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12. Can you describe your experience working with supply chain or manufacturing systems such as SAP, WMS, MRP, or other database management platforms?

> **Inventory from the record.** One line of what I did, one line of what data I owned. I did not configure these systems. Engineers and the business owners did.

SYSTEM	WHAT I DID IN IT	DATA I OWNED
SAP S/4HANA and ISM	Inventory supply-management integration. I owned the policy and the integration governance for one definition of available-to-sell. The publish-switch plan: the order system reads ISM available-for-sale and never computes. ERP and ISM engineers owned configuration. On the Utah program, SAP was one finance system on the location re-point; I owned the requirement row.	Exit bars: no SKU with two published AFS values; Negative AFS clean at T+1. January go-live, reserve rules live against demand; May go-live, execution. Zero customer-facing oversell incidents on that signed definition. record
Blue Yonder WMS	On the Utah conversion the warehouse system was upgraded to Blue Yonder while the building moved to one omni pool. That upgrade is part of the Utah program, not a migration I tell on its own. The team configured the warehouse system. I owned the requirement row, the gate, and the council pack.	One pool, two selling identities, three reserve buckets. No historical conversion. Go or no-go off counted rows. record
Manhattan OMS / WMS	Peak allocation across the network. I owned the operating rule and the bridge when the scheduler and the floor disagreed, not the product configuration.	3 distribution centers and 100 stores. Fill 99% against a 95% goal. Stockouts down 20%. About $6 million was on the line when the page fired; about $8 million of that day's sales was protected. record
Elementum	Requirements and the analytical pull for a supply-and-demand view, a lane program, the Austria routing guide, and a carrier bid. Planners left a Google Sheet for the supply view.	Forecast accuracy 70% to 85% at the planning grain; about $2 million of safety stock released. Routing-guide quality about 55% true to about 99% within a quarter. Planner collaboration up 50% — not an effort cut. record
Snowflake	I wrote the business requirements and the acceptance gate for four systems into one pipeline. Infrastructure built the path into Snowflake. I did not build it, and I will not say I did.	Seven frozen metrics. System accuracy 60% to 90%. Parity within 1% for consecutive weeks before cutover. record
Locus robotics APIs	Workstream owner. The integration decision on the track was sockets rather than REST, plus tote-type mapping. I did not write the API. Locus and our warehouse engineers did.	The readiness gate and the bot-versus-cart dial. 98% published; week two at 94%; full automation in week three. record

SYSTEM	WHAT I DID IN IT	DATA I OWNED
Flexport's platform	Shipment Visibility rollout and the forwarding application's sequence. I owned graduation criteria, the weekly review, and which workflow went first. Engineering owned the build.	Events in versus events out at each system boundary, and a trace id on the load. Customer counts and cost-to-serve sit in the main packet if they ask; I will not stack them onto this list. record-from-docs on the boundary count.

How I learn a new system. I sit with the person who runs the screen. I take one transaction and walk it backward to the row. I count what entered a boundary and what left it before I trust a dashboard. That is how I learned the S/4 and ISM publish without configuring either system, and how I found a missing trace id by reading application logs across six layers. The gate comes before my confidence does.

> **VERIFY — MRP as a configured module.** Planning grain — forecast, safety stock, a lane promise — is on Elementum and against SAP extracts. Hands-on MRP configuration is not in the record. Safe sentence: "I can tell you what the planning input has to mean, and I can reconcile it. I will not tell you I configured MRP."

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13. Tell me about a time you identified an operational inefficiency or process gap. What steps did you take to improve it, and what was the outcome?

> **Story:** Elementum, the semi-annual carrier bid. The gap was reconciliation dressed up as optimization. user anchor on the cycle time — it is not in the Apple workbook. No award dollar. If they want a document-backed gap with no duration claim, use the Customs ownership note at the bottom.

> **SAY THIS FIRST**
>
> A transportation team ran a carrier bid twice a year and called it an optimization problem. Hundreds of lanes, dozens of rate files, constraints like volume commitments and incumbency floors, about six weeks a cycle. I did not own the award, and Transportation, Procurement, and engineering did not report to me. I sat through one full cycle and logged the hours. The six weeks were not the optimizer. They were reconciling lane names and mode codes that meant different things in every carrier file. I wrote a lane master — one origin, one destination, one mode — and I turned every tribal constraint into one written rule. Before any scenario ran, the rules had to reproduce the last two real awards with under a five percent mismatch, and each scenario had to be explainable in one sentence. We parallel-ran one region. The cycle went to about three weeks, and other regions adopted it without a mandate. I claim no dollar on the award.

S — Semi-annual bid. The room, my sponsor, and engineering all assumed the bottleneck was the solver. Origin, destination, and mode were spelled differently per carrier. user anchor

T — Cut the cycle without touching the award, and without authority over the people who owned the rates. user anchor

A — Hour log on one live cycle, then a lane master and explicit constraint rules. The gate was a back-test against the last two awards, under five percent mismatch, before the optimizer was allowed to run. One region in parallel. I refused to publish a "savings" number on the award. user anchor

R — About six weeks to about three. The pilot region was early; other regions picked it up without a mandate. No award dollar, in the story or in this answer. user anchor

Data-integrity angle. The bad input was identity: the same lane under more than one name, and a constraint that lived in someone's head. The owner of a word in a rule had to be the owner of that field. The control was the back-test — the new rules had to reproduce history inside a written mismatch line — which is the same shape as

a go/no-go read from counted rows. A process gap that is actually a data gap does not get fixed by a faster meeting.

Failure / learning. Procurement joined after design sign-off, and “incumbency floor” meant two different things in the room. Every function that owns a word in a rule is in discovery on day one.

If they want the version with a signed page and no duration. At Flexport, Customs Platform and Freight Forwarding Applications were both building customs-declaration work. I stopped the organization argument and mapped the 2023 roadmap at work-item grain: 15 items, 12 export and 3 import. The agreement was that Freight Forwarding Applications owned all of them except ISF filing. The documented result is decision closure and an explicit interface, not a cycle-time or a dollar. record-from-docs My first framing was “Customs owns customs,” which cannot be configured. The grain that works is the grain where the work changes hands.

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14. How familiar are you with the Apple ecosystem, its internal systems, and the way different teams collaborate across Apple's supply chain and operations?

PUBLIC KNOWLEDGE — VERIFY CURRENT FIGURES BEFORE THE CALL. NO INTERNAL SYSTEM IS NAMED BELOW.

Say this, and then stop for their correction. “I do not have inside knowledge of Apple’s internal systems; here is my model of how the CapEx lifecycle runs across GSM, TPM, AME, CM, Finance and the Systems PM — correct me where I’m wrong.”

What I know from the outside is in question 4, and I will not re-read it unless they ask. The one figure I will repeat if they ask for scale: reported capital expenditures on the order of ten billion dollars a year in recent years. Public knowledge — verify the latest 10-K before the call. Manufacturing largely at contract manufacturers, with Apple owning much of the specialized tooling and test equipment placed at those partners, is why a CapEx record across those partners is a real discipline. Supplier diversification — India and Vietnam alongside China — and the annual Supplier Responsibility report are public. The role names GSM, TPM, AME, and OPM are the names in this requisition and on the interviewer’s public profile. I do not have an org chart.

THE MODEL, IN ONE BREATH — THE PICTURE IS THE SOLUTION-DESIGN SECTION OF THE MAIN PACKET

A program needs capacity. Capacity becomes a list of tools and fixtures. Finance appropriates it. The Global Supply Manager sources it. The supplier builds it. Someone accepts it, ships it, and installs it at the contract manufacturer. Finance capitalizes it: asset identity, placed-in-service date, where it sits, who is custodian. The line ramps. Later the same asset is reused, retrofitted, or disposed of. The Technical Program Manager and the Advanced Manufacturing Engineer own the technical need. The contract manufacturer lives with the tool. Finance owns the book. The Systems PM owns the record. The Data Integrity PM owns whether the input that was typed is the input the next team can trust. The fields I would watch first are the ones that break quietly: purchase order to asset, placed-in-service date, quantity versus serialized units, site, cost center, and which program a shared tool belongs to. That swimlane, the ownership table, and the onboarding runbook are drawn in the solution-design section of the main packet. I will walk them only if you want the picture, and I will mark anything you correct.

Collaboration, said without an internal map. From the outside, these teams meet on a launch: engineering wants a capability, finance has a number, the supply manager has a supplier, the contract manufacturer has a floor, and the system of record is what keeps those four from each keeping a private workbook. I have done that meeting. I have not done it inside Apple. Change without escalation, on this team, means the owner of the field agrees the field before anyone escalates the person.

Answer → story → number

#	QUESTION, SHORT	STORY	NUMBER TO LAND
1	Tell me about yourself	Utah omni pool, then S/4 publish-switch. 60 seconds.	Zero incidents; +15% DC Ops; no SKU with two published AFS values; Negative AFS clean at T+1
2	Day to day, recent project	Utah program week. Two-block status, issues log, counted-row gate.	836 rows, 85.4% closed; 44 rows / 23 systems; 331 / 77 / 36
3	Why Apple	This team: system of record, change without escalation, pilot to playbook.	No new number. Do not quote a tooling dollar.
4	What do you know about Apple	Public only. CEO through operations; CM tooling; India and Vietnam; annual supply-chain report; NPI roles.	Order of ten billion a year — verify the latest 10-K
5	Program management in operations	Utah dual inventory. Discovery, council, two go-lives.	Jul 26 planned, Aug 14 signed; three reserve buckets; +15%; zero incidents
6	Systems onboarding, data, adoption	S/4 HANA and ISM. Counted-row go/no-go. Failure alternate: Elementum UAT.	January reserve rules, May execution; oversell recurring to zero. Five users × ten scenarios; concurrency failed two days before sign-off
7	Discrepancies across systems	Austria routing guide. Events-in versus events-out as the mechanism.	~55% true vs ~40% reported → ~99% within a quarter
8	Stakeholders resistant to a new process	20-person cadence. Gate owned by the skeptic.	6-week to 4-week in 6 weeks; critical defects about −50%
9	Excel	S/4 old-versus-new AFS; issues workbook; Sheets WBR; SQL defect census.	Negative AFS clean at T+1; 836; 60%→90%; 8–10 hrs a week. No Tableau claim.
10	Many timelines, competing priorities	Same Utah program, schedule lens. Do not retell question 5.	Nov 22 / Dec 20 / Jan 15→end of January / May 5; ~30 build tracks
11	External vendors	Locus, Salt Lake City. Statement of work, gate, open slip.	94% vs a 98% gate; ~30% to carts; full automation in week three
12	SAP, WMS, other platforms	S/4 and ISM, Blue Yonder inside Utah, Manhattan, Elementum, Snowflake, Locus, Flexport. One line each above.	One number per system, already in the table. No MRP-configuration claim.
13	Process gap	Carrier bid. Backup: Customs ownership at work-item grain.	~6 weeks → ~3; no award dollar. Backup: 15 work items, 12 export + 3 import
14	Apple ecosystem and how teams collaborate	Honest line, then the lifecycle model. Picture is the main packet's solution-design section.	Same 10-K line as question 4, only if asked. No internal tool name.

VERIFY list — safe sentences

- **Steering clock.** "I will not invent a meeting name or a clock time. The cadence I can point to is a weekly two-block status, thirty minutes IT and thirty business, and a daily blocker read in hypercare."
- **Decision log versus risk register.** "One risk register carried the decisions. I will not claim a second system."
- **Tableau and Power BI.** Not in the sources. "I have built the metrics behind weekly reviews in Excel and SQL; Tableau I would pick up quickly — I learn tools fast, which is how I became the go-to on three enterprise systems."
- **MRP configuration.** "I can tell you what the planning input has to mean, and I can reconcile it. I will not tell you I configured MRP."
- **Carrier-bid cycle.** Not in the Apple workbook. "My program anchor is about six weeks to about three. I claim no award savings."
- **Austria duration.** Workbook and the other canon differ. "Data quality reached about 99 percent within a quarter."

- **Elementum UAT.** “Five key users passed ten scenarios. Concurrent access crashed the system two days before sign-off. I missed the date. Load is now an entrance criterion.”
- **S/4 annual dollar.** “If they ask for money on the inventory definition, about one million of support cost retired is my program anchor, and the rest of the annual figure is my model. I do not open with it, and I do not say a bare total.”
- **Trace-id speedup.** “Two to four hours to locate a blocker is in the ingestion note. ‘About half the time’ is my estimate, not a measured result.”
- **Functional-design date.** “Planned July 26, signed August 14. I do not say the gate chain held to the day.”
- **Resume aggregate.** Do not say it. “I will use each program’s number with its noun.”
- **Marketplace solver workbook.** “A design case I worked, not a client result.”
- **Utah hypercare.** “The counted spike is 271 log rows in June, then 65, 21, 14, 4. A daily bar of under two blockers, sustained, is the bar I would describe as mine. I will not invent the clock time.”

Source trace

Number, then story, then where it lives. “Apple v11” is the candidate’s workbook. “Council deck” is the July 31, 2024 Architecture Council sign-off. “Issues log” is the consolidated Utah log. “Delta” is the Apple numbers-check add-on. Match means the say-aloud line agrees with that source’s safe wording.

NUMBER	STORY	SOURCE	MATCH
Zero customer-facing incidents; +15% processing efficiency	1, 5, 10	Apple v11 C1; resume; delta numbers-check	Y — DC Ops’ measurement
Publish switch: order system reads ISM AFS and never computes. Exit bars: no SKU with two published AFS values; Negative AFS clean at T+1. January reserve rules live against demand; May execution. Oversell recurring → zero.	1, 6, 9, 12	S/4 end-to-end add-on; S1 cutover phase in the panel pack	Y — record design. No cutover dollar in the say-aloud
Five users × ten scenarios; concurrency failed two days before sign-off	6	Apple delta, Elementum track-and-trace UAT	Y. The date was missed
836 rows; 714 closed = 85.4%; Blocker+Critical 441	2, 5, 9	Apple v11 C2; issues log recompute in the numbers canon	Y
Go-live 331 / fast-follow 77 / enhancement 36	2	Apple v11 C2; issues log	Y — category counts, not a split of 836
44 requirement rows; 23 systems; seven families	2, 5	Council deck pp. 9, 13–14	Y. “About a dozen” remains the domain count
Reserve-copy invariant: ISM reserve equals the plan at the nightly cut	6, 9	S/4 reserve-copy invariants	Y — record design
FSD planned Jul 26, signed Aug 14; councils May 15 and Jul 31	2, 5, 10	Council deck p. 10; Apple v11 C4	Y. Never “held to the day”
Jan 15 and May 5; retail E2E to end of January; Nov 22 / Dec 20	5, 10	Council deck p. 10; Apple v11 C1/C3	Y
Three reserve buckets; no historical conversion	5	Council deck p. 4 and requirement rows	Y
~30 build tracks; weekly two-block status; 60+ / ~80 names	2, 10	Apple v11 C1/C2/C4	Y. Not “fifteen tracks”
~55% true vs ~40% reported → ~99% within a quarter; 8–10 hrs/week	7, 9	Apple v11 F2; delta numbers-check	Y — “within a quarter”
Events in vs events out; 2–4 hours per blocker	7, 12	Shipment Visibility 3PL ingestion note	Y. Half-time claim is an estimate
20 people; 6-week → 4-week in 6 weeks; defects about −50%	8	Apple v11 D3; resume for the defect figure	Y

NUMBER	STORY	SOURCE	MATCH
60% → 90%; seven metrics; parity within 1%; ~4% drop; $103,680 priced	9, 12	Apple v11 A3; Zendesk-to-Snowflake BRD	Y — priced, not saved
98% gate; week two 94%; ~30% throttled; full automation week three; 86 of 836	11	Apple v11 C3; Locus SOW v1.4 (Nov 19, 2024); issues log	Y
3 DCs + 100 stores; 99% vs 95%; stockouts −20%; ~$6M on the line; ~$8M protected	12	Apple v11 M1; resume; delta numbers-check	Y — nouns kept apart
Forecast 70% → 85%; about $2M safety stock; collaboration +50%	12	Apple v11 F3; resume	Y. Not an effort cut. Not added to any other dollar
~6 weeks → ~3; under 5% back-test; no award dollar	13	Numbers canon S17; panel process pack. Not in Apple v11	User anchor. No award dollar
15 work items; 12 export + 3 import; all except ISF filing	13	Customs / FFA ownership proposal	Y — decision closure, no outcome dollar
Order of ten billion dollars a year	4, 14	Public company filings, not a candidate record	Labelled. Verify the latest 10-K

Sources: CapEx brief; numbers canon; Apple delta numbers-check and resume map (the resume text file and the addendum were not on disk); Utah council pack; S/4 and ISM end-to-end add-on; Locus statement of work; Elementum process pack (Austria, cadence, carrier bid, UAT failure); Flexport (Customs, Snowflake, trace-id). Main packet files were not edited.

APPLE · 30-MINUTE CALL · CAPEX SYSTEMS ONBOARDING DATA INTEGRITY · ELEMENTUM PAIR · PART 5

Two Elementum stories for a data-integrity interviewer

Rajesh Rishiyur Chandrasekaran · interviewer: Capital Expenditure Systems Lead. Speak in first person. Every number keeps its noun and its basis: record Apple v11 / resume, user anchor earlier packet only, illustrative worked example, modeled. The eight domain names below are the list in the Hoerger S6 packet. Field-level owners the packet does not name are marked **my reconstruction — confirm**.

Story 1 — The Class A finished-goods shortage alert: eight domains of data, one trustworthy number

Where: Elementum, a six-site manufacturing customer. I owned definitions, the data-integrity program and delivery. SAP teams and integration engineers built the feeds. Planners owned the forecast; I owned the measurement.
Grain: finished goods, SKU × location × date, scoped class-A SKUs. The packet's supply rule excludes unfirmed planned orders. This walk does not convert a finished good into components.
Headline, said apart: forecast accuracy 70% → 85% on scoped class-A SKUs record ; about $2M of safety stock released under a both-errors rule record, one-time ; alert product live in 3 months record . The 200-lane lead-time program is a different result: 99% lead-time accuracy, 580,000 units, about $1.5M of inventory carrying cost released, zero stockouts record . I never add the two releases and I do not annualize the carrying-cost release.

60-SECOND PITCH — STOP ON THE CLASS-A ACCURACY

Planners at a six-site manufacturer learned about a shortage when the line stopped. About eight data domains lived in SAP and the site systems and never met, so "which orders are at risk?" had no path. I took the definitions and the data integrity. Supply, at finished-goods grain, was on-hand plus in-transit due plus production due plus released planned orders only — unfirmed orders change every planning run, so I refused to count them. Each domain went live behind a data-quality gate, inventory and production first, with a nightly full load to re-baseline and intraday deltas so receipts and shipments stayed current, and a daily reconciliation back to the SAP stock and requirements lists before anyone trusted the alert. The alert fired when days of cover fell to the lane's measured 90th-percentile lead time — a lead time early, not at line-down. Safety stock came out only where forecast error and lead-time error had both improved. Forecast accuracy on the scoped class-A SKUs went from 70% to 85%. About $2M of safety stock was released, one time, under that signed rule.

20-SECOND VERSION

Eight domains, one definition of supply at finished-goods grain, a nightly re-baseline plus intraday deltas, and an alert when cover falls inside the lane's measured lead time. Class-A forecast accuracy 70% to 85%. About $2M of safety stock released only where both errors had improved.

STAR — FIRST PERSON

S — S&OP ran on private spreadsheets. Shipment data was incomplete. The eight domains — supplier, material, bill of materials, inventory, production, transportation, sales order and planning — sat in SAP and never joined. Leadership could not say which customer orders were at risk this quarter. The line found out at line-down. Buffers were sized on that noise.

T — Nobody owned the seam. I took the grain, the definition of supply, the quality bar and the delivery. The product was a shortage a planner could act on a lead time early, and a number Finance would sign.

A — I profiled every domain for completeness, latency and whether it would join. I wrote the definition before any screen: grain SKU × location × date; supply equals on-hand, plus in-transit due by that date, plus production due by that date, plus released planned orders only. I sequenced waves by what planners already believed — inventory and production, then purchasing and the advance ship notice, then transportation — and each wave had a published quality gate the domain owner could fail. Feeds were a nightly full load, which re-baselines and heals a missed change, and intraday deltas on receipts and shipments, plus carrier status as an event. Every morning the landed quantities were reconciled to the SAP lists (stock requirements, production orders, sales orders, warehouse stock) before the alert was treated as true. A bad row was fixed in the source system, not edited in the warehouse. False alarms were classed each week as a threshold effect or a data defect. The release rule, signed with Planning and Finance before the first unit came out, was both errors improved: forecast error and lead-time error.

R — Forecast accuracy on scoped class-A SKUs 70% → 85% record. About $2M of safety stock released, one time, under that rule record. The alert product was live in three months record. Planner collaboration increased 50% record — Apple v11 wording. Adoption rose eight times record in the Hoerger packet. “Which orders are at risk?” was answered in the live planning meeting from the platform.

TRADE-OFF I MADE. A blanket buffer cut would have looked like a win in one meeting. I refused it. Capital came out only on the class-A families where forecast error and lead-time error had both improved, and lanes with a fatter measured tail kept or raised their buffer. What it cost: a slower release, and a sponsor who wanted the whole number. What it bought: a release that survived audit, and zero stockouts on the separate 200-lane watch.

Failure / learning. The manufacturer-part cross-reference gated longer than the plan because I did not open that worklist on day one. Austria had already taught me the rule I should have led with: measure the data before blaming the process. Reported quality there was 40%; at the real grain it was about 55%; it reached about 99% within a quarter, and manual reconciliation fell from 8–10 hours a week to 1–2 record. I do not say a month count inside that quarter.

Dive Deep the supply definition. Earn Trust the both-errors rule signed before the release. Deliver Results the alert in three months behind gates, not a mandate off the spreadsheet.

SAFE-SAY. “Accuracy is 1 minus MAPE on the scoped class-A SKUs, bias published beside it. Planners owned the forecast; I owned the measurement and the inputs. About $2M is safety stock, one time. The lane program is separate: about $1.5M of inventory carrying cost released, and I do not annualize it. The worked SKU below is illustrative. The eight domain names are the packet’s list; the person I name as source owner is my reconstruction unless I say otherwise.”

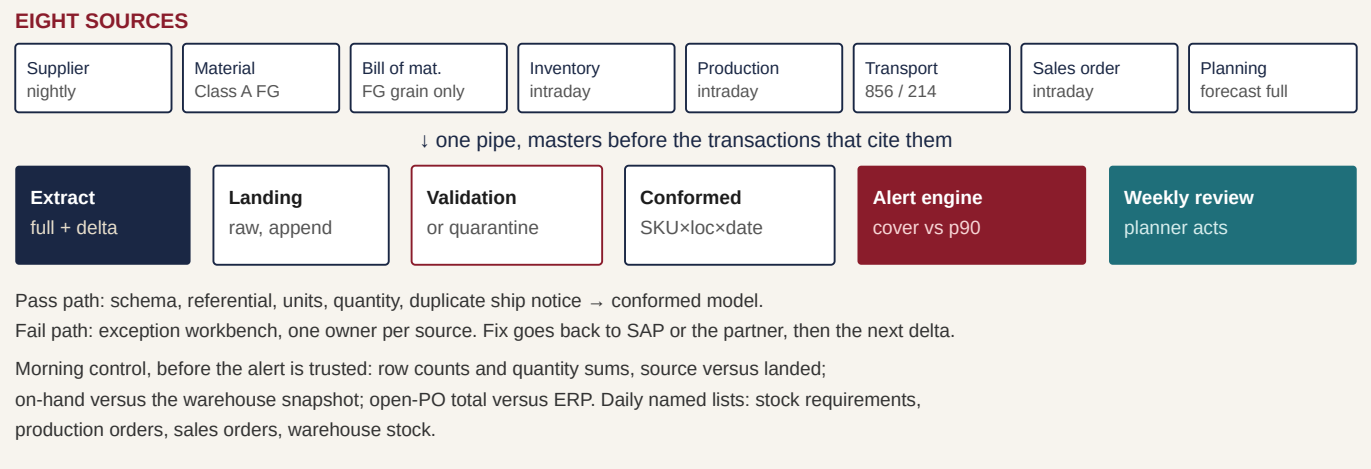
The eight domains, who owns each at the source

The Hoerger S6 packet enumerates the eight: supplier, material, bill of materials, inventory, production, transportation, sales order, planning user anchor — “about eight”. The solution-flow diagram names the SAP objects and the nightly-versus-delta pattern. It does not name a person per domain. The owner column is my reconstruction — confirm.

DOMAIN (PACKET)	WHAT THE ALERT READS	SOURCE OWNER — RECONSTRUCTION, CONFIRM	HOW IT ARRIVES
1 · Supplier	Id, calendar	Procurement steward	Nightly full
2 · Material	Class A finished good, unit	Material steward	Nightly full
3 · Bill of materials	Ingested. Not used to convert this alert.	Manufacturing engineering	Nightly full
4 · Inventory	On-hand by location	Warehouse	Intraday; morning snapshot
5 · Production	Due quantity; released orders only	Production planning	Intraday confirmation
6 · Transportation	856 ship notice; 214 status	Logistics; carrier owns the 214	Event, not a nightly batch

DOMAIN (PACKET)	WHAT THE ALERT READS	SOURCE OWNER — RECONSTRUCTION, CONFIRM	HOW IT ARRIVES
7 · Sales order	Open orders as demand	Order management	Intraday delta
8 · Planning	Forecast; planned vs actual lead time	Planners own the forecast; I owned the measurement	Forecast nightly; lane from receipts and 214s

Open POs and acknowledgements are a feed in that same flow, not a ninth domain. Lane lead time sits in planning. Calendars and order minimums sit on supplier and material. Confirm those field homes.



Source → extract → landing → validation → conformed model → alert → weekly planning meeting. “Quarantine” is my name for the packet’s exception workbench.

Nightly full loads and delta feeds, and how a bad row gets corrected

The full load re-baselines masters and the forecast and heals a missed delta. The intraday delta keeps receipts and in-transit true the same day. The 856 and the 214 are events; they wait for neither batch.

	NIGHTLY FULL	INTRADAY DELTA AND EVENTS
What the packet shows	Planning forecast as a nightly full. Master IDocs on change pointers, nightly.	Goods movements, PO history and production confirmations as intraday deltas. Deliveries and schedule lines the same way. EDI 856 and EDI 214 as events through the value-added network or AS2.
Why	A missed delta would otherwise live forever. The morning number has to match SAP.	A receipt or a shipment that lands at noon changes cover today.
Order	Masters before the transactions that cite them — my reconstruction of referential integrity; confirm. Program sequence in the record is trust order: inventory and production, then purchasing and the ship notice, then transportation.	
Change key	Packet word: change pointers. “Watermark” and an idempotent upsert on SKU, location, document and line are my words — confirm. A late ship notice applies on the next event and is healed again by the nightly full.	

ERROR CORRECTION — WHAT IS RECORD, WHAT I AM RECONSTRUCTING

Record. A published data-quality gate per domain. Daily reconciliation to the stock-requirements list, the production-order list, the sales-order list and the warehouse stock list. False alarms worked as defects, weekly, with a root cause. Domain owners could fail their own wave. The exception workbench was staffed in hypercare.

My reconstruction — confirm the names. At landing I reject, and do not silently fix: schema, a missing SKU or location, a unit mismatch, a negative or impossible quantity, a duplicate ship notice. The reject sits with the source owner. Before the alert is trusted that morning I count rows source versus landed, the sum of quantities, the open-PO total versus ERP, and on-hand versus the warehouse snapshot. The fix goes back to SAP, the supplier or the carrier, and returns on the next delta. I do not patch the warehouse. That is the Austria rule: measure the data before blaming the process — about 55% true against 40% reported, then about 99% within a quarter.

How the shortage alert is calculated — finished goods, values substituted

Two illustrative examples from the Hoerger packet. They are not the same SKU. The rule is record; the quantities are illustrative. Program results stay the record numbers above.

Example A — timing, not volume. Finished-good SKU ROUTER-100 at DC-East. On-hand 400. In-transit 600 lands day 5. Production 800 lands day 9. Demand 500 on day 3, 700 on day 7, 400 on day 10. Projected position on a date = on-hand + in-transit due by that date + production due by that date − demand through that date. Released planned orders would add the same way; this example has none. Unfirmed planned orders are excluded on purpose.

DAY	SUPPLY DUE	DEMAND TO DATE	PROJECTED POSITION	WHAT THE PLANNER HEARS
3	400	500	**−100**	Short 100. In-transit has not landed. Second tier: position has crossed zero.
5	400 + 600 = 1,000	500	+500	Ship notice lands. Same inputs; the packet's checkpoints are days 3, 7 and 10.
7	1,000	1,200	**−200**	Short 200. Day-9 production is too late. Fix window = 9 − 7 = 2 days.
10	1,800	1,600	+200	The quarter is long, the week is short. This is a timing shortage, not a volume shortage.

Example B — the alert rule, and safety stock, on the packet's other illustrative lane (demand 100 units a day, not ROUTER-100). Days of cover = on-hand ÷ daily demand = 1,200 ÷ 100 = 12. Lane 90th percentile = 14 days → 12 is inside 14 → the alert fires today, a lead time early. Same SKU on a lane whose 90th percentile is 9 days: 12 > 9 → no alert. That is how a measured lane kills a false alarm the brochure would have raised.

Safety stock on that lane, the form I use to explain the buffer, not a screen I claim was in production: SS = z × d × σ of lead time = 2.33 × 100 × 4 = 932 units. If measured σ of lead time falls from 4 days to 2, SS = 2.33 × 100 × 2 = 466, and 466 units are the release — only if forecast error and lead-time error have both improved. Position falls below that 932-unit safety stock on day 3 (1,200 − 300 = 900). Position hits zero on day 12. Two tiers: below safety stock inside the lead time (re-plan), and at zero (the line-down we used to discover). z = 2.33 because the signed service level was 99%.

The formula I actually shipped on the lanes replaced that single root with two terms a planner can read, because the old root was dominated by a lead-time swing nobody had measured. Illustrative Lane A: demand 500 a day, σ of demand 150, brochure lead time 18 days, ERP default σ of lead time 8 days (never measured), measured mean 21 days, measured σ 4 days, ranked 99th percentile 34 days, about 400 shipments.

STEP	FORMULA WITH THIS LANE'S VALUES	RESULT	BASIS
Old demand piece	$18 \times 150^2 = 405{,}000$	97.5% of the variance was the unmeasured 8-day swing	illustrative
Old lead-time piece	$500^2 \times 8^2 = 16{,}000{,}000$. $\sqrt{(405{,}000+16{,}000{,}000)} = 4{,}050$. $\times 2.33 = 9{,}437$. Plus a 7-day pad × 500 = 3,500.		illustrative
Before	9,437 + 3,500	12,937 units (25.9 days)	illustrative
After, two terms	Lead-time cover $500 \times (34 - 21) = 6{,}500$. Demand cover $2.33 \times 150 \times \sqrt{21} = 2.33 \times 150 \times 4.583 = 1{,}602$. Pad = 0.	8,102 units. Release 4,835.	illustrative lane
What the bell curve would have done	$2.33 \times \sqrt{(21 \times 150^2 + 500^2 \times 4^2)} = 4{,}928$ units, a plan of 30.3 days	14 of 400 shipments ran past it — a "99%" plan that held 96.5%	illustrative

Planned lead time versus the 200-lane actuals — false alerts and misses

WHAT CHANGED	BEFORE	AFTER	BASIS
Example lane, plan vs scans	Brochure 18 days. 240 of 400 shipments (60%) slower than the plan. ERP σ = 8 days.	Mean 21, σ 4, ranked 99th percentile 34. Late against that plan: 4 of 400, 1%.	illustrative lane; the pattern behind the program
Lane A promise misses	7 in 100 late against brochure-plus-pad	1 in 100 late against the ranked 99th percentile	same illustrative lane (28 of 400 → 4 of 400)
Shortage-alert false alarms	Brochure lead time alerted fast lanes. Defects (bad on-hand, missing receipt, unfirmed order counted) were mixed in.	At the 90th percentile, illustrative precision 71.2% and recall 98.6% — about 29% of alerts are the price of the window. At the 99th, precision 58.8%, recall 99.9%: more false alerts, almost no misses saved. Defects were burned down weekly. No program count of that burn-down.	rule record; rates illustrative
Program	Brochure lead time on 200 lanes	99% lead-time accuracy at a 99th-percentile promise. 580,000 units. About $1.5M of inventory carrying cost released. Zero stockouts in the watch. I do not annualize it.	record

The alert and the promise read the same distribution and make different decisions. The alert is a re-plan trigger at the 90th percentile, so the planner still has a window. The promise is a commitment at the 99th percentile, the tail you can defend.

ALERT PAYLOAD, AND WHO ACTS IN THE WEEKLY PLANNING MEETING

- **SKU, location, day of breach, shortfall.** On the illustrative finished good: ROUTER-100, DC-East, day 3 short 100 and day 7 short 200, with a two-day window before production lands.
- **The input that moved.** Apple's own write-up of this program: every alert carried the input that moved, so a planner could check it in about thirty seconds. That is record of the design, not a field list I can quote from a screenshot.
- **Recommended action.** Re-plan inside the window first. Expedite only if the window is already gone. Pull a released order forward if one exists. Reallocate only against stock the definition counts. I describe those four; confirm the menu labels.
- **Who acts.** The planner, in the weekly planning meeting (the meeting the packets call S&OP). Domain owners take data defects, not the planner. Planning and Finance act on the buffer release, not on a single SKU row. I would call that meeting a weekly business review if the room uses those words.

DATA-INTEGRITY ANGLE FOR THIS INTERVIEWER — THE SAME MECHANISM, DIFFERENT NOUNS

What I ran is not a planning trick. It is eight source domains, one owner each, a nightly re-baseline plus deltas, a quarantine row with an owner, and counted rows — source versus landed, sums, open documents versus the book of record — before anyone trusts a number. On a capital-equipment program the nouns change and the mechanism does not: the equipment list, the supplier, the purchase order, the ship notice, the acceptance, the asset tag, the site, the placed-in-service date. Each input has one owner. A nightly snapshot heals a missed change from the contract manufacturer's tracker. An intraday delta keeps a receipt current. A row that fails referential checks waits, with an owner, and the fix goes back to the source spreadsheet or the vendor, not into a side file that makes the dashboard green. I would not put a capitalization number, or any later model, on top of a reconciliation that has not been counted. This is my analogy from the manufacturing program. It is not a description of Apple's tool.

SIX PROBES — ANSWER THE ONE ASKED, THEN STOP

1. Why both a nightly full load and a delta?

The full load heals a missed change pointer so masters match SAP in the morning. The delta is the clock: a noon receipt or 214 changes cover today. Masters before the transactions that cite them. “Watermark” and “upsert” are my words for the change pointer and the business key.

2. Why exclude unfirmed planned orders?

They change every planning run and were inflating supply. Released orders only, agreed before the first screen.

3. Walk one Class A finished good.

ROUTER-100, illustrative. On-hand 400, in-transit 600 on day 5, production 800 on day 9, demand 500 / 700 / 400. Position −100 on day 3, −200 on day 7, +200 on day 10. Timing, not volume. The other illustrative lane: cover 12 inside a 14-day 90th percentile, so the alert fires now; below 932 units of safety stock on day 3; zero on day 12.

4. Why the 90th for the alert and the 99th for the promise?

One distribution, two decisions. The alert is a re-plan trigger: illustrative recall 98.6% at 71.2% precision. The promise sits at the ranked 99th. A bell-curve 99th on Lane A held 96.5%.

5. How do you separate a false alert from bad data?

About 29% of alerts at the 90th percentile are the price of the window, illustrative. A defect is the data lying — wrong on-hand, missing receipt, an unfirmed order counted — and that kind is burned down weekly. I have no program count of the burn-down.

6. What would you refuse to automate until the number is trustworthy?

The buffer release, and any model on top of it. Austria was 40% reported and about 55% true. I release capital only where both errors improved, and I do not put a capitalization or a model on a reconciliation I have not counted.

Story 2 — Procure-to-pay in Elementum: bringing supplier and carrier EDI into one record

Where: the same Elementum years. I owned the onboarding standard and the scope cut. Engineers built the maps. Transportation owned the carrier relationships. I did not own a payment run.
Sets in the record: X12 850, 855, 856, 940, 945, 214, 997, plus an XML or portal floor. **Standard cycle I am reconstructing — confirm before you say the customer exchanged them:** 860/865, receipt as a goods movement, 810, 820, and the carrier set 204, 990, 210.
Numbers I will say: a thousand-lane program across 25 European locations record ; completeness 60% → 95% record ; cohort gate at least 95% valid ASN record . A thousand lanes were asked; 597 first user anchor . No supplier count, interchange volume, match rate or days-to-onboard is in the notes.

60-SECOND PITCH

Inbound visibility on a thousand lanes, across 25 European locations, was asked for in two months. Completeness was 60%. I owned the standard. The order went out as an 850, the acknowledgement came back as an 855, the ship notice as an 856, the warehouse pair as 940 and 945, carrier status as a 214, and a 997 closed every interchange. No EDI meant the same fields by XML or a portal. A cohort went live only after at least 95% valid ship notices. One cohort failed twice on a timestamp; I held it. A thousand lanes were asked; 597 went first. Completeness went from 60% to 95%. That 856 and that 214 are the in-transit the alert reads.

20-SECOND VERSION

I joined the purchase order, the ship notice and the carrier status into one shipment, behind a 95% valid-ASN gate. A thousand lanes were asked across 25 European locations; 597 went first. Completeness 60% to 95%. I will not quote a supplier count or a match rate I do not have.

STAR — FIRST PERSON

S — Inbound visibility was a pile of files. The ERP had the purchase order. The acknowledgement lived in email. The ship notice was late, partial, or missing. Carrier status lived in a portal the planner did not join. Completeness was 60%. The ask was a thousand lanes in two months, scope and date and quality all frozen.

T — Make one record a planner and the shortage alert could trust: the order, the acknowledgement, the shipment, the milestone. Influence an executive two levels up. No line authority over Transportation, Procurement or the carriers.

A — I ranked lanes by spend, criticality, mode and volume and took a phased cut up with Transportation and Procurement already signed on, so no lane owner could contradict it in the room. I wrote the document set and published the gate before the first cohort: at least 95% valid advance ship notices in test, 997s flowing, tracking numbers joinable to the order. Engineers built a parameterized harness — one map shape, not a new map per carrier. Test files, then certify, then go-live, then a weekly scorecard. A lane that slipped went back to fix, not live. For partners who could not send X12, the same fields arrived as XML or a portal upload. The procure-to-pay story I tell around that standard is the full cycle below; I mark the transaction sets the notes do not list.

R — Completeness 60% → 95% on a thousand-lane program across 25 European locations record. A thousand lanes were asked; 597 first user anchor — I confirm what actually went live before I call 597 the delivered count. The gate held: one cohort failed twice on a timestamp format user anchor. In my earlier notes, freight cost on the onboarded lanes was down about 10%, about $2M of freight cost in six months user anchor — not annualized, and not the safety-stock release. The ship notice and the 214 fed the shortage alert's in-transit. I claim no award dollar on the later carrier bid.

TRADE-OFF I MADE. All thousand lanes in two months would have automated dirty data and manufactured confident wrong answers. I phased by value — 597 first in my earlier notes, the remainder behind the same gate — and I held a cohort that missed the bar. What it cost: a date on the long tail, and an executive who had asked for everything. What it bought: a number the alert could read.

Failure / learning. The harness should have existed for cohort one, not cohort two; early maps were more bespoke than they needed to be. The gate itself was right: a timestamp format failed twice and I held the cohort both times rather than relaxing the bar. Same lesson as Austria — measure the data before blaming the carrier. Reported quality there was 40%, true quality about 55%, about 99% within a quarter.

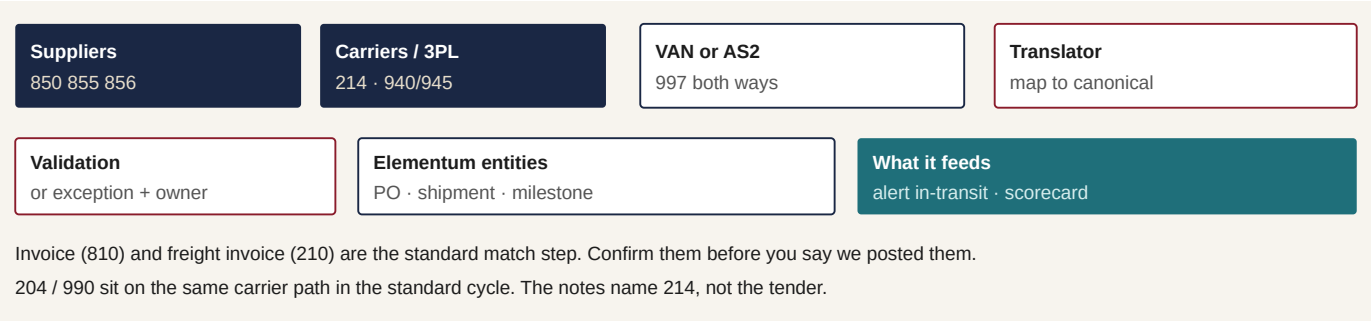
Insist on the Highest Standards the gate, held in public. Have Backbone the re-scope, two levels up, then full commitment to the phased plan. Dive Deep the spend curve before the debate.

SAFE-SAY. "It was a thousand-lane program across 25 European locations. I phased by value. The 597-first figure is my earlier user anchor, not a settled delivered count. Completeness 60 to 95 is the record, and so is the 95% valid-ASN gate. About $2M of freight cost in six months is a different number from the safety-stock release, and it is a user anchor. I do not have a supplier count, an interchange volume, a three-way match rate, or days to onboard. I will not invent them."

The cycle, and which sets the notes actually name

STEP	SET	IN THE NOTES?
Requisition to PO	850	Yes — ERP out via VAN or AS2

STEP	SET	IN THE NOTES?
Acknowledgement	855	Yes — back from the supplier
Change	860 / 865	Standard set — confirm; not in the list I published
Ship notice	856	Yes — the in-transit the alert nets
Receipt	ERP goods movement	Yes as a delta. No EDI receipt code in the notes.
Invoice, match, pay	810, three-way match, 820	Standard close — confirm. No match rate. I will not invent one.
Warehouse	940 in, 945 out	Yes, for the 3PL
Carrier tender	204, 990	Standard carrier set — confirm
Status	214	Yes. Late events apply on the next delta. No late-rate.
Freight invoice	210	Standard set — confirm. Austria used paid freight; that is not a claim we ingested 210.
Functional ack	997 both ways	Yes. A reject stops the map. No 997 percentage in the notes.



Supplier or carrier → VAN or AS2 → translator → validation → Elementum entities → the alert and the scorecard. Exception path is the validation box.

ONBOARDING A NEW SUPPLIER OR CARRIER — THE PLAYBOOK I ACTUALLY RAN

- **Test files.** The partner sends a file. The harness classifies defects as format, timestamp, missing segment, or join key, each with an owner.
- **Map.** One parameterized template onto the canonical purchase order, shipment and milestone. Not a new integration per name.
- **Certify.** Valid advance ship notices at or above 95% in test, 997s flowing both ways, tracking number joinable. The gate was published before cohort one. A gate that holds only in private is a suggestion.
- **Go live.** Activate only off that gate. The first wave, in my earlier notes, was the 597 lanes. I do not quote a days-to-onboard figure — **[your number] days to onboard a supplier** is not in the notes. Safe line: “The first wave was the spend, not a stopwatch per supplier.”
- **Monitor.** Weekly scorecard: events received over events expected, the 997 rate, joinability, completeness. Below the line, the lane returns to fix.

ERROR CLASS	WHAT THE NOTES SUPPORT	WHO CLOSED IT
997 reject	The scorecard watched the 997 rate. No percentage is in the notes.	The partner, on their map. We did not activate over a reject.
Timestamp format	One cohort failed the gate twice user anchor.	I held it. The carrier fixed the format. Then it activated.
Missing segment, join key	Named in the cohort test-day classification.	Owner per defect. Fix at the source file.
Duplicate or partial ASN; quantity or unit mismatch; unknown PO line; late 214	Not counted in the notes. Same control as the landing rules on the shortage alert: quarantine, one owner, fix at the source, next delta heals it.	Confirm these classes with your own examples before you tell one. Do not invent a rate.

ERROR CLASS	WHAT THE NOTES SUPPORT	WHO CLOSED IT
Five spellings of one carrier	Austria defect census record. A name map, not a new carrier.	Fix the master. Do not default a carrier on a no-hit.

What the record fed. The alert's in-transit is the 856 and the 214. Transportation watched the per-lane scorecard. Austria — about 55% true to about 99% within a quarter — shows a carrier comparison is only as good as the promise; I do not claim that scorecard set an award. The bid used a lane master (one origin, destination and mode) so dozens of rate files could be compared: about six weeks to about three, back-test under 5% against the last two awards user anchor. Hundreds of lanes, dozens of rate files, no carrier count, no award dollar. **[your number] suppliers onboarded**, **[your number] interchanges a week**, **[your number] three-way match rate** are not in the notes. Safe line: "I can defend completeness and the gate. I cannot defend a volume I did not write down."

SIX PROBES

1. Which documents did the customer actually send?

850, 855, 856, 940, 945, 214, and 997 both ways, plus XML or a portal for partners without EDI. 860, 865, 810, 820, 204, 990 and 210 are the standard cycle I use to explain procure-to-pay. I confirm those before I say we exchanged them.

2. What does 95% valid ASN mean?

The cohort had to clear it in test before activation. That bar is record. Cohort sizes are not in the notes, so I do not quote a sample size or a confidence bound as if I had run one. A lane under the scorecard went back to fix.

3. A partner's file was wrong. What did you do?

Classified it — format, timestamp, missing segment, join key — named an owner, and held the cohort. Twice, on a timestamp. I did not relax the gate and I did not patch the warehouse.

4. How does this reach the shortage alert?

The 856 is in-transit due, dated. The 214 is the milestone that says whether that date is still true. The alert's projected position adds that quantity on the due date and does not add it before. A duplicate or an unknown line never reaches that sum; it waits on the exception owner.

5. How many suppliers, and how fast did you onboard them?

I don't have those counts in my notes. A thousand lanes were asked across 25 European locations; 597 went first in my earlier record; completeness went from 60% to 95%. That is the scale I can defend. **[your number]** stays a blank until you fill it.

6. Did the carrier bid use these scorecards?

The bid used a lane master so rate files shared one key, and a back-test under 5% against the last two awards. The EDI scorecard made the shipment record trustworthy. I will not say the scorecard set the award. I claim no award dollar. The cycle, about six weeks to about three, is a user anchor.

Close — the control table, the 90-second versions, what to verify

Eight domains → field → owner → control

Domain names are the Hoerger list. The field is what I would defend in the room. The owner is my reconstruction unless the cell says the packet named the role. The control is the one I would refuse to skip.

DOMAIN	FIELD THE NUMBER USES	OWNER	CONTROL BEFORE THE NUMBER IS TRUSTED
Supplier	Id, calendar	Procurement — confirm	Nightly full. Unknown id quarantined. Five spellings mapped to one carrier.
Material	Class A finished-good id, unit	Material steward — confirm	Nightly full. Unit mismatch rejected. Finished goods only.
Bill of materials	Ingested; not used to convert this alert	Manufacturing engineering — confirm	Nightly full. Walk stays at finished-goods grain.
Inventory	On-hand by location	Warehouse — confirm	Intraday. Morning sum vs warehouse list. Negative qty quarantined.
Production	Due qty, released only	Production planning — confirm	Intraday. Unfirmed orders excluded by rule.
Transportation	856 qty and date; 214; lane	Logistics; carrier owns the 214	Event plus 997. Duplicate notice quarantined. ≥95% valid ASN before go-live.
Sales order	Open-order demand	Order management — confirm	Intraday. Reconciled to the sales-order list.
Planning	Forecast; planned vs actual lead time	Planners own the forecast. I owned the measurement — packet.	Forecast nightly. Lane from receipts and 214s. Both errors improved before a release.

SAY IT IN 90 SECONDS — STORY 1

We found shortages when the line stopped, because eight domains never met and supply counted orders that changed every planning run. I defined supply at finished-goods grain before any screen: on-hand, plus in-transit due, plus production due, plus released orders only. Masters came in on a nightly full load, so a missed change healed overnight. Receipts, ship notices and carrier status came as deltas and events, so cover was true the same day. A bad row — schema, reference, unit, quantity, duplicate ship notice — waited with an owner, and the fix went back to the source. Each morning I counted rows and quantities against SAP before the alert was trusted. On the illustrative finished good, position went negative on day 3 and again on day 7 even though total supply covered total demand: a timing shortage, two-day window. The rule fired when cover fell inside the lane's measured 90th-percentile lead time, and a second tier fired at zero. Measured lanes — 200 of them, promise at the 99th percentile, 99% lead-time accuracy — made that threshold honest and released about $1.5M of inventory carrying cost, which I do not annualize. Separately, class-A accuracy went from 70% to 85%, and about $2M of safety stock came out only where both errors improved. I never add the two releases. Austria sits underneath: measure the data before you blame the process.

SAY IT IN 90 SECONDS — STORY 2

Procure-to-pay became one record only where the documents shared one pipe. The order went out as an 850, the acknowledgement returned as an 855, the ship notice as an 856, the warehouse pair as 940 and 945, carrier status as a 214, and a 997 closed every interchange both ways. Partners without EDI sent the same fields by XML or a portal. I confirm 860, 865, invoice 810, remittance 820, and carrier tender 204, 990 and 210 before I claim them — that is the standard cycle, not the list I published. Onboarding was test file, map, certify, go live, monitor. Certify meant at least 95% valid ship notices, published in advance, and I held a cohort twice on a timestamp. A thousand lanes were asked across 25 European locations; 597 went first in my earlier notes; completeness went from 60% to 95%. That ship notice and that status are the in-transit the shortage alert adds. The same lane key later let a bid compare rate files, and I claim no dollar on the award. I have no supplier count, match rate, or days-to-onboard written down, so I will not make one up.

Verify — fill these, and the sentence until you do

BLANK	SAFE SENTENCE
[your number] suppliers; interchanges a week; days to onboard	“A thousand lanes, phased by value. Completeness 60% to 95%. I will not quote a supplier count, a daily volume, or days per supplier.”
[your number] three-way match rate	“Invoice match closes procure-to-pay. I do not have a match rate, so I will not invent one.”
597 as a delivered count	“A thousand lanes were asked; 597 first, in my earlier notes. I confirm what went live.”
Quarantine, watermark, upsert; owner names	“The packet shows change pointers, a nightly full, intraday deltas, and an exception workbench. Those words, and the person per domain, are my reconstruction — confirm.”
860, 865, 810, 820, 204, 990, 210	“Standard cycle. The sets I published were 850, 855, 856, 940, 945, 214 and 997.”
Duplicate ASN, unit mismatch, unknown PO line, late 214	“Same control: an owner, fix at the source. I do not have a rate.”
Watch-window length; Austria months	“Zero stockouts in the watch. About 99% within a quarter. I do not quote the window or a month count.”
Freight about $2M; bid six weeks to about three	“About $2M of freight cost in six months is a user anchor, not the safety-stock release. The bid cycle is a user anchor. No award dollar.”

Source trace

NUMBER	STORY	SOURCE FILE	MATCH
Eight domains, named	S6	hoerger-kimi-k3.html S6	User anchor
Supply definition; unfirmed orders out	S6	same, action and table 2	Record rule
Nightly full, change pointers, intraday deltas, 856/214	S6	same, solution flow	Record design
Daily recon to MD04, COOIS, VA05, MB52; trust-order waves; alert in 3 months	S6	same	Record
70% → 85%; about $2M of safety stock; collaboration +50%	S6 / F3	Hoerger S6; apple delta §5; master S6	Record. Say collaboration increased 50%.
ROUTER-100 position −100 / −200 / +200	S6	Hoerger table 2	Illustrative
Cover 12 vs p90 14; SS 932 → 466	S6	Hoerger table 2	Illustrative values; rule is record
Precision 71.2% / recall 98.6%; at p99, 58.8% / 99.9%	S6	Hoerger alert table	Illustrative
Lane A 18 / 21 / σ 4 / p99 34; 60% → 1%; 12,937 → 8,102; bell curve held 96.5%	S8a, S8b	Hoerger	Illustrative lane
200 lanes; 99% lead-time accuracy; 580,000 units; about $1.5M of inventory carrying cost released; zero stockouts	S8 / F4	numbers-reconciliation S8; apple delta §5	Record. Do not annualize.
Austria 40% reported, about 55% true, about 99% within a quarter; 8–10 h → 1–2 h	S10	cabreros-stories-b; apple delta §5	Record
Thousand lanes; 25 European locations; 60% → 95%; gate ≥95% valid ASN	S9 / F1	apple delta §5; Hoerger S9	Record
597 first; about 80% of spend; about $2M of freight cost in six months; timestamp held twice	S9	Hoerger S9; numbers-reconciliation S9	User anchor
850, 855, 856, 940, 945, 214, 997; XML or portal floor	S9	Hoerger S9; josyula S9	Record design
Bid about six weeks → about three; under 5%; hundreds of lanes; dozens of rate files	S17	cabreros-stories-b; numbers-reconciliation S17	User anchor. No award dollar.
Alert carried the input that moved	F3	apple delta JΔ3	Record of the design

Files under /workspace/amazon/: panel/amazon-nacf-panel-2-hoerger-kimi-k3.html, amazon-l6-scm-master-packet.html, panel/amazon-nacf-panel-4-cabreros-stories-b-star-kimi-k3.html, panel/amazon-nacf-apple-delta-addon-star-kimi-k3.html §5, numbers-reconciliation.md. Not a claim about Apple’s capital system.

USE CASE · REAL-WORLD ASSET MANAGEMENT · FOR THE CONVERSATION WITH THE CAPEX SYSTEMS LEAD

What actually happens to a machine the brand owns and does not hold

> **Illustrative storyline built from public knowledge of how capital equipment is managed at contract manufacturers and from the job description — not inside knowledge of Apple.** Names, prices, and dates below are the practice extract for program TS-LINE-RAMP. They are there so you can point at a row. They are not Apple's books.

> **Who this is for.** You, explaining the use case in plain words to Apple's Capital Expenditure Systems Lead.
> **The practice program.** Test-station line ramp, **TS-LINE-RAMP**. 40 purchase-order lines, CM site A and CM site B, one internal plant, three suppliers, 12 fixtures reused. As-of **16 March 2026**. People: Elena Vasquez (AME), David Cho (TPM), Priya Raman (GSM), Wei Chen (site A), Jordan Hale (site B), Morgan Ellis (Finance). No row for you. You run the load, the scorecard, and the queue.

1. The real-world setting

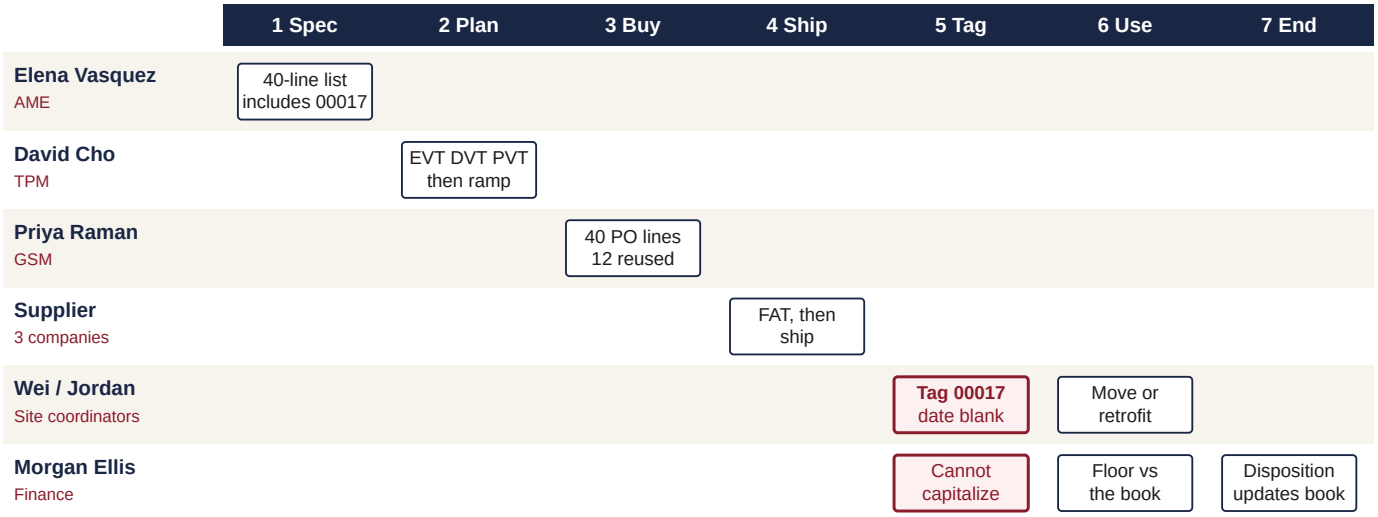
A company designs a product. Contract manufacturers — other companies' factories — build it. To build it and to test it, the line needs capital equipment: test stations that check each unit, handlers that move it, fixtures and jigs that hold it, instruments that measure it, and automation cells that repeat a motion without a person at every step.

Much of that equipment is bought by the brand and stays on the brand's books, while the machine sits in the contract manufacturer's factory, often in another country. The brand owns a very large number of assets it does not physically control, across factories, programs, and suppliers. Order, pay, ship, install, move, reuse, scrap: the record can be wrong at each step. That is the asset-management problem. This role makes the record match the machine. The **ERP** is the main business system, where the purchase order and Finance's book live. The **system of record** is the one official place this program's equipment data lives after onboarding. Wei's spreadsheet and Jordan's spreadsheet are sources the load reads.

WORD	WHAT IT MEANS ON A FACTORY FLOOR AND IN THE BOOK
CapEx	Capital expenditure: money for a thing used for years. A test station counts. Parts consumed this week do not.
Capital appropriation	The formal yes: an amount, a cost center (the accounting bucket), and a program. Here the yes is the 40-line equipment list. Prices sum to **$197,500**. lab list
PO	Purchase order. One line is one buy: what, how many, the price, which program. Line **00017** is one test station.
Asset tag	The label and its number. How someone on the floor points at one machine. Line 00017's tag is **AST-TS-0017**.
Asset master	The official list. One row per tagged asset, with the fields Finance and an auditor rely on.
Placed-in-service date	The day the asset starts its job. That day starts capitalization and depreciation. A financial control. In this lab: not the ship date, and not the day the crate was unpacked.
Capitalization	Recording the cost on the balance sheet as a long-lived asset.
Depreciation	Charging a share of that cost to each year of use. A blank date means it does not start.
Fixed-asset ledger	Finance's book: cost, in-service date, depreciation taken, and where the asset is supposed to be.
Custodian	The named person accountable for the physical machine. Here, Wei at site A and Jordan at site B. The extract stores the field owner, not a separate custodian column.
FAT / SAT	Factory acceptance test: the supplier proves it works before ship. Site acceptance test: the factory proves it works after install. Two signatures, two dates.
Retrofit / reuse	Using a machine the company already owns, as-is or modified. Twelve ramp rows point at fixtures from program Fixture reuse 2024.
Disposition	The end for that asset: transfer, hold, sell or return, or scrap. The ledger records the same fate.
Ghost asset	On the books and not findable on the floor. Or on a record with no purchase order behind it.
Physical-to-book audit	Walk the floor, read the tags, compare to the ledger. The lab reconciles purchase order to tag. That check sits beside the walk; the walk still happens.

2. One asset’s life, step by step

Follow **line 00017**: a Northline test station, PO-55022, CM site B, cost center CC-4100, lab price **$12,400**, ordered **20 January 2026**. The program around it is 40 lines — 8 test-station lines, 6 handlers, 7 instruments, 7 new fixtures, 12 reused fixtures; 22 for site A, 18 for site B. lab



Read left to right. The box sits in the lane of the person who does that step. Red is line 00017, asset AST-TS-0017: tagged at CM site B on 12 February 2026, placed-in-service left blank, so Morgan cannot start depreciation. You, the Data Integrity PM, are not a builder of the asset. You arrive when this picture is already wrong.

STEP	WHO	WHAT HAPPENS
a. Spec	Elena, AME	Units per hour × shifts × yield becomes the 40-line list (equipment_list_ts_line_ramp.csv). Line 00017 is a site B test station on it.
b. Plan	David, TPM	Install dates for **EVT** (early units: can we build the design?), **DVT** (does the process repeat?), **PVT** (does the line run as production will?), then ramp. He owns program allocation. In the last 90 days he is the only person who has written it on this ramp.
c. Buy	Priya, GSM	Northline, stations and handlers, PO-55022. Helix, instruments, PO-55023. Atlas, new fixtures PO-55021 and reused fixtures PO-55030. She checks owned assets first. Lines **00027–00038**, all site B, are 12 reused fixtures at **$0**, each pointing at **AST-FR-0001** through **AST-FR-0012** from Fixture reuse 2024 at the internal plant. That search is what the interviewer’s profile credits with millions in retrofit opportunities. This file stores the link, not a savings figure.
d. Ship	Supplier	Build, factory acceptance test, ship. A ship notice carries serials. This extract has no ship-notice file. Line 00017 is received **12 February 2026**.
e. Tag	Jordan at B, Wei at A	Install, site acceptance test, then the tag: asset, serial, PO line, site, line, station, custodian, cost center, program, reuse source, placed-in-service date. Jordan tags **AST-TS-0017**, serial **SN-TS-0017**, on 13 February, and leaves the date blank. His sheet says site **B** and program **ts-line-ramp**.
f. Use	Factory, then Finance	The station runs, moves, is retrofitted, or sits idle. Each of those is a data event. Finance and audit walk the floor against the ledger.
g. End	David, with Morgan	Transfer, hold, sell or return, or scrap. The ledger records the same fate.

3. Who enters which data, where, and why it breaks

Wei's file and Jordan's file are both "the tracker," and the headers do not match. Site A has PO Line, Site, In Svc, Reuse From. Site B has po_line, Location, pis, prior_asset. Dates, costs, and the program name are typed in more than one format. Each sheet can be accurate about what that person saw, the ERP accurate about what was ordered, and Finance still unable to capitalize.

EVENT	WHO ENTERS IT	WHERE TODAY	THE FIELDS	WHY THIS INPUT EXISTS	WHAT GOES WRONG
Capacity becomes an equipment list	Elena Vasquez, AME	Engineering spec, then the approved equipment list	Line, type, site, quantity	The buy has to match what the line needs to hit the rate.	The model changes after the list is copied. The purchase order still says 40.
Launch plan and the approval	David Cho, TPM	The program record. In the lab, program allocation on the asset.	Program, need-by dates for EVT, DVT, PVT, ramp	Spend and the asset have to belong to a program someone can name.	One person holds the allocation rule. In the last 90 days David is the only editor of that field on this ramp.
Reuse check, then the PO	Priya Raman, GSM	ERP for the purchase order. A mailbox or a private sheet for the reuse check, unless the source asset id is a real field.	PO line, supplier, price, cost center, reuse-source asset	The PO is the commitment. The reuse link is how the company avoids buying a fixture it already owns.	A fixture at the internal plant is invisible, so a new one is bought. In the lab the 12 links exist, at $0, only if someone keeps AST-FR-0001 and the rest filled in. Elena owns that field.
Build, FAT, ship	Supplier	Supplier email or portal. This extract starts at the received date.	Serial, ship date	The serial is what ties a physical unit to a PO line before the company tags it.	The serial stays in an inbox. The tracker gets a typed copy, and one value is entered on two rows.
Receive, SAT, tag	Wei Chen (site A), Jordan Hale (site B)	Two CM Excel trackers, different headers	Asset, serial, PO line, site, cost center, program, reuse source, placed-in-service date	So Finance can start depreciation, and so an auditor can find the machine.	The date is left blank. The site is spelled five ways. A serial is repeated. One PO line arrives as two tagged assets. One asset is keyed with no PO line.
Move, retrofit, idle	Coordinator; Elena if the tool is retooled	The same spreadsheet, sometimes nowhere	New site or line, program, reuse flag	The asset's life continues after the first install. The book has to follow.	The factory moves it over a weekend. The ledger still has the old line. A code that exists can still be the wrong site.
Count, then disposition	Morgan Ellis, with the custodian; David for the end state	Count sheets, then the fixed-asset ledger	Found or not found; transfer, hold, sell, or scrap, and the date	The company has to show the book and the floor are the same set of machines.	Paid for, and not on the floor: a ghost asset. Scrapped on the floor, still depreciating. Or the reverse of a ghost: on a record, and in neither tracker, with no PO.

WHAT THOSE BREAKS COST, IN THE REAL WORLD

- **Ghost assets.** Paid for, and not on the floor at audit. Lines **00039** and **00040** are in transit and still count as bought. **AST-TS-0040**, serial **SN-TS-ORPHAN-9001**, is the other ghost: keyed at site B with a date and no purchase-order line, in neither spreadsheet. Priya owns the link.
- **The in-service date.** Twelve tagged assets have it blank, so Morgan cannot capitalize them. Line **00019** / AST-TS-0020 has the date **11 February 2026** and a receipt of **18 February**, so depreciation would start before the tool arrived. Either one is a financial-control finding.
- **Missed reuse.** A fixture that could be retooled sits in another factory, off the new tracker, and Priya buys a new one. The lab's counter is the reuse-source field, scored only on the 12 reuse lines. Elena owns it. Without that id, Finance can capitalize a tool the company already owns.
- **Budget and forecast.** The list is a $197,500 commitment. Twelve tagged assets are not in service, and two lines have not arrived. The purchase-order forecast and the asset book describe different sets of machines.
- **One person's head.** In the 90 days to 16 March, a single editor holds placed-in-service (Morgan), program allocation (David), and reuse source (Elena). Cost center is shared by Priya and Morgan. On the older fixture program, allocation was written by a prior TPM who is not in today's owner list.
- **Both spreadsheets are "right."** Jordan spells site B as **B**, **Site B**, **SITE-B**, **site b**, and B with a trailing space. Wei's sheet does the same for site A, and it has line **00901**, "Mystery box," site SITE-Z, which is not on the equipment list. Jordan's sheet has a row with no line id, cost **abc**, and the date **13/32/2026**. The sheets, the 40-line list, and Finance's rule on a blank date are three true statements, and still no single picture of the line.

4. The storyline, on the extract's calendar

Speak it as a sequence: spec, approve, buy, ship, tag, reconcile. The extract does not number those as week 0, week 2, week 3. It stores dates. The as-of date is **16 March 2026**. The picture below is the open queue on that morning, before anything is closed.

WHEN	WHAT THE FILES SHOW
June 2025	**Fixture reuse 2024** at internal plant PLANT-01: 28 fixtures. Twelve become the reuse sources. Those edits sit outside the 90-day window.
6 Jan–27 Feb 2026	Priya's 40 lines. 00001 on 6 January, 00017 on 20 January, 00040 on 27 February. The equipment list is the authority. Rule R1: one missing line is a no-go.
2–23 Feb 2026	**38 of 40 lines** have a tag. **00039** (Helix instrument, site A) and **00040** (Atlas fixture, site B) have no receipt and no asset. They stay in the denominator. 38/40 is 95 percent.
12 Feb 2026	Line **00017** arrives at site B. Jordan tags AST-TS-0017 and leaves the date blank: one of **twelve**. The other eleven are **00006–00016** at site A, left by Wei. Morgan owns the field. EX-PIS-00017 opens 12 March, age 4 days.
The same tag pass	**SN-DUP-4421** on 00020 and 00021 (Jordan; the second entry failed). **SN-DUP-4422** on 00022 and 00023 (Wei). Line **00018** is quantity 2: assets AST-TS-0018 and AST-TS-0019, a fan-out, not a duplicated PO id. Line **00019** is dated before receipt. Three real site codes are on the wrong factory: **00024** bought for A, tagged B; **00025** bought for B, tagged A; **00026** bought for A, tagged PLANT-01.
1 Mar 2026	Tolerance published before the load. R1: zero missing list lines. R2: a repeated PO id is a no-go; two assets on 00018 are allowed. R3: quarantine rate at or under 5 percent; a fully blank Excel row is not a submission. R4: alert if this program is under 98 percent. The file calls 98 percent a proposed lab alert. A blend can sit at 98 percent and stay quiet while this program at 95 percent should page. Pack cell expansion is clean on purpose, so a blended rate looks finished.
16 Mar 2026	You run the reconciliation. **21 open exceptions**, all go-live, no closed date. Aging **7 / 7 / 7** in 0–7, 8–30, and 31+ days. The orphan is 60 days. Line 00040 is 32. First entry passed on **21 of 40** assets.

WHAT YOU DO, IN THIS ORDER

1. **Publish the tolerance before the load.** R1–R4 are dated 1 March. You state them before you trust the file. You do not choose 98 percent after seeing 95.
2. **Load** both trackers onto the 40-line list. Quarantine the mystery-box row and the row whose cost is "abc." A fully blank Excel row is not a submission.
3. **Counted-rows go/no-go.** Loaded PO ids against the 40. A missing id stops the load. A repeated id stops the load. Line 00018's two assets stay: quantity 2, both serials kept.
4. **One exception per defect, named owner.** Twelve blank dates, Morgan. Duplicate serials: Jordan on 00021, Wei on 00022. Wrong sites: Wei, Jordan, Wei. Date-before-receipt: Morgan. Two in-transit lines and the orphan: Priya. Twenty-one rows. No second list in email.
5. **Owner-per-field scorecard.** Morgan sees the date. Priya sees the PO link, transit, and the orphan. Wei and Jordan see serial and site on the rows they submitted. Elena sees reuse source, on the 12 reuse lines only. David sees program allocation.
6. **Chase the twelve dates with Wei, Jordan, and Morgan. Do not escalate.** Show the ledger impact: the twelve are tagged, and Finance cannot capitalize them, so depreciation does not start. Agree the rule: the date is the day SAT is signed — not the ship date, and not the unpack date. Put line 00019 on the table as the early-date case. An unsigned SAT stays open under Morgan's name. You do not type a guess to turn the tile green.
7. **Close when the field is corrected.** Closed date is filled because the serial, the site code, or the SAT date is now in the record. On 16 March every closed date is empty. That is the before picture.
8. **Write the playbook rule: "placed-in-service date required at tagging, validated at entry."** The next tracker that omits it fails at entry. Filling it later leaves first-time-right at 21 of 40. That measure is the first entry, which is what the playbook is judged on.

5. What this role does, and the output you leave behind

Elena specs the station, David dates the launch, Priya places the buy, Wei and Jordan live with the tool, Morgan capitalizes it. You make the record trustworthy enough that Morgan can start depreciation, Priya can find the fixture next time, and an auditor can walk to the tag. You name an owner for every field, publish the tolerance before the load, count the rows, and close exceptions with the person who has the fact. The next program inherits the rule. The output on the table is a scorecard and a queue.

TILE	WHAT IT COUNTS ON THIS EXTRACT	THE REAL-WORLD FAILURE IT IS THERE TO CATCH
Reconciliation %	38/40 tagged, 95 percent. 00039 and 00040 stay in the denominator. Read it on TS-LINE-RAMP. The clean pack-cell program can hold a blend at 98 percent and the alert stays quiet.	Money committed, asset not in service. Paid for and not tagged. A company-wide percent hides the program you are onboarding.
Exception aging	7 / 7 / 7 in 0–7, 8–30, and 31+ days. EX-PIS-00017 is 4 days. The orphan is 60. Line 00040 is 32.	The date someone meant to fill after SAT, which became a quarter. A control finding gets old quietly.
Completeness by field, by owner	Twelve blank dates. Field owner: Morgan. Wei typed eleven blanks, Jordan typed 00017. Reuse source is scored on the 12 reuse lines only.	One field and one owner, instead of “the program is messy.” A single editor on program allocation is how shared tools get split by memory.
First-time-right %	21 of 40 assets passed on first entry. A later fix does not flip the flag.	Repeated serials, a date before receipt, a valid site code on the wrong factory, a blank date. The playbook is judged on the next file’s first pass.
Person-dependent fields	90-day window: date = Morgan, program = David, reuse source = Elena. Cost center is shared by Priya and Morgan.	The rule leaves with that person. The prior program still has a TPM on allocation who is not in today’s list.
Exceptions queue	21 rows, one defect, one owner, go-live, no closed date.	Both sheets can look right until a named person has to change one field. The queue is that ask, with an age.

How the lab data maps to this story

No README was in the lab folder. Columns are the CSV headers. Every defect below is a row in planted_defects.csv or the rule that file is demonstrating.

REAL-WORLD EVENT	CSV FILE / COLUMN	PLANTED DEFECT	METRIC THAT CATCHES IT
Approved equipment list	equipment_list_ts_line_ramp.csv · po_line_id, qty, unit_cost	40 lines. Sum of unit cost × quantity is $197,500. A missing line fails R1.	Counted-rows go/no-go
Buy and receipt	fact_po_line.csv · po_line_id, supplier_id, site_id, received_date	38 lines received between 2 Feb and 23 Feb. 00039 and 00040 blank.	Reconciliation %. In transit stays in the denominator.
Tag, and the in-service date	fact_asset.csv · asset_id, placed_in_service_date. fact_exception.csv · EX-PIS-*	12 blanks: 00006–00016 (Wei, site A) and 00017 / AST-TS-0017 (Jordan, site B). Owner OWN-FIN, Morgan Ellis.	Completeness by field by owner
Date before the tool arrived	placed_in_service_date against received_date	00019 / AST-TS-0020: 2026-02-11 before 2026-02-18. EX-DATE-00019.	First-time-right. Exception aging.
Serial typed twice	fact_asset.csv · serial_number. Raw: raw_cm_tracker_siteB.csv · SN	SN-DUP-4421 on 00020 and 00021 (Jordan). SN-DUP-4422 on 00022 and 00023 (Wei).	First-time-right. Duplicate-serial exceptions.
Site spelled three ways	raw_cm_tracker_siteB.csv · Location. Site A column is “Site ”.	Same factory written B, Site B, SITE-B, site b, and B plus a space. Line 00017 is stored as B.	Load conform, before any rate. A filter on “Site B” would drop line 00017.
Right code, wrong factory	fact_asset.site_id vs fact_po_line.site_id. Sites in dim_site.csv: CM-A, CM-B, PLANT-01.	00024 A bought, tagged B. 00025 B bought, tagged A. 00026 A bought, tagged PLANT-01.	First-time-right. Completeness would pass: the code exists.
One PO, two serialized assets	fact_po_line.qty · fact_asset rows for 00018	Quantity 2. AST-TS-0018 and AST-TS-0019. Rule R2: this is a fan-out, not a duplicate id.	Go/no-go must allow it. Reconciliation has to count assets, not assume one tag per line.
Asset with no PO	fact_asset.po_line_id (blank). EX-ORPHAN-0040.	AST-TS-0040, SN-TS-ORPHAN-9001, site B, in neither tracker. Owner Priya Raman.	Orphan on the queue. Reconciliation from the asset side, not only from the PO.

REAL-WORLD EVENT	CSV FILE / COLUMN	PLANTED DEFECT	METRIC THAT CATCHES IT
Reuse instead of a new buy	fact_asset.reuse_source_asset_id · lines 00027–00038 · field_owner_map.csv	12 rows, $0, PO-55030, sources AST-FR-0001–0012 from FIXTURE-REUSE-24. Elena owns the field.	Completeness of reuse source. Denominator is the reuse lines only.
Who actually typed the field	fact_field_edit.csv · field_name, edited_by. Window in lab_parameters.csv: 90 days, as-of 2026-03-16.	Single editors: Morgan, David, Elena, on the three fields above.	Person-dependent fields
The open queue	fact_exception.csv · owner_id, opened_date, closed_date, triage_class	21 open, all go-live, buckets 7 / 7 / 7. Owners in dim_input_owner.csv.	Exception aging. The queue itself.

6. How to say it to the hiring manager

ABOUT 90 SECONDS

The brand designs the product. Contract manufacturers build it. The test stations, handlers, fixtures, and instruments on those lines are often the brand's property, on the brand's books, in a factory the brand does not run. That is the asset problem: a large number of machines you own and do not physically control.

On the program I would walk you through, the manufacturing engineer specs 40 purchase-order lines across two factories and three suppliers. The supply manager reuses 12 fixtures from the prior program and points each one at the asset it came from. The factories tag what arrives in two spreadsheets that do not share column names. Thirty-eight lines have a tag. Two are still in transit. Twelve tags have no placed-in-service date, so Finance cannot start depreciation. Line 00017 at site B is one of those twelve.

I publish the pass rules before I load the file, count the rows against the 40-line list, and open one exception per defect with the owner that field already has. For the twelve dates I sit with Finance and the site coordinator. I show them that a blank date means the book cannot capitalize the asset. We agree the date is the day the site test is signed. Then I write the rule down: placed-in-service date required at tagging, validated at entry.

The sentence that shows why this team cares. Apple cares because these are assets the company owns inside factories it does not control, and the placed-in-service date is a financial control, not a formality.

ONE LINE EACH, FROM WORK I HAVE DONE

- **Sephora, Utah DC.** One inventory record, two channels; 44 requirement rows across 23 systems; issues log 331 go-live / 77 fast-follow / 36 enhancement.
- **Sephora, S/4 HANA.** The go/no-go was counted rows; the line we could stand on was "Negative AFS clean at T+1."
- **Elementum, Austria.** Data quality about 55% to about 99% within a quarter; measure before blaming the process.
- **Flexport.** Events in versus events out at every boundary.

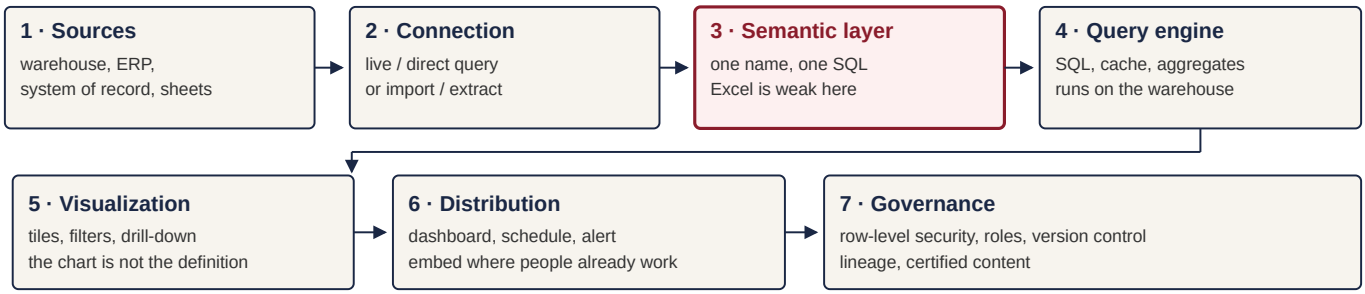
Counts are from the lab CSVs as of 2026-03-16. $197,500 is unit cost × quantity on the 40-line equipment list. 38/40 and 21/40 are TS-LINE-RAMP. No README was in the folder. Files: equipment list, fact_po_line, fact_asset, fact_exception, fact_field_edit, planted_defects, field_owner_map, published_tolerance, lab_parameters, the dim_* tables, and both raw trackers.

How a BI tool works, using Looker as the example

Call. CapEx Systems Onboarding Data Integrity Program Manager, with the Capital Expenditure Systems Lead. Thirty minutes. Tableau is a nice-to-have; Power BI or Tableau is preferred. Explain the tool. Do not pretend to be the BI engineer.
How to use it. Section 1 is the model. Section 2 is Looker, only as deep as a follow-up. Section 6 is what you say, including the honest proficiency line.

1. The seven layers of any BI tool

Looker, Tableau, Power BI, and a careful spreadsheet are the same stack. The chart is layer 5. Integrity is decided in the semantic layer and in governance, and it is lost in the connection when nobody knows how fresh the copy is.



LAYER	WHAT IT IS	WHERE INTEGRITY BREAKS
1 · Sources	The warehouse, the ERP, the system of record, and the spreadsheet a team still keeps.	A tile with no source system cannot be reconciled to a PO or an asset.
2 · Connection	**Live / direct query** asks the warehouse each time: fresh, as fast as the source, and the number is the source's number. **Import / extract** copies rows and refreshes on a schedule: fast, and only as fresh as the last refresh.	Trust follows the refresh. A missed refresh is a silent old truth. People believe the tile, not the timestamp.
3 · Semantic layer	One definition per metric. Joins and grain live here. A **dimension** is what you group or filter by (site, program, owner). A **measure** is the aggregation (count, sum, a ratio).	Two formulas with the same name. This is the layer a data-integrity program exists to own.
4 · Query engine	Turns the picked fields into SQL, checks a result cache, and may answer from a pre-aggregate instead of the raw rows.	A cache with no invalidation trigger still serves yesterday after the load has finished.
5 · Visualization	Tiles, filters, drill-down from the number to the rows that made it.	A filter default can hide the exceptions the scorecard is for.
6 · Distribution	The dashboard, a scheduled delivery, an alert when a tile crosses a threshold, an embed in the tool people already open.	Two emailed files diverge the moment someone edits a formula in one of them.
7 · Governance	Row-level security, roles, version control, lineage from the tile back to the column, and a certified mark on the content people should trust.	Without lineage you cannot answer why a field was filled in. That question is the job.

Excel is weak at the semantic layer. A formula in a cell binds nobody. Copy the workbook and the definition forks — five files, five meanings of reconciled — with no row-level security that survives a forward, no lineage back to the source column, and no certified version. Excel can hold rows during a migration. It cannot stop the next copy from changing the rule.

2. Looker specifically

> **What you are allowed to claim.** You can explain this. You have not shipped LookML. Section 6 has the sentence. Looker here means Looker on Google Cloud, the product with LookML. **Looker Studio** (formerly Data Studio) is a different, lighter product: reports on top of connectors, no LookML, no Git-backed semantic model. If they say “Looker,” ask which one in the first sentence, then talk about this one.

Looker does not store the business data. At query time it generates SQL and runs it on the warehouse you point it at — BigQuery, Snowflake, Redshift, and the usual others. What Looker itself keeps is the LookML, the saved content, users and permissions, and a result cache. A persistent derived table is still a table in the warehouse; Looker builds and refreshes it there. The book of record never moves into Looker.

LOOKML OBJECT	WHAT IT IS, IN THE ORDER YOU MEET IT
Project	A Git repository of LookML. This is the unit you branch, review, and merge.
Model	Which warehouse connection, which explores are exposed, and who may open them.
View	One table, or one derived table, plus its dimensions and measures. The definition of a column and of a metric lives here.
Explore	A starting view plus its joins. This is the self-serve surface: the field picker a GSM or a finance partner actually uses. They never see the SQL unless someone opens the SQL tab.

What the model contains

PIECE	THE SENTENCE YOU NEED
Dimension	An attribute. Site, program, supplier, a status. You group by it and filter on it. You do not sum it.
Measure	An aggregate: count, sum, count_distinct, or a ratio written sql: ${a} / NULLIF(${b}, 0). count_distinct on the primary key is the fan-out-safe habit. A raw count counts joined rows.
Dimension group	One timestamp becomes a date, a week, and a month (type: time, with timeframes). Aging buckets are built on that date, not retyped in every tile.
Join	Declared on the explore, with a cardinality: one-to-one, many-to-one, one-to-many, many-to-many. Looker can rewrite a sum so a one-to-many join does not double-count, but only if the cardinality is declared correctly. A ratio you write by hand is not protected for you. Wrong cardinality, wrong total.
Derived table	A view defined by SQL you write, or by a native derived table (Looker builds the SQL from an explore). Use it when the grain you need is not a raw table — one row per line per field, for completeness.
PDT	A persistent derived table: the same idea, materialized in the warehouse and rebuilt on a trigger (a datagroup, often “rebuild when max(loaded_at) changes”). It is a pre-aggregate for speed. It is still warehouse data. A PDT that does not rebuild is a stale extract with a better name.
Aggregate awareness	You declare rollup tables. Looker uses the smallest rollup that can answer the question, and falls back to the base table when you ask for a field the rollup does not have. That is how a weekly total stays fast and a drill to the PO line still works.
Caching	Identical SQL can be served from the result cache. Tie the cache to the same datagroup as the load. No trigger means a Monday tile can contradict a query run a minute later.

What a person touches, and what makes the definition auditable

An **Explore** is the self-serve surface. A **Look** is one saved query. A **dashboard** is a layout of tiles; each tile is a query, not a picture. Filters narrow a tile. drill_fields are the columns you land on when you click a number. A link opens the same record in the system of record — the PO, the asset — so the tile is not a dead end. A scheduled delivery sends the dashboard on a clock (Monday 07:00). An alert fires when a tile crosses a threshold, which a schedule does not do. A Looker Action sends rows onward (a ticket, a webhook) when a person or a schedule asks it to.

Row-level security is a **user attribute** plus an access_filter. The attribute (the program ids this GSM or this CM may see) is set on the user. The filter is injected into the SQL and the user cannot remove it. Content sits in

folders; roles (viewer, explorer, developer, admin, or a custom role) decide who may look, explore, or edit LookML. None of that is a substitute for warehouse permissions. It is the application of them.

LookML in Git is the audit trail. A developer branches, Looker validates the LookML, a pull request is reviewed, and production is the merge. The commit is who changed the definition of reconciled, when, and from what. A Look saved in a personal folder is not that. The metric people steer by has to be the merged one.

```
view: po_lines {
  sql_table_name: capex.po_lines ;;
  dimension: po_line_id {
    primary_key: yes
    type: string
    sql: ${TABLE}.po_line_id ;;
  }
  dimension: site_id {
    type: string
    sql: ${TABLE}.site_id ;;
    description: "Receiving site. Blank means the input owner has not set it."
  }
  dimension: received_date {
    type: date
    sql: ${TABLE}.received_date ;;
  }
  measure: po_line_count {
    type: count_distinct
    sql: ${po_line_id} ;;
    description: "Lines received. Distinct so an asset join cannot double-count."
  }
  measure: null_site_count {
    type: count_distinct
    sql: ${po_line_id} ;;
    filters: [site_id: "NULL"]
    description: "Lines with no site. A completeness exception, not a hidden filter."
  }
  measure: reconciliation_rate {
    type: number
    sql: 1.0 * ${assets.tagged_asset_count} / NULLIF(${po_line_count}, 0) ;;
    value_format_name: percent_1
    description: "Assets tagged / PO lines received. One definition of reconciled."
  }
}

view: assets {
  sql_table_name: capex.assets ;;
  dimension: asset_id { primary_key: yes  type: string  sql: ${TABLE}.asset_id ;; }
  dimension: po_line_id { type: string  sql: ${TABLE}.po_line_id ;; }
  measure: tagged_asset_count {
    type: count_distinct
    sql: ${asset_id} ;;
  }
}

explore: po_lines {
  join: assets {
    type: left_outer
    relationship: one_to_many
    sql_on: ${po_lines.po_line_id} = ${assets.po_line_id} ;;
  }
}
```

Teaching snippet, not a model you have run. Three dimensions on po_lines (po_line_id, site_id, received_date), three measures (po_line_count, null_site_count, reconciliation_rate), one join. The assets view exists only so the ratio has a real numerator. Cardinality is one PO line to many assets; both counts are count_distinct.

3. How one query flows, end to end

1. The user opens the Explore and picks fields — program, site, reconciliation rate — and a filter.
2. Looker reads the LookML and builds the SQL: the columns, the join, the aggregates, and any access filter.
3. It checks the result cache. A hit returns the stored rows and does not touch the warehouse.
4. A miss runs the SQL on the warehouse (or on a PDT, which is still a warehouse table).
5. The warehouse returns the aggregated rows.
6. Looker renders the tile.
7. The user drills. Drill fields return the row-level detail behind the number, still under the same access filter.
8. A link opens that record in the system of record — the PO, the asset — where the input owner can fix the field.

WHERE IT GOES WRONG	WHAT THE INTEGRITY PROGRAM DOES ABOUT IT
Stale extract	A Friday refresh, read on Monday, misses a weekend receipt. The rate looks finished. Require a visible “loaded at,” and treat a cache with no datagroup trigger as the same defect.
Two measures, one name	Both called reconciliation rate. One divides tagged assets by lines received; the other divides by lines ordered. The semantic layer allows one of them in production.
A join that fans out	One line, two assets. A plain count counts the line twice, the denominator inflates, and the rate looks healthier than the book. Declare one-to-many and count distinct on the line id.
A filter default nobody sees	The explore opens on status = Received. Open and in-transit lines never appear, so the scorecard is green because the exceptions were filtered out. Defaults are visible, or they are a bug.
A field with no description	Placed-in-service date with a blank description. Nobody knows whether it means acceptance or the day depreciation starts. The description is the “why” behind the input, written where the field is picked.

4. Power BI, Tableau, and Looker on one table

Same seven layers. The words change. If they say Tableau, stay on this table and do not switch into LookML unless they ask how a definition is audited. If they say Power BI, same table. The integrity idea does not change: one definition, stored where a second copy cannot silently fork it.

	POWER BI	TABLEAU	LOOKER
Where the rows are at query time	Import: copied into VertiPaq, in memory. DirectQuery: still in the source; Power BI sends a query.	Live: still in the database. Extract: copied into a Hyper file and refreshed on a schedule.	Still in the warehouse. A PDT is a table Looker maintains there. The cache holds results, not the book of record.
Modeling language	Power Query (M) shapes the tables. DAX defines the measures.	VizQL draws the sheet. Calculated fields and level-of-detail expressions define the numbers.	LookML. The explore is the only surface a user queries.
Where the metric definition is stored	In the semantic model, the published dataset. A local file can drift from it.	In the workbook, unless someone published a data source and pointed the workbook at it.	In LookML, in Git. A Look stores a query. It does not store the definition.
Row-level security	Roles in the model. A DAX filter decides which rows a role sees.	User filters, or a row-level policy on the published data source.	A user attribute plus access_filter, written into the SQL the user cannot remove.
Versioning	The dataset people query lives in a workspace. Deployment pipelines promote it. Git is optional, not the default audit trail.	Revision history on the server. Certification is a badge a steward applies, not a pull request.	Branch, LookML validation, pull request, merge. The merge is who changed “reconciled,” and when.

	POWER BI	TABLEAU	LOOKER
Best for	A Microsoft shop, a fast in-memory model, apps handed to a large set of viewers.	Visual analysis on a published data source someone has certified, so ten workbooks do not each invent the ratio.	One audited definition, queried live, when the failure mode is five spreadsheets and five formulas.

A calculated field pasted into ten Tableau workbooks is the same fork as five spreadsheets. Publishing the data source and certifying it is how Tableau tries to stop that. LookML makes the stop structural: production is the merge.

5. Worked example — the CapEx data-integrity scorecard in Looker

> **Illustrative, end to end.** This is the scorecard from the product idea, drawn as a Looker explore so you can walk it. It is not Apple's tool, and none of the counts below are a program you ran. Every figure on this page is **ILLUSTRATIVE**.

Tables behind the explore: **programs**, **purchase_orders**, **po_lines**, **assets**, **suppliers**, **sites**, **input_owners**. The explore starts at po_lines, one row per received line. It joins **assets** one-to-many (a line can tag more than one asset — the fan-out), **programs** and **sites** many-to-one, and **input_owners** many-to-one on the line's owner. **purchase_orders** and **suppliers** are many-to-one parents, joined only when a tile needs the PO number or the supplier name. They are not on the fan-out path. Reconciliation measures are count_distinct on the line id and on the asset id. A plain count would count a line once per asset and make the rate look healthier than the book.

Field completeness is a different grain: one row per line per field (placed-in-service date, site, cost center, program allocation, reuse source asset), because each field has its own owner. That derived table is one-to-many off the line. Its measures count distinct lines *inside one field*. Joining it into the reconciliation tile would multiply the denominator. Two tiles, two grains, one explore only if the joins are the safe cardinality.

CapEx data-integrity scorecard · scheduled Monday 07:00 · row-level security on program

EVERY NUMBER IN THIS WIREFRAME IS ILLUSTRATIVE

94.2%
PO ↔ asset reconciliation
113 assets tagged / 120 lines received. Assumes one asset on each tagged line.

5 · 3 · 1
Exception aging, open
0–7 days · 8–30 · 31+. Nine open. The 31+ row is a close-out, not a color.

81%
First-time-right
17 of 21 required fields passed on first submit. Two newly onboarded programs.

3
Person-dependent fields
Edited by one user only in the window. The knowledge leaves with that person.

FIELD	INPUT OWNER	FILLED	NULL	COMPLETE	FLAG
Placed-in-service date	Finance	108	12	90%	One editor — person-dependent
Site	Contract manufacturer	116	4	97%	
Cost center	GSM	111	9	93%	
Program allocation	TPM	102	18	85%	One editor — person-dependent
Reuse source asset	AME	40 of 48 reuse lines	8	83%	One editor — person-dependent

ILLUSTRATIVE throughout, including the 98% alert line below. The picture assumes one tagged asset per tagged line, so "assets tagged / lines received" equals "lines with an asset / lines received." If a line can carry two assets, certify the line version. The asset version then reads high and hides untagged lines. Say that before anyone builds the tile. 113 / 120 = 94.2%; 17 / 21 = 81%; 40 / 48 = 83%.

```
explore: po_lines {
  access_filter: {
    field: po_lines.program_id
    user_attribute: allowed_program_ids
  }
  join: assets {
    type: left_outer
    relationship: one_to_many
    sql_on: ${po_lines.po_line_id} = ${assets.po_line_id} ;;
  }
  join: programs {
    relationship: many_to_one
    sql_on: ${po_lines.program_id} = ${programs.program_id} ;;
  }
  join: sites {
    relationship: many_to_one
    sql_on: ${po_lines.site_id} = ${sites.site_id} ;;
  }
  join: input_owners {
    relationship: many_to_one
    sql_on: ${po_lines.field_owner_id} = ${input_owners.owner_id} ;;
  }
}
```

Each GSM and each CM gets a user attribute, allowed_program_ids. The access filter is in the SQL, and they cannot clear it. A scheduled delivery goes out Monday at 07:00 to the input owners on the completeness tile — the people who can fill the null, not a general list. An alert, separate from the schedule, fires when reconciliation falls below the team's threshold. Ninety-eight percent in the sketch is illustrative; you do not invent the real line. The description on placed-in-service date says whether the field means acceptance or the day depreciation starts. That is the why behind the input, sitting where the field is picked.

What changes in behavior. A null site is the contract manufacturer's row, not a shared inbox. An exception has an age, and the one past thirty-one days is a close-out conversation with an owner. "Reconciled" is the merged measure. A side spreadsheet that divides tagged assets by lines ordered is a different metric, and it cannot replace the tile without a pull request.

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6. How to say it in the call

SIXTY SECONDS — HOW A BI TOOL WORKS

A BI tool is seven layers, and the chart is only the fifth. The rows stay in the warehouse, the ERP, or the system of record. The connection is either live, meaning the tool asks the source every time, or an extract, a copy that is only as fresh as its last refresh. The layer that protects data integrity is the semantic layer: one name, one formula, the join and the grain written down, so reconciled means one thing on every tile. The query engine turns that into SQL, checks a cache, and sometimes uses a pre-built aggregate. People see a tile, filter it, and drill to the rows. Distribution is the dashboard, the Monday email, and an alert. Governance decides who sees which programs, and whether that definition is versioned. Excel can hold the rows. It does not hold a definition. The next copy of the file is a second meaning of reconciled.

WHAT THE SOURCES ACTUALLY SUPPORT — READ THIS BEFORE YOU IMPROVISE

/tmp/sources was not on disk when this primer was written. There was no LSA workbook, no BRD text, and no WBR deep-dive text to quote. What follows is what the numbers canon records about those files (/workspace/amazon/numbers-reconciliation.md, rows S11.24, S16, S20), not a fresh read of the files. Do not upgrade it.

LSA_WBR_2022, Instructions rows 6–9. A weekly-review format: input to output, a five-to-six-week window, red or green when the week-over-week move is more than 5% relative, week running Monday to Sunday. The canon calls it

a format copied from the LSA review. It is an operating-review convention in a spreadsheet. It is not a Looker model and not a Tableau workbook.

BRD for Zendesk data in Snowflake, pages 1–5. You wrote the requirements. The as-is was manual pulls into spreadsheets, an hour and a half to two hours for one week-over-week snapshot of seven metrics. The document names Looker SQL Runner as the reporting surface and Snowflake as the warehouse, and it prices the manual work. The infrastructure team built the pipeline. You froze the seven definitions and the acceptance gate — a parallel run within 1% for consecutive weeks — and you called the cutover. You did not write LookML. The BRD names no table or column.

WBR deep dive (the Tech Ops prep book the canon tags S20). Your own interview-prep notes. The sample weeks are marked illustrative. Not a dashboard you shipped. Not evidence of Tableau, Power BI, or Looker.

SQL, narrower than "I built the warehouse." The interview workbook's Austria routing-guide story says you pulled three years of shipment history in SQL yourself, to show true data quality near 55% rather than the 40% that had been reported. That is a pull against a source. It is not a semantic layer. The companion packet's reading of the resume is the same shape: Tableau and Power BI do not appear in the mapped experience bullets.

THE PROFICIENCY SENTENCE — SAY THIS, THEN STOP

I would pick up Looker or Tableau quickly; I have learned three enterprise systems to go-to level.

If they ask what you have actually done. "I built the metrics behind the weekly review in spreadsheets, and I wrote the definitions and the parity gate for the Snowflake pipeline. Looker SQL Runner was the surface the requirements named. I did not build the Looker layer." If they ask which three systems, do not count Looker, Tableau, or Power BI. Name systems you can walk: the warehouse system you took through cutover, and SAP. Offer a third only if you can open it from memory. This primer does not freeze that third name.

THE LINE THAT LANDS WITH A DATA-INTEGRITY TEAM

The semantic layer is where you stop five spreadsheets from each having their own definition of reconciled.

Ten terms, if they use one

TERM	ONE SENTENCE
Semantic layer	The place a metric is defined once — name, SQL, grain, join — so every tile uses that definition.
Dimension	Something you group or filter by: site, program, owner, a date. You do not sum it.
Measure	An aggregate: a count, a sum, a count distinct, or a ratio of two of those.
Explore	The self-serve surface in Looker: one starting view, its joins, and the field picker.
LookML	Looker's modeling language. It lives in Git. Production is a merge, which is what makes a definition auditable.
PDT	A persistent derived table: a pre-aggregate Looker builds and refreshes in the warehouse, not inside Looker.
Fan-out	A one-to-many join repeats the left-hand row. A plain count then double-counts. Count distinct on the key, or declare the cardinality and mean it.
Row-level security	A filter the tool applies for you, from who you are. In Looker that is a user attribute plus an access filter. The user cannot take it off.
Extract vs live	An extract is a scheduled copy, fast and possibly stale. Live, or direct query, asks the source at query time and is only as fast as that source.
Certified content	The dataset or dashboard a steward has marked as the one to trust. In Tableau that is a badge on a published source; in Power BI, an endorsement. In Looker the equivalent is LookML merged to production, not a Look in a personal folder. Looker Studio is a different product — do not invent a Looker badge.

APPLE CAPEX · 30-MINUTE CALL · PART 4 · LOOKER END-TO-END FLOW · PRACTICE PROGRAM TS-LINE-RAMP

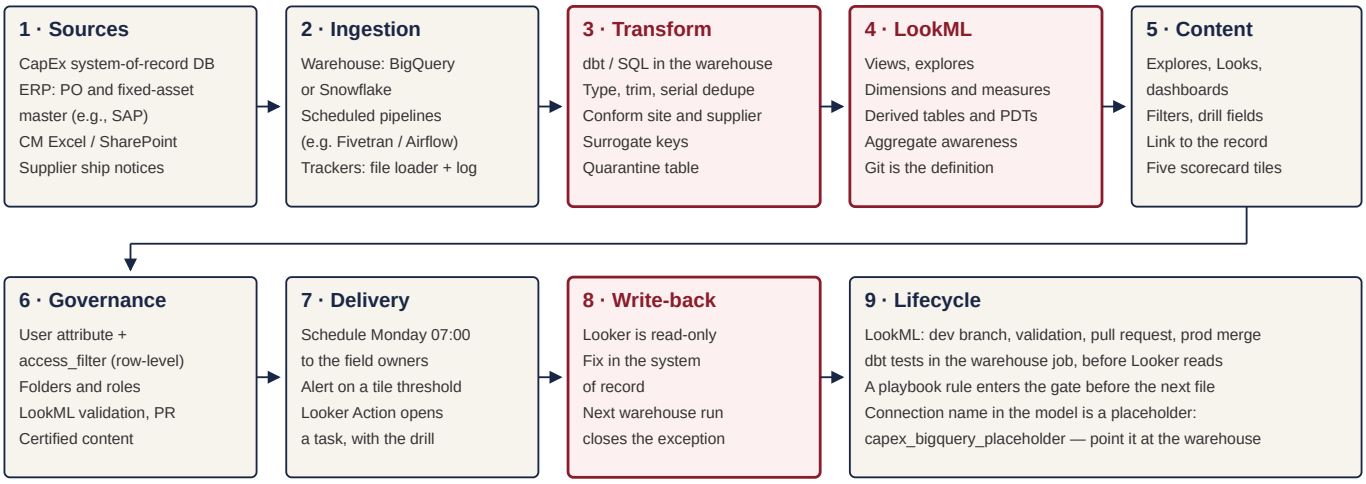
From the tracker to the closed exception, in Looker

> **Who this is for.** You, walking the Capital Expenditure Systems Lead through one technical flow. The tool is **Looker on Google Cloud** (LookML in Git). Looker Studio is a different product. This packet does not use a Microsoft BI tool.
> **Already written, do not re-teach.** Part 3 is the seven layers, LookML object names, and the Power BI / Tableau / Looker comparison. This part is the path for one program: sources, the warehouse, the lab's LookML, one PO line, the load gate, and what you say.
> **Banner.** The flow is my model of how a company like this would run CapEx data integrity — public knowledge, the job description, and the interviewer's profile. It is not inside knowledge of Apple's proprietary system. In the room: "Here is how I would wire it. Correct me where your stack differs." Every count below is the lab's answer key unless a line says `record` or `record-from-docs`.

> **The practice program, exact.** **TS-LINE-RAMP**, test-station line ramp, as-of **16 March 2026**. Forty PO lines. Sites **CM-A**, **CM-B**, and internal **PLANT-01**. Suppliers Atlas Fixture Co, Northline Test Systems, Helix Instruments. Owners: Elena Vasquez (AME), David Cho (TPM), Priya Raman (GSM), Wei Chen (site A), Jordan Hale (site B), Morgan Ellis (Finance). The row you walk is **PO line 00017**, asset **AST-TS-0017**, Northline, PO-55022, CM-B, received 12 February 2026, placed-in-service blank. LookML files: capex_integrity.model.lkml, fact_po_line.view.lkml, fact_asset.view.lkml, fact_exception.view.lkml, fact_field_edit.view.lkml, the five dim_*.view.lkml files, and capex_integrity_scorecard.dashboard.lookml. `lab`

1. Architecture — nine stages, left to right

Looker does not hold the book of record. The warehouse does. Looker stores LookML, users, saved content, and a result cache, and it generates SQL at query time. A persistent derived table is still a table in the warehouse. The fix never lands in Looker.



Read the top row left to right, then the bottom row left to right. Red boxes are where a data-integrity PM owns the definition: the tests, the measure, and the rule that the tool must not write the fix. Aggregate awareness sits in stage 4 so a program total can use a rollup and a drill to the PO line still hits the base table. The lab model does not declare a rollup; the five tiles each query the explore whose grain matches the question.

2. What happens at each stage

Owner means the person accountable in this model. You, the data-integrity PM, do not type the asset row and you do not merge LookML you have not reviewed. The pipeline clock in this flow is **06:00**. The scorecard delivery is **Monday 07:00**, after that run. model schedule The as-of date the lab freezes is 16 March 2026. lab

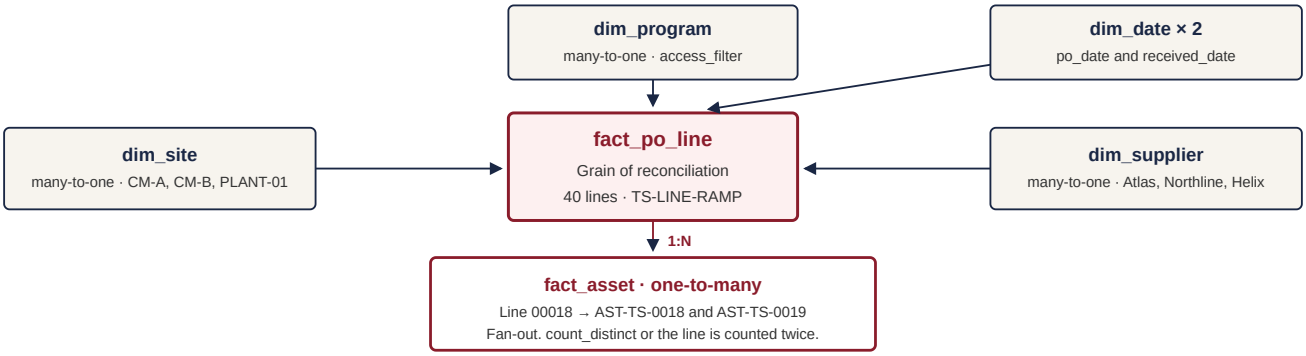
STAGE	WHAT HAPPENS TECHNICALLY	OWNER	WHAT GOES WRONG	CONTROL
1 · Sources	Four feeds. The CapEx system-of-record database. The ERP for the PO and the fixed-asset master (e.g., SAP). Wei's and Jordan's Excel or SharePoint trackers, with different headers. Supplier ship notices, which this extract does not contain as a file; the received date is already on the PO line.	Field owners: GSM, Finance, CM coordinators, supplier. Systems PM owns the system of record.	Two trackers, two header languages. A serial lives in an inbox and is typed twice. The floor moves a tool over a weekend and the book keeps the old site.	One system of record after onboarding. The tracker is a source, not a second book. Ship notices land as rows, not as email.
2 · Ingestion	Scheduled pipelines (for example Fivetran or Airflow) into BigQuery or Snowflake. ERP and the system of record are table syncs. Trackers go through a file loader that writes a load log: file name, row count, arrival time, hash. A fully blank Excel artifact is not a submission. The lab's connection string is capex_bigquery_placeholder.	Data platform runs the job. You own the load log and the go / no-go call.	A silent miss: the file arrived and the log does not. Monday's tile then describes Friday. A second copy of the same file double-counts if the loader is not idempotent on PO line id.	The 06:00 job fails closed when the log row is missing. Idempotent load on po_line_id. The log is what you count before you say go.
3 · Transformation	dbt or SQL in the warehouse, the checks in sql/02_quality_checks.sql. Trim values. Map site tokens (A, SITE A, CM A) to CM-A, CM-B, PLANT-01. Parse three date formats. Conform supplier codes to SUP-ATLAS, SUP-NORTH, SUP-HELIX. Surrogate keys on the dimensions. Rows that fail the structural gate go to a quarantine table and do not become facts.	Analytics engineer builds the models. You publish the tests and the tolerance before the file arrives.	A code that exists can still be the wrong site. A second asset on one PO line is a fan-out, not a duplicate id. Blank placed-in-service is not in today's structural gate, so twelve blanks load.	Quarantine reasons are written, not dropped. Referential tests against dim_program, dim_site, dim_supplier. Serial uniqueness is its own test. Section 5 is the counted-rows gate.

STAGE	WHAT HAPPENS TECHNICALLY	OWNER	WHAT GOES WRONG	CONTROL
4 · LookML	The Git project. capex_integrity.model.lkml sets the connection, the datagroup capex_daily, and five explores. Views declare dimensions, measures, and descriptions. exception_aging and field_completeness are derived tables. The aging one is a PDT, rebuilt on the datagroup. Aggregate awareness is how a later rollup would answer a program total and fall back to fact_po_line when the drill asks for the line id. This model does not declare that rollup yet.	A LookML developer branches. You own the certified measure and the description — the why behind the input.	A hand-written ratio is not protected from fan-out. Two measures with one name. A description left blank, so nobody knows that placed-in-service is the day depreciation starts.	One certified reconciliation_rate. The unsafe counts are hidden: yes. Production is the merge. LookML validation runs before the pull request can merge.
5 · Explores, Looks, dashboards	Each tile is a query. Reconciliation uses explore po_lines. Aging uses exceptions. First-time-right uses assets, so the orphan stays in the denominator. Completeness uses completeness. Person-dependent fields use field_edits. Filters are Program (default TS-LINE-RAMP, visible) and Site. drill_fields return the rows. A link on po_line_id and asset_id opens the system of record.	Content steward builds the dashboard from the merged explores. You refuse a tile whose grain is wrong.	A hidden “received only” filter turns 38/40 into 38/38. Joining completeness onto the PO explore multiplies the denominator. The site filter means the PO site on one tile and the asset site on another.	Default program filter is published, not hidden. No default on received date. Dashboard notes in capex_integrity_scorecard.dashboard.lookml say which site the filter is.
6 · Governance	access_filter on dim_program.program_id bound to user attribute program_ids. The user cannot remove it. Priya: all three programs. Jordan: TS-LINE-RAMP only. Morgan: TS-LINE-RAMP and PACK-CELL-EXP. Folders and roles (viewer, explorer, developer) decide who may open a Look or edit LookML. Certified content is the LookML merged to production, not a Look in a personal folder.	Looker admin sets the user attribute from user_program_map. You specify which programs a CM may see.	An attribute that is empty shows nothing, or, if the filter is omitted, shows every program. A personal Look copies a formula the merge no longer uses. Warehouse grants are still required; the access filter is not a substitute.	The filter is in the generated SQL. Jordan’s test is on the PO explore, where the other programs exist. LookML validation plus PR review is the change control.

STAGE	WHAT HAPPENS TECHNICALLY	OWNER	WHAT GOES WRONG	CONTROL
7 · Delivery	A schedule sends the dashboard Monday at 07:00 to the owners named on the completeness tile. An alert is a different object: it fires when reconciliation_rate is below 0.98. That 0.98 is lab rule R4, a proposed line, not a threshold this team has adopted. A Looker Action sends the drilled row onward — a task — with the link to the record.	You name the recipients and the threshold. You do not invent 98% as Apple's number.	A schedule with no alert looks fine while the rate sits at 95%. An alert on the wrong measure pages Finance for an in-transit line they do not own. The cache can still be Friday's if nothing invalidates it.	Schedule and alert stay separate. R4 is labelled proposed. The datagroup trigger plus max_cache_age: "24 hours" bounds staleness. Say the 98% is the lab's line.
8 · Action and write-back	Looker is read-only. The Action opens a task to the field owner with the drill link. Morgan enters the placed-in-service date in the system of record. The next 06:00 run reads that row, sets closed_date on EX-PIS-00017, and the PDT drops it from the open queue.	The field owner writes the system of record. You own the queue until the next run shows the close.	Someone "fixes" the tile, or a side file, and the book is unchanged. A correction that does not move a load watermark leaves MAX(created_at) untouched, so the datagroup does not rebuild and the tile still shows 12 missing until the 24-hour cap.	The link goes to the system of record, not to a spreadsheet. The warehouse job writes a load watermark the datagroup watches. You close the task only when the next run shows the exception gone.
9 · Lifecycle	LookML: a dev branch, Looker validation, pull request, prod merge. dbt tests run in the warehouse job before Looker is allowed to read the new tables. The playbook rule — blank placed-in-service with a received date is quarantined — is added to the gate before the next tracker, in playbook/RULE-pis-required-at-tagging.md. It is not in force on the load you just scored.	Developer and data platform ship. You review the PR for the definition, and you publish the new gate before the file arrives.	A rule edited after the file arrives is not a gate. A prod explore pointed at a dev schema. A merge that changes the denominator without a note on the tile.	Dev and prod connections. dbt tests are red or green in the job log. The rule's owner (Finance) writes the definition; you put it in the gate and in the tracker instructions.

3. Data model and the lab's LookML

Reconciliation is counted at the **PO line**. An asset with no line, and a line with two assets, are both real, and neither is allowed to change the denominator by accident. The explores below are the ones in capex_integrity.model.lkml. They are not a new model.



The star is the po_lines explore. Program, site, and supplier are many-to-one and are not on the fan-out path. dim_input_owner is not joined here: the owner is per field, and putting the owner on this explore would sit on the asset join. Three other explores stay off this star so they cannot multiply the 40: exceptions (the aging PDT), completeness (one row per asset per field), and assets (first-time-right, so orphan AST-TS-0040 stays in the denominator). field_edits is a fourth, one row per program per field, and it does not listen to the site filter.

Fan-out and the symmetric aggregate. Line 00018 has quantity 2 and two assets. A plain count on the joined explore is 41 for the program. That measure is in the view as plain_row_count_do_not_certify with hidden: yes. Dividing assets by lines is 39/40 = 97.5%, because the second asset on 00018 hides one of the two in-transit lines (00039 and 00040). That measure is hidden too. The certified rate counts distinct lines on both sides: lines with at least one asset, over lines. Looker can protect a count_distinct of the primary key from the fan-out. A ratio you write by hand is not protected, so both the numerator and the denominator are count_distinct. There is no default filter on received date. Hiding nulls would turn 38/40 into 38/38. From the PO explore the orphan disappears and first-time-right becomes 21/39. The assets explore keeps 21/40. lab

```
# fact_po_line.view.lkml — both sides count_distinct. TS-LINE-RAMP is 38/40 = 95.0%.
measure: lines_with_an_asset {
  type: count_distinct
  sql: ${po_line_id} ;;
  filters: [fact_asset.asset_id: "-NULL"]
}
measure: reconciliation_rate {
  type: number
  sql: 1.0 * ${lines_with_an_asset} / NULLIF(${po_line_count}, 0) ;;
  value_format_name: percent_1
  description: "Certified definition: distinct PO lines with at least one asset, divided by distinct PO
lines. TS-LINE-RAMP is 38/40 = 95.0%. Not assets divided by lines."
}
```

```
# capex_integrity.model.lkml — same block on po_lines, exceptions, completeness, field_edits, assets.
# Priya: TS-LINE-RAMP, PACK-CELL-EXP, FIXTURE-REUSE-24. Jordan: TS-LINE-RAMP only.
access_filter: {
  field: dim_program.program_id
  user_attribute: program_ids
}
join: fact_asset {
  type: left_outer
  relationship: one_to_many
  sql_on: ${fact_po_line.po_line_id} = ${fact_asset.po_line_id} ;;
}
```

The aging PDT. Age is not a column on fact_exception. exception_aging in fact_exception.view.lkml is a derived table Looker persists in the warehouse. It keeps closed_date null, subtracts opened_date from the lab as-of (16 March 2026, not the wall clock), and buckets the day count. Open counts before the fix are 7 / 7 / 7. Closing EX-PIS-00017 removes one row from 0–7 on the next rebuild, so the counts become 6 / 7 / 7. The datagroup is sql_trigger: SELECT MAX(created_at) FROM capex.fact_asset and max_cache_age: "24 hours". A correction of an existing asset does not by itself move created_at. The warehouse job has to touch a watermark that trigger reads, or the tile waits for the 24-hour cap.

```
# exception_aging PDT — open rows only. Age 7 is still 0-7. Age 8 is 8-30.
DATE_DIFF(p.as_of_date, e.opened_date, DAY) AS age_days,
CASE
  WHEN DATE_DIFF(p.as_of_date, e.opened_date, DAY) <= 7 THEN '0-7'
  WHEN DATE_DIFF(p.as_of_date, e.opened_date, DAY) <= 30 THEN '8-30'
  ELSE '31+'
END AS aging_bucket
... WHERE e.closed_date IS NULL
```

The description is the why. Every input the tile can pick has a description in the view. That sentence is what a GSM reads in the field picker. These are the ones that decide the scorecard, quoted from the LookML and from field_owner_map.csv.

FIELD	OWNER	DESCRIPTION — THE WHY
po_line_id	GSM, for the link	"The PO line. Reconciliation counts this, not the asset and not the tracker row. Line 00017 is the record the lab closes. Line 00018 is the fan-out."
received_date	CM coordinator	"Null means in transit. A default filter that hides nulls drops lines 00039 and 00040 and makes reconciliation 38/38."
placed_in_service_date	Morgan Ellis, Finance	"The day depreciation starts. Not the ship date and not the day the CM unpacked the crate. Null means Finance has not capitalized the asset. Twelve assets on TS-LINE-RAMP are null. A date before received_date is a different defect, on line 00019."
site_id on the asset	Wei or Jordan	"Where the asset physically is. A code in the site list can still be the wrong site." Completeness of site is 22/22 and 18/18. Three mismatches still sit on the queue. Complete is not valid.
reuse_source_asset_id	Elena Vasquez, AME	"Required only when the PO line is a reused fixture. Twelve of twelve are filled. Dividing 12 by all 40 assets is the wrong denominator."
passed_validation_on_first_entry	The gate, not the editor	"1 if the first submit passed every check that existed that day. A later correction does not flip this. PO line 00017 stays 0 after Finance enters the date."

4. One record, end to end — PO line 00017

Jordan Hale tags **AST-TS-0017**, serial SN-TS-0017, at CM-B on 13 February 2026. The PO line was received 12 February. Placed-in-service is blank. Cost center CC-4100 and program TS-LINE-RAMP are filled. The asset is linked, so this row is already inside the 38. The defect is the date, which Finance owns. Exception **EX-PIS-00017** opened 12 March, age 4 days on the as-of date, bucket 0–7. lab

1. **06:00 warehouse run.** The loader writes the load log. dbt builds fact_po_line, fact_asset, and fact_exception. The structural gate does not yet quarantine a blank placed-in-service date, so the row lands and the validation flags it. EX-PIS-00017 is open. passed_validation_on_first_entry is 0.
2. **Looker live query.** Cache miss, or a hit only if the datagroup still matches this load. Explore assets, program TS-LINE-RAMP: **missing placed-in-service = 12**, completeness **28/40 = 70.0%**. Explore po_lines: reconciliation **stays 38/40 = 95.0%**. A blank date does not remove the asset link. The two in-transit lines are the gap, and they were already the gap before this flag.
3. **Alert.** Rule R4 fires because 95.0% is below 98%. R4 is the lab's proposed alert, published in published_tolerance.csv on 1 March 2026, not a company threshold. It pages on the reconciliation tile. It does not by itself tell Morgan which date is blank. PACK-CELL-EXP and FIXTURE-REUSE-24 are each 100% on purpose; an unfiltered blend would look finished and stay quiet. The program filter is why this page happens.
4. **Looker Action.** From the missing-date drill — asset id, PO line, site, who keyed it — a Looker Action opens a task to Morgan Ellis with the link on asset_id (fact_asset.view.lkml: "Open the asset in the system of record"). Looker does not write the date.

5. **Fix in the system of record.** Finance enters placed-in-service **17 February 2026**, which is after received 12 February, so the date-logic check passes. The next 06:00 run sets closed_date on EX-PIS-00017 to 16 March 2026.
6. **What the next query shows.** Missing dates **11**. Completeness **29/40 = 72.5%**. Aging **6 / 7 / 7**. Reconciliation **still 38/40 = 95.0%**, so R4 still fires. First-time-right **still 21/40 = 52.5%**. The flag is not updated when someone fills a blank later. lab, after_fix

> **Playbook rule, and why first-time-right does not recover.** playbook/RULE-pis-required-at-tagging.md: a tracker row with a received date and a blank placed-in-service date is quarantined and does not load. Finance writes that definition. You put it in the structural gate and in the tracker instructions before the next file arrives. A rule edited after the file arrives is not a gate. Entering the date closes EX-PIS-00017. It does not flip passed_validation_on_first_entry. First-time-right stays 21 of 40. Say that before anyone treats the close as proof the onboarding got cleaner.

CapEx data-integrity scorecard · capex_integrity_scorecard.dashboard.lookml · Monday 07:00 · program filter TS-LINE-RAMP, visible · as-of 16 March 2026

FIVE TILES, FIVE EXPLORES. NUMBERS ARE THE OPEN STATE, BEFORE FINANCE FILLS LINE 00017. LAB, NOT A PROGRAM YOU RAN.

95.0%	7 · 7 · 7	52.5%	12	3
Reconciliation po_lines. 38/40. Alert on, below 98%.	Exception aging exceptions PDT. After the close: 6 · 7 · 7.	First-time-right assets. 21/40, orphan in. Stays 52.5%.	Missing placed-in-service completeness. 28/40 = 70%. After: 11, 72.5%.	Person-dependent field_edits. Morgan, David, Elena. No site filter.

FIELD	OWNER	APPLIES TO	FILLED	MISSING	NOTE
Placed-in-service date	Morgan Ellis, Finance	40 assets	28	**12**	Becomes 29 / 11 when 00017 closes. R4 does not clear.
Cost center	Priya Raman, GSM	40	40	0	Complete. Complete is not the same word as valid.
Program allocation	David Cho, TPM	40	40	0	One editor in 90 days. Person-dependent.
Reuse source asset	Elena Vasquez, AME	12 reused fixtures	12	0	Denominator is 12, not 40. One editor.
Site	Wei 22, Jordan 18	40	40	0	Filled, and three site mismatches remain on the queue.

Drill on 95.0% lands on po_line_id, site, received date, asset id, serial. Drill on a bucket lands on the exception row, including the detail sentence. The site filter is the PO receiving site on reconciliation and on exceptions, and the asset site on the asset and completeness tiles. Person-dependent does not listen to it: a site split would flag a field the program as a whole does not.

5. Excel-to-system onboarding — the gate before the scorecard

The file can be a **GO** and the book can still be wrong. Rules R1–R3 were published on **1 March 2026**, before the trackers are treated as arrived (data/published_tolerance.csv). Looker does not run them. sql/02_quality_checks.sql does, in the warehouse, and the answer key is data/expected_outputs/go_nogo.csv. The same counts hold after the line 00017 fix, because that fix is not a reload. lab

RULE	PUBLISHED BAR	THIS LOAD
R1	Missing equipment-list lines = 0. The approved list is 40. One missing line is a no-go.	**0 missing. Pass. 40 distinct lines accepted.**
R2	Duplicate PO line ids in the load = 0. Two asset rows on one id are a fan-out, not a second id.	**0 duplicate ids. Pass. One id (00018) has two asset rows.**
R3	Quarantined rows / submitted tracker rows ≤ 0.05. A fully blank Excel artifact is not a submission.	**2 / 43 = 4.65%. Pass.**

RULE	PUBLISHED BAR	THIS LOAD
R4	Alert when the certified reconciliation rate is below 98%. Proposed lab line, not a load gate and not a company threshold.	95.0% fires. Stays firing after 00017 is filled.

Submitted 43, quarantined 2, accepted 41, distinct PO lines 40, decision **GO**. The two quarantined rows: site A line **00901**, site token SITE-Z — unknown site and not on the equipment list; site B, blank PO line, date **13/32/2026** — blank key and a date that will not parse. What the gate did not catch, because the rule was not in force: twelve blank placed-in-service dates, two duplicate serials (SN-DUP-4421, SN-DUP-4422), three site mismatches, line 00019 with a date before receipt, and orphan AST-TS-0040, which is not in either tracker. Those are post-load exceptions. A go on the file is the count against the published bar. It is not a clean scorecard.

The dbt tests that implement the file are named in that SQL: trim and drop fully blank artifacts; canon_site to CM-A, CM-B, PLANT-01; parse_lab_date for three formats; quarantine reasons written, not dropped; referential checks to program, site, supplier, and reuse-source asset; duplicate serials; site on the asset versus site on the PO; placed-in-service on or after received. The playbook rule is the test you add before the next tracker: a received date with a blank placed-in-service date does not load.

LIVED EVIDENCE — ONLY THESE FOUR. DO NOT BORROW A WAREHOUSE-CUTOVER STORY.

Sephora, S/4 HANA. record The go/no-go was a count of rows. Drift is ISM's copy of S/4 available minus MB52 at one watermark; a non-zero row is a severity, tolerance zero. The exit bar I say out loud is "Negative AFS clean at T+1." OMS reads the ISM available-for-sale and never computes its own. S/4 keeps the physical book. I owned the definitions, the tolerance, and the go/no-go recommendation; the engineers built the extracts. Any unit count in a one-SKU walk is illustrative. Do not quote it. Zero customer-facing oversell through both go-lives is the record.

Sephora, UDC dual inventory. record-from-docs 44 requirement rows across 23 systems. The issues log's own categories: 331 go-live, 77 fast-follow, 36 enhancement. Two go-lives, zero customer-facing incidents. The test on a row was whether the customer's promise becomes false without it.

Elementum, Austria. record Routing-guide data quality was about 55% true — the report had said 40% — and reached about 99% within a quarter. I pulled three years of history in SQL myself. Finance signed the savings method before any figure was claimed. The scorecard was released only after automated on-time matched a manual sample.

Flexport. record Reading application logs, I found no id tying a load event to the screen, and I specified counting events in versus events out at each system boundary. On the Zendesk-to-Snowflake pipeline I wrote seven metric definitions and specified the Looker SQL Runner queries. The infrastructure team built the pipeline. Cutover was a parallel run within 1% for consecutive weeks. I did not write LookML. The weekly-review metrics I actually ran were in Excel and in SQL.

6. How to say it

SIXTY SECONDS — WALK THE FLOW, THEN STOP

TS-LINE-RAMP is forty PO lines leaving two CM trackers for one system of record. At six a.m. dbt lands them, trims and types, conforms site codes, and quarantines against a tolerance published before the files arrived. Looker stores LookML, not the rows. Reconciliation is distinct lines with an asset over distinct lines: 38 over 40, 95 percent. Line 00018 has two assets, so a raw count would lie. Line 00017 is tagged with a blank placed-in-service date: twelve missing, completeness 70 percent, reconciliation still 95. Under the lab's 98 percent line the alert fires, and a Looker Action opens Finance a task with the drill link. Looker cannot write the date. The next run shows eleven missing and aging six, seven, seven. First-time-right stays 21 of 40. The playbook quarantines a blank date on the next file.

PROFICIENCY — SAY THIS, THEN STOP

I wrote seven metric definitions and specified the Looker SQL Runner queries for the Zendesk-to-Snowflake pipeline. The infrastructure team built it. The weekly-review metrics I ran were in Excel and in SQL. I have not built a dashboard hands-on in Looker, Tableau, or Power BI.

Ask, if they use a BI tool

Is the stack Looker with LookML in Git, or Looker Studio — and where does the certified definition of reconciled live today?

Ask, about freshness

Is the warehouse BigQuery or Snowflake, and what invalidates the Monday cache: a datagroup on the load watermark, or a clock?

Ask, about change control

Who may change a measure, and is that a LookML pull request with validation, or a Look saved in a personal folder?

TERM	ONE SENTENCE, ON THIS FLOW
Grain	The row the metric counts. Reconciliation counts PO lines. Completeness counts one asset and one field.
Symmetric aggregate	Both sides count_distinct on the line id. Line 00018 counts once: 38/40, not 39/40.
Fan-out	A one-to-many join repeats the left row. Line 00018 and two assets make a plain count 41.
access_filter	Row-level security. User attribute program_ids is in the SQL. Jordan cannot take it off.
PDT	A derived table Looker builds in the warehouse. Aging buckets live there, open rows only.
Datagroup	capex_daily rebuilds on MAX(created_at) and never serves a result older than 24 hours.
Aggregate awareness	The smallest rollup that can answer; a drill for the line falls back to fact_po_line. None is declared yet.
Drill fields	The columns behind a click. The link opens that PO or asset in the system of record.
Looker Action	Sends the row out as a task. It does not update the book. The next run closes EX-PIS-00017.
First-time-right	21/40 = 52.5%, orphan included. A later correction does not flip the flag.

LOOKER LAB · CAPEX SYSTEMS ONBOARDING · DATA INTEGRITY PROGRAM MANAGER

Work one program from the tracker to the scorecard

Role. You make the inputs accurate when a program moves off Excel trackers onto the CapEx system of record, and you do the change with GSM, TPM, AME, the contract manufacturers, and Finance without escalating it.
Tool. Looker on Google Cloud, the product with LookML. Excel is the must-have. This lab is how you explain the scorecard, not a claim that you have shipped LookML.
Time. Path A is about 45 minutes on this machine. Path B is a real Looker or Looker Studio session and takes longer.

Illustrative, and labeled that way. Not Apple's system and not a program you ran. People, suppliers, and sites are lab labels. Every rate below is from `python3 run_local.py` against these files. The lived numbers you are allowed to put next to it are named where they appear: a counted-rows go/no-go and "Negative AFS clean at T+1"; 44 requirement rows across 23 systems; an issues log of 331 go-live, 77 fast-follow, 36 enhancement; Austria data quality about 55% to about 99% within a quarter; Shipment Visibility events-in versus events-out; a Zendesk-to-Snowflake requirements document where you wrote seven metric definitions and named Looker SQL Runner. You did not write the Looker layer on that pipeline.

Part 0 — What we are doing and why

A test-station line, `TS-LINE-RAMP`, is still living in two contract-manufacturer trackers. The team's job is to get that program onto the system of record with inputs someone can defend: the PO line matches an asset, the placed-in-service date is the day depreciation starts, the site code is the site the tool is actually at, a reused fixture points at the asset it came from, and each of those fields has a named owner. The scorecard is how you see that without opening the tracker. The queue is how you close it without a side file.

Two other programs are already loaded and clean, on purpose. `PACK-CELL-EXP` has 32 lines. `FIXTURE-REUSE-24` has 28, including the twelve source assets the ramp reuses. If you forget the program filter, reconciliation is 98 of 100, which is 98.0%. The ramp alone is 38 of 40, which is 95.0%. The blend looks finished. The program you are onboarding is not.

What the lab proves. You can read a messy tracker, publish a tolerance before you load, quarantine what fails a structural rule, and still find the business defects the structural rules did not cover. You can define "reconciled" once, show three other formulas that print a healthier number, and refuse them. You can put an owner on each field, age the open defects, and show that fixing one record closes the exception and does not repair first-time-right.

The final output is the data-integrity scorecard for `TS-LINE-RAMP`, as of 2026-03-16, plus the exception list behind it.

TILE	RESULT FROM THE SQL RUN
Reconciliation	38 / 40 = 95.0%. Alert fires. The proposed line is under 98%.
Exception aging, open	7 in 0–7 days, 7 in 8–30, 7 in 31+. Twenty-one open, all go-live.
Field completeness by owner	Placed-in-service 28 / 40 = 70%, owner Finance. Reuse 12 / 12. Site 100% complete and not 100% valid.
First-time-right	21 / 40 = 52.5%.
Person-dependent fields	3. Placed-in-service (Morgan Ellis), program allocation (David Cho), reuse source (Elena Vasquez).

That output is this role. The accuracy of the inputs is the job. The owner on the field is how you ask for a correction without escalating. The playbook rule at the end is how the next tracker does not repeat the miss. A chart with no owner is a status meeting. This scorecard is a close-out list.

WHAT YOU SAY, IF THEY ASK WHAT YOU ARE ABOUT TO SHOW

I would walk one program off the tracker and onto a scorecard where reconciled means one thing, every blank has an owner, and a late fix does not pretend the first entry was right.

Part 1 — Two paths, honestly

Path A — this machine, about 45 minutes

You do not need Looker, a cloud account, or a card. DuckDB runs the same checks the answer key was built from. The dashboard you can hold up is `dashboard_mock.html`, drawn from those query results. It is a static page, not a Looker session. Say that in the first sentence if you show it.

```
cd /workspace/apple/looker-lab
python3 generate_data.py
python3 run_local.py
```

`generate_data.py` rewrites the CSVs from a fixed seed. You can skip it; the files are already there. `run_local.py` loads them, runs `sql/02` and `sql/03`, writes `data/expected_outputs/`, applies the step-9 fix into `data/expected_outputs/after_fix/`, and leaves `/tmp/lookerlab-scratch/lab.duckdb` on the open queue so a query matches the answer key. Open `dashboard_mock.html` in a browser. Diff any CSV you compute against `expected_outputs/`.

If DuckDB is missing: `pip install duckdb`. The SQL that creates the tables also exists as BigQuery in `sql/00_create_tables.sql`. The file you execute locally is `sql/00_create_tables_duckdb.sql`.

Path B — a real Looker, or Looker Studio if that is what you have

1. **BigQuery sandbox.** Google's sandbox is free and does not ask for a card. It has storage and query caps this lab will not touch. Confirm the current cap on Google's page before you depend on it. Create a dataset named `capex`.
2. **Load the CSVs** with the column types in `sql/00_create_tables.sql`. `po_line_id` is STRING. If the loader guesses a number, `00017` becomes `17` and every join fails. Do not load the raw trackers into the fact tables. They are the file the gate reads. `sql/01_load.sql` has the `bq load` shape.
3. **Connect Looker.** Looker, the product with LookML, needs a Google Cloud Looker instance or a trial. There is no free Looker core that simply attaches to the sandbox. **Looker Studio is a different product.** It is free, it can sit on these BigQuery tables, and it can draw the same five tiles. It does not run the LookML in `lookml/`. If they say "Looker," ask which one, then stay on the one they named.
4. **If you have Looker:** new project, connection name whatever your instance uses (the model file says `capex_bigquery_placeholder`), paste the LookML, validate, and explore. Set user attribute `program_ids` from `user_program_map.csv`. Jordan's value is `TS-LINE-RAMP` only.
5. **If you have Looker Studio:** build the five tiles as five charts on the BigQuery tables, with the SQL from `sql/03_metrics.sql` as the definition you refuse to fork. You will not have an access filter or a Git-backed measure. Say that. The grain and the tolerance are still yours.
6. The checks in `sql/02` and `sql/03` are DuckDB, because that is what was executed. On BigQuery, `date_diff('day', opened, as_of)` becomes `DATE_DIFF(as_of, opened, DAY)`, and the CSV reader becomes an external table. The counts do not change.

> **WHAT YOU SAY ABOUT THE TOOL**
>
> I have not shipped LookML. On the Snowflake pipeline I wrote seven metric definitions and named Looker SQL Runner as the surface. The infrastructure team built the pipeline. I can specify the grain, the owner, and the tolerance, and I would want your Looker developer on the model until I have one in production.

Part 2 — Step by step

1. Inspect the raw trackers and list the defects by eye

WHAT. Open `data/raw_cm_tracker_siteA.csv` and `data/raw_cm_tracker_siteB.csv` in a text editor, not by saving them out of Excel. Excel will turn `00017` into `17`.

WHY IT MATTERS. The must-have skill is Excel, and the failure mode of Excel is a silent type change. You list what a person can see before you trust a loader. You will not see all of it. That is why the checks exist.

What is visible without SQL:

- Two header languages. Site A's header has trailing spaces: `PO Line ` , `Site ` , `Qty ` , `Unit Cost ` , `Notes ` . Site B uses `po_line`, `PO#`, `Location`, `date received`.
- Three date shapes: `2026-01-21`, `01/08/2026`, `12-Feb-2026`. One impossible date: `13/32/2026`.
- Site text: `A`, `Site A`, `SITE-A`, `site a`, `Site B`, `SITE-B`, `B`. Line `00024` sits in the site A file and says `Site B`. Line `00026` says `PLANT-01`. Line `00025` says `A` in the site B file.
- Line `00018` appears twice, quantity 2, two serials.
- `SN-DUP-4421` on lines `00020` and `00021`. `SN-DUP-4422` on `00022` and `00023`.
- Blank in-service cells. You may not count all twelve by eye. The SQL says twelve.
- Line `00019`: in-service `11-Feb-2026`, received `2026-02-18`.
- Lines `00039` and `00040`: no asset tag, no received date.
- Row `00901` with site `SITE-Z`. A row with a blank PO line. A fully blank row at the bottom of the site B file.
- Trailing space on serial `SN-TS-0005 ` and asset tag `AST-TS-0005 ` .

What you will not see: asset `AST-TS-0040`. It was keyed into the system of record and is on neither tracker.

EXPECTED. A written list in those words. The blank Excel row is not a defect and not a submission.

> **WHAT YOU SAY**
> I read the file before I load it, and I write down what I can see, because the loader will hide the rest inside a green row count.

2. Load

WHAT. Run `sql/00_create_tables_duckdb.sql` and `sql/01_load_duckdb.sql`, or just `python3 run_local.py`. The fact tables are the system of record. The equipment list is the authority for which PO lines exist. The trackers are not copied in as facts. Values are trimmed. Empty strings become null. Ids stay strings.

WHY IT MATTERS. A load that "succeeds" with the wrong type is a bad load. The counted-rows read after load is 100 PO lines, 100 assets, 21 exceptions, and 40 lines on `TS-LINE-RAMP`.

EXPECTED. `SELECT COUNT(*) FROM fact_po_line` → 100. `WHERE program_id = 'TS-LINE-RAMP'` → 40. `fact_asset` → 100. `fact_exception` → 21. `SELECT po_line_id FROM fact_po_line WHERE po_line_id = '00017'` returns the five-character id, not 17.

> **WHAT YOU SAY**
> I count the rows after the load against the count I published before it, and I check that the line id is still text.

3. Quality checks, and the counted-rows go / no-go

WHAT. Read `data/published_tolerance.csv` first. The rules are dated 2026-03-01. Then run `sql/02_quality_checks.sql`. The tolerance is not edited after you see the file.

WHY IT MATTERS. On the Sephora S/4 HANA work, the go / no-go was a counted-rows read, and the bar I still use the morning after a load is the one we called "Negative AFS clean at T+1": the business rules, not the file shape, have to be clean the next day. Here the structural gate can pass and the T+1 rules can still be red. Both results belong in the room.

CHECK	VALUE	BAR, PUBLISHED 2026-03-01	RESULT
Submitted tracker rows	43		
Quarantined	2		
Quarantine rate	2 / 43 = 4.65%	R3 ≤ 5%	Pass
Accepted rows	41		40 lines plus the second asset row on 00018
Distinct PO lines loaded	40	Equipment list 40	Pass
Missing list lines	0	R1 = 0	Pass
Duplicate PO line ids in the load	0	R2 = 0	Pass. The fan-out is not a duplicate id.
Decision	GO		The file may load. The book is not clean.

Quarantine detail, from `quarantine_rows.csv`: site A row `00901`, reasons `unknown_site` and `not_on_equipment_list` (the raw site is `SITE-Z`). Site B row with a blank PO line, reasons `blank_po_line` and `bad_po_date` (the raw date is `13/32/2026`). The blank artifact row is ignored. `tracker_vs_fact.csv` is 0 received-date mismatches and 0 linked-asset mismatches. The orphan is the one asset the tracker does not have.

What the structural gate did not stop, and the T+1 checks did:

CHECK	ROWS
Duplicate serials	2 serials, each on 2 assets. SN-DUP-4421 on AST-TS-0021 and AST-TS-0022. SN-DUP-4422 on AST-TS-0023 and AST-TS-0024.
Orphan	1. AST-TS-0040, no PO line, created by Jordan, site CM-B.
Site mismatch	3. 00024 PO CM-A / asset CM-B. 00025 PO CM-B / asset CM-A. 00026 PO CM-A / asset PLANT-01.
Placed-in-service before receipt	1. Line 00019, received 2026-02-18, date 2026-02-11, seven days early.
Referential, other	0 bad programs, sites, suppliers, owners, or reuse sources. The only referential failure is the orphan.

EXPECTED. Decision **GO** on the file, and a queue that is not empty. A gate you tighten after you see the quarantine rate is not a gate.

> **WHAT YOU SAY**
>
> I publish the tolerance before the file arrives. This file is a go on counted rows, and it is not clean: the rules I had not written yet are the queue.

4. Define the metrics once

WHAT. Read the view files in `lookml/` next to `sql/03_metrics.sql`. A **dimension** is something you group or filter by: program, site, owner, a date. You do not sum it. A **measure** is an aggregate: a count distinct, or a ratio of two of those. The **primary key** on `fact_po_line` is `po_line_id`. `count_distinct` of that key is a symmetric aggregate: Looker will not let the one-to-many asset join multiply it. `type: count` counts joined rows. A ratio you type by hand is not protected.

WHY IT MATTERS. Two spreadsheets with two formulas is how “reconciled” forks. On Shipment Visibility, the useful watch was events-in versus events-out, both sides, not a blended status. Same shape here: lines in, assets out. The requirements I wrote for Zendesk data in Snowflake froze seven metric definitions and named Looker SQL Runner. The acceptance gate on that pipeline was a parallel run within 1% for consecutive weeks. I did not write the LookML. This lab’s bar is the counted-rows tolerance, a different file. The habit is the same: one definition, written down before the tile.

Line `00018` is quantity 2, assets `AST-TS-0018` and `AST-TS-0019`. It is not a defect. It is the fan-out.

ON TS-LINE-RAMP	RESULT
Joined rows after the asset join	41

ON TS-LINE-RAMP	RESULT
Plain count of po_line_id	41. The line with two assets is counted twice.
Count distinct po_line_id	40
Count distinct asset_id on the join	39. The orphan is not on this join.
Line 00018 alone	plain count 2, distinct lines 1, assets 2

DEFINITION	RATE	CERTIFY?
Lines with at least one asset / lines	38 / 40 = 95.0%	Yes. The two in-transit lines stay in the denominator.
Assets on a line / lines	39 / 40 = 97.5%	No. The extra asset on 00018 hides one missing line.
Lines with an asset / lines that have a received date	38 / 38 = 100%	No. A default filter dropped 00039 and 00040.
Assets allocated to the program / lines	40 / 40 = 100%	No. The orphan fills the hole and is not a reconciliation.
First-time-right, all assets on the program	21 / 40 = 52.5%	Yes.
First-time-right, dropping the orphan	21 / 39 = 53.85%	No. The PO explore cannot see an asset with no line.
Reuse source filled / every asset	12 / 40 = 30%	No. The field applies to 12 reuse lines, and 12 of 12 are filled.

The descriptions on the LookML fields are the “why” behind the input. Placed-in-service says it is the day depreciation starts, not the ship date and not the unpack date. A field with a blank description is how two people capitalize on two different days and both think they followed the rule.

EXPECTED. You can point at `reconciliation_rate` in `fact_po_line.view.lkml` and say the other three rates are hidden measures, present so you can show them and then not ship them.

> **WHAT YOU SAY**
> Count distinct on the PO line is the fix for the fan-out. A plain count on this join says 41 lines. There are 40.

5. Build the explore and run reconciliation

WHAT. In Looker, open explore `po_lines`. The base is the PO line. Assets join `left_outer`, relationship `one_to_many`. Program, site, and supplier join `many_to_one`. There is no default filter on received date. Pick program, the reconciliation rate, and filter program to `TS-LINE-RAMP`. On Path A, the same query is `metric_reconciliation_by_program` and `metric_scorecard`.

WHY IT MATTERS. The explore is the only surface a user queries. A Look someone saved in a personal folder is not the definition. The definition is the merged view.

EXPECTED. One row. 40 lines, 38 with an asset, rate 0.95. By program, the other two are 32/32 and 28/28, each 1.0. All programs together, which is what Priya sees with the filter off: 98 / 100 = 98.0%.

> **WHAT YOU SAY**
> I start the explore at the PO line so a line with no asset is still in the denominator. If I start at the asset, the two in-transit lines disappear.

6. Build the five tiles

WHAT. The dashboard file is `lookml/capex_integrity_scorecard.dashboard.lookml`. On Path A, open `dashboard_mock.html`. Five tiles, filters for program and site, drill to the exception list. Default program is `TS-LINE-RAMP`, and that default is visible. It does not hide the in-transit lines, because those lines are inside the program. The site filter is the PO receiving site on the reconciliation tile and the exception tile. On the asset tile and the completeness tile it is the asset site. Say that before you turn it. Person-dependent does not listen to site: Wei alone writes site A, and splitting by site would call the field person-dependent when the program has two editors.

WHY IT MATTERS. Each tile is a query. The owner on the completeness tile is who you ask, which is the change-management part of the job. The same habit, at a larger scale, was 44 requirement rows across 23 systems, each

row pre-wired to an owner before the room saw it. The queue uses the same three classes as that issues log — 331 go-live, 77 fast-follow, 36 enhancement. Those are category counts, not a split of some other total. Every one of these 21 rows is go-live, because the book is wrong without the fix. A nicer chart would be an enhancement, and this queue has none.

TILE	NUMBER
Reconciliation	95.0% (38/40). Alert fires, because 95% is under the proposed 98%.
Exception aging	7, 7, and 7. Age 7 is still 0–7. Age 8 is 8–30. EX-PIS-00017 is age 4.
Completeness	Finance, placed-in-service, 28 filled, 12 missing, 70%. GSM cost center 40/40. TPM program allocation 40/40. AME reuse 12/12. Wei 22/22 sites. Jordan 18/18 sites. Eleven of the twelve blank dates were submitted from site A. The demo row is the one on site B.
First-time-right	52.5% (21/40). The 19 misses are the 12 blank dates, 2 duplicate serials, 3 site mismatches, 1 date-order failure, and 1 orphan.
Person-dependent	3 fields, one editor each, window 2025-12-16 through 2026-03-16.

Click the aging number and you land on `exception_open.csv`: exception id, PO line, asset, field, owner, age, and the sentence. That is the drill. A tile that does not drill is a poster.

EXPECTED. The five numbers above, and you can name the owner of the blank date without opening the tracker. It is Morgan Ellis, Finance, even though Wei or Jordan keyed the asset.

> **WHAT YOU SAY**
> Complete is not valid. Site is 100 percent filled, and three of those codes disagree with the PO. Those three are on the queue, not in the null count.

7. Row-level security, tested as the CM user

WHAT. The model sets `access_filter` on `dim_program.program_id` using user attribute `program_ids`. An admin loads the attribute from `user_program_map.csv`. The user cannot clear it. On Path A, `metric_rls_by_user` is that filter written as SQL.

WHY IT MATTERS. A contract manufacturer does not get the other programs by clearing a dashboard filter. Content folders are not a substitute. The filter has to be in the SQL.

USER	PROGRAMS	LINES VISIBLE	LINES WITH AN ASSET	RATE
Jordan Hale, CM site B	1	40	38	95.0%
Morgan Ellis, Finance	2 (ramp and pack, not the prior program)	72	70	97.22%
Priya Raman, GSM	3	100	98	98.0%

Jordan asking for `PACK-CELL-EXP` returns **0** lines. That is `rls_jordan_pack.csv`. His security is on program, not on site, so he can still see site A's rows of this program. Say that. Site-level restriction is a follow-on, and it is not in this model.

EXPECTED. Jordan sees 95%, not 98%. He cannot produce Priya's blend, and he should not be able to.

> **WHAT YOU SAY**
> I test the access filter as the CM user, and I try to open a program he does not have. Zero rows is the pass. A dashboard filter he can clear is not security.

8. Schedule and alert

WHAT. Two different mechanisms. A schedule sends the dashboard Monday at 07:00 to the owners on the completeness tile — Morgan for the blank dates, not a general list. An alert fires when the certified reconciliation

rate is strictly below 0.98. That 0.98 is rule R4, proposed for this lab and dated before the tile. It is not a number I am attributing to this team. Replace it with theirs when they have one.

WHY IT MATTERS. A schedule does not notice that the rate crossed a line. An alert does not explain the rows. You want both, and you want the datagroup: the cache rebuilds when `max(created_at)` on the asset table moves, and it is never older than 24 hours. A Friday cache read on Monday is how a weekend receipt goes missing. Looker keeps the result cache. The rows stay in the warehouse.

EXPECTED. On `TS-LINE-RAMP`, 0.95 is below 0.98, so `alert_fires` is true. On Priya's unfiltered 0.98, strictly-below is false, so the blend stays quiet. That is the bug you are preventing with the program filter. Morgan's unfiltered book is 0.9722 and would still fire, because she cannot see the prior program that makes Priya's blend land on the line.

> **WHAT YOU SAY**
>
> The Monday email goes to the person who can fill the null. The alert is a separate object, and I do not let an unfiltered blend sit on the line and stay quiet.

9. Fix one record

WHAT. In the system of record, not in the tracker and not on the tile, set the placed-in-service date on the asset for PO line `00017` (`AST-TS-0017`) to `2026-02-17`. That is five days after received `2026-02-12`, so you do not create a new date-order defect. Set `closed_date` on `EX-PIS-00017` to `2026-03-16`. Insert one field edit by `usr_morgan_fin`. Do not update `passed_validation_on_first_entry`. The SQL is `sql/04_rls_and_fix.sql`. Path A already ran it into `expected_outputs/after_fix/`. The CSVs in `data/` are still the open queue. Re-run `00` and `01` if you applied the update and want the open queue back.

WHY IT MATTERS. The write path leaves Looker. The owner fixes the system of record. The next refresh closes the row. If Jordan had typed the date, first-time-right would still not recover, and placed-in-service would pick up a second editor and stop being person-dependent. The fix path is the field owner.

	BEFORE	AFTER
Missing placed-in-service	12	11
Completeness of that field	28/40 = 70%	29/40 = 72.5%
Aging, open	7 / 7 / 7	6 / 7 / 7
Reconciliation	95.0%	95.0%
First-time-right	21/40 = 52.5%	21/40 = 52.5%. The flag on AST-TS-0017 is still 0.
Person-dependent fields	3	3. Morgan is still the only editor of the date.
Alert	fires	fires
EX-PIS-00017	open, age 4	closed 2026-03-16, gone from the open bucket

`after_fix/tracker_vs_fact.csv` shows one linked-asset mismatch. The tracker still has a blank, and the system of record does not. You do not reload the tracker over the correction.

EXPECTED. The table above, from the two scorecard CSVs, not from memory.

> **WHAT YOU SAY**
>
> Closing the exception is the refresh. First-time-right does not move, and I do not want a measure that lets it move.

10. The playbook rule that prevents it next time

WHAT. Write the rule, then compare it to `playbook/RULE-pis-required-at-tagging.md`. A tracker row with a received date and a blank placed-in-service date is quarantined. It does not load. The date cannot be earlier than receipt. Finance owns the definition. The CM coordinator sees the rule in the instructions before the file is due, not in a note after the load.

WHY IT MATTERS. The scorecard found a miss the gate was not asked to catch. Austria was the same shape of work at a different company: data quality about 55% to about 99% within a quarter, by making the rule explicit and staying with it. This lab's first-time-right is 52.5%. I am not saying this file is that program. I am saying the rate moves when the rule is in the gate, and it does not move when someone backfills a cell. On the next tracker, those twelve blanks would be part of the quarantine count. If that pushed the rate over 5%, the decision would be no-go, and that would be the gate working.

EXPECTED. One short rule with an owner, a reject condition, and a sentence that a late fix does not restore first-time-right.

> **WHAT YOU SAY**
>
> The playbook entry is the reject, written down before the next file. The scorecard is how I see it if the rule is not in the gate yet.

Part 3 — Explaining it to the hiring manager

About three minutes. You are walking a model of the work, and you say so. You have not shipped this LookML.

> **THE NARRATION**
>
> A program is still in two trackers. My job is the accuracy of the inputs when it moves onto the system of record, and the change with the GSM, the TPM, the AME, the two CM coordinators, and Finance.
>
> I publish the tolerance before the file shows up. Missing lines against the equipment list: zero. Duplicate line ids: zero. Quarantine under 5 percent. This file is 2 quarantined out of 43, and 40 lines loaded. That is a go. The morning after, I still recount the business rules. Two serials are duplicated, one asset has no PO line, three sites disagree with the PO, one depreciation date is before receipt, and twelve assets have no placed-in-service date. The file gate was not asked to catch those. They are the queue, and each one has an owner.
>
> The model starts at the PO line. Assets join one-to-many, because one line can carry two units. I count distinct line ids. A plain count on this data says 41. There are 40. The certified rate is lines with an asset over lines, 38 over 40, 95 percent. Assets over lines prints 97.5, and a filter that keeps only received lines prints 100. I do not certify either. The definition lives in the view, in Git. A personal report does not get its own formula.
>
> The scorecard is five queries. Reconciliation 95 percent, so the alert fires under the line I proposed at 98, which I would replace with the team's number. Aging is 7, 7, and 7. Completeness by owner: Finance is at 70 percent on the date that starts depreciation, and site is 100 percent filled with three codes that are still wrong. First-time-right is 21 of 40. Three fields were edited by exactly one person in the last 90 days.
>
> I fix one record in the system of record, PO line 00017, the blank date. The exception closes. Missing dates go from 12 to 11. Reconciliation stays 95. First-time-right stays 52.5, because the flag is about the first submit. Then I write the rule that quarantines a blank date on the next file, so the scorecard is not where we notice it the second time.
>
> Row-level security is a user attribute the CM cannot clear. Jordan sees this program only. He sees 95 percent, and a program he does not own returns no rows.

Five follow-ups

Why is the PO line the grain? Because the failure you are managing is a buy with no asset, and an asset-grain table cannot show a row that was never tagged. Two assets on one line are still one line. The orphan is a different check, on the asset, and it does not get to fill the reconciliation hole.

Why Looker rather than Excel for this? Excel holds the rows. It does not hold a definition. The next copy of the file is a second meaning of reconciled, and it will turn `00017` into a number if someone saves it. Looker, if that is the tool they have, keeps the measure in Git and queries the warehouse. Looker Studio can draw the tiles and will not give you that Git history. I would not pretend those are the same product. I have not shipped either model. I have frozen definitions and a parity gate, with SQL Runner named as the surface.

How do you stop two definitions of reconciled? One measure in the view, count distinct on both sides, reviewed like code. The asset version and the received-only version stay in the repo as hidden measures so someone can see the trap, and they are not on the tile. The Monday schedule and the alert both point at the certified measure. A side file that divides assets by lines ordered is a different metric, and it does not replace the tile without a change to the view.

What would you not automate yet? I would not let a model fill placed-in-service, the reuse source, or the program allocation. Those are judgments, and this program's first-time-right is 52.5 percent. I would not train anything on a queue that still has duplicate serials and an orphan. I would automate the gate that is already a rule: trim, parse the date, map the site, reject a blank key, reject an unknown site, reject a line that is not on the equipment list. Capitalization and reuse stay with Finance and the AME until the scorecard says the input is trustworthy.

How does this map to onboarding playbooks? The pilot is this program. The playbook entry is the reject we added after the scorecard found the blank date. The next program inherits the rule, the owner map, the tolerance,

and the same five tiles. What stays local is the equipment list and the CM's site codes, which still have to be mapped. I would ask, on the pilot that became their playbook, which lesson teams actually followed and which one they had to keep enforcing.

Part 4 — Glossary and the checklist

TERM	IN THIS LAB
Dimension	Program, site, owner, a date. Group and filter. Do not sum.
Measure	count_distinct, or a ratio of two. The reconciliation rate is a measure.
Primary key	po_line_id on the line, asset_id on the asset, exception_id on the defect.
Fan-out	Line 00018, one line, two assets. Plain count becomes 2. Count distinct stays 1.
Symmetric aggregate	count_distinct of the primary key. Looker will not let the one-to-many join multiply it. A hand-written ratio is not protected.
Explore	po_lines for reconciliation. exceptions for the queue. assets for first-time-right, so the orphan stays in. completeness and field_edits because those grains must not join into the line.
LookML	The model in lookml/. Production would be a merge. This lab is the files, not a merge you have done.
PDT	exception_aging and person_dependent_fields. Derived tables Looker builds in the warehouse, refreshed by the datagroup. Not stored inside Looker.
Access filter	User attribute program_ids on dim_program.program_id. Jordan cannot take it off.
Datagroup	capex_daily. Trigger on the newest asset timestamp, and never older than 24 hours.
First-time-right	Passed validation on the first submit. A later correction does not flip it.
Person-dependent	A field written by exactly one user in the 90 days ending 2026-03-16, inside the program.
Looker Studio	A different product. Free, no LookML, no Git-backed measure. Fine for the five tiles. Not this model.

Files and the numbers to expect after `python3 run_local.py`. Before the fix unless the path says `after_fix/`.

FILE	EXPECT
data/expected_outputs/go_nogo.csv	43 submitted, 2 quarantined, 40 lines, rate about 0.0465, decision GO
quarantine_rows.csv	00901 unknown site; blank PO line and bad date
tracker_vs_fact.csv	0, 0, and 1 orphan not in the tracker. After the fix: 1 linked-asset mismatch
duplicate_serials.csv	2 serials
orphans.csv	AST-TS-0040
site_mismatches.csv	3 rows
date_logic.csv	1 row, line 00019
referential.csv	orphan 1, everything else 0
reconciliation_by_program.csv	ramp 0.95, pack 1.0, fixture 1.0
reconciliation_traps.csv	0.95 certified; 0.975; 1.0; 1.0
fanout_program.csv	41 joined rows, 40 distinct lines, 39 assets
fanout_line_00018.csv	qty 2, plain count 2, distinct lines 1
missing_pis.csv	ramp 12 of 40. After the fix, 11
completeness_by_field_owner.csv	placed-in-service 28/40. After the fix, 29/40
reuse_denominator_trap.csv	certified 1.0 on 12; trap 0.3 on 40
exception_aging.csv	7, 7, 7. After the fix, 6, 7, 7
first_time_right.csv	21/40 = 0.525, unchanged after the fix

FILE	EXPECT
first_time_right_trap.csv	21/39 = 0.5385
first_time_right_all.csv	81/100 = 0.81 with the filter off. Not the same 81 as any other sketch
person_dependent_fields.csv	3 true. Without the window, program allocation is false (2 editors)
rls_by_user.csv	Jordan 0.95, Morgan 0.9722, Priya 0.98
rls_jordan_pack.csv	0
scorecard.csv	0.95, missing 12, 7/7/7, 0.525, 3 person-dependent, alert true
after_fix/scorecard.csv	0.95, missing 11, 6/7/7, 0.525, 3, alert true
fix_proof_00017.csv	Before: blank date, flag 0, closed blank. After: 2026-02-17, flag 0, closed 2026-03-16
dashboard_mock.html	The five tiles, from the open queue, plus the one-line after
lookml/	The model you would commit. Not a model you have run in Looker unless you did Path B

Regenerate with `python3 generate_data.py && python3 run_local.py` from this directory. If a number in this guide and a CSV disagree, believe the CSV. The SQL is the definition.